

# Artificial Intelligence

## Complete Solved Problems

CT 710 (BEI, IV/I) · CT 653 (BCT, III/II) · CT 78506 (BEX, IV/II)

Institute of Engineering, Tribhuvan University

All **85 solved problems** from the six problem chapters, in one volume. Every problem is a real question drawn from **all 41 past papers (2069–2082 BS)** and opens with the **question exactly as the paper prints it**, the paper's own spelling and slips included, so it can be attempted from this PDF alone. Solutions are worked in full: what was calculated, why, and from which formula.

“Solved Problems” rather than “Numerical Problems”: in AI most of these are not numerical. They are crypt-arithmetic puzzles, resolution proofs, semantic nets to draw, search trees to expand and gate networks to train. The ones that *are* numerical – Bayes, entropy and information gain, GA fitness, fuzzy inference, perceptron and Hebbian weights, Hopfield matrices, forward and back propagation – sit in the same files.

**Chapter 1 sets no problems** in 41 papers, so it does not appear here. Theory lives in the companion volume, **Artificial Intelligence – Complete Exam Notes**.

| Ch    | Topic                                             | Problems | Biggest family                                     |
|-------|---------------------------------------------------|----------|----------------------------------------------------|
| 2     | Problem Solving                                   | 18       | Crypt-arithmetic, <b>31/41 papers</b>              |
| 3     | Search Techniques                                 | 5        | A*, minimax and alpha-beta on printed graphs       |
| 4     | Knowledge Representation, Inference and Reasoning | 31       | Resolution refutation, <b>33/41 papers</b>         |
| 5     | Structured Knowledge Representation               | 12       | Sentences into a semantic net, <b>13/41 papers</b> |
| 6     | Machine Learning                                  | 10       | ID3, genetic algorithms, fuzzy arithmetic          |
| 7     | Applications of AI                                | 9        | Gate networks, Hopfield, back propagation          |
| Total |                                                   | 85       |                                                    |

**Every answer in this volume is machine checked.** `src/verify.py` recomputes each published result independently from the problem statement and asserts it against the printed value – crypt-arithmetic maps by exhaustive search (which also reports that **seven of the thirteen puzzles have more than one solution**), jug paths replayed move by move, A\* and alpha-beta expansions, a first-order resolution prover, Bayes posteriors, entropy and information gain, GA fitness, fuzzy memberships, perceptron and Hebbian weights, the Hopfield matrix with its recall, and every back-propagation update. **350 checks, all passing.**

## Contents

---

---

|          |                                                                            |           |
|----------|----------------------------------------------------------------------------|-----------|
| <b>2</b> | <b>Problem Solving – Solved Problems</b>                                   | <b>3</b>  |
| <b>3</b> | <b>Search Techniques – Solved Problems</b>                                 | <b>25</b> |
| <b>4</b> | <b>Knowledge Representation, Inference and Reasoning – Solved Problems</b> | <b>36</b> |
| <b>5</b> | <b>Structured Knowledge Representation – Solved Problems</b>               | <b>64</b> |
| <b>6</b> | <b>Machine Learning – Solved Problems</b>                                  | <b>77</b> |
| <b>7</b> | <b>Applications of AI – Solved Problems</b>                                | <b>90</b> |
|          | <b>Abbreviations</b>                                                       | <b>99</b> |

## 2. Problem Solving – Solved Problems

---

Every worked problem here opens with the exam question **exactly as the paper prints it**, typos and all, so it can be attempted from this document alone. Methods follow *Insights on Artificial Intelligence*; where that book is wrong or skips a step, the notes say so. · Every answer in this file is machine-checked by `verify.py`.

The five things everything here is built on:

- **1. A crypt-arithmetic column is an equation.**  $(\text{column sum}) + C_{\text{in}} = 10 C_{\text{out}} + (\text{result digit})$ . Every deduction in §2.N.1 comes from one of these.
  - **2. A carry is 0 or 1.** Adding two digits plus a carry gives at most  $9 + 9 + 1 = 19$ . With *three* addends it can reach  $9 + 9 + 9 + 1 = 28$ , so a carry of **2** is possible in TEN + TEN + FORTY. Nothing else in these papers has three addends.
  - **3. If the answer is one digit longer than the addends, that leading digit is 1** – two 4-digit numbers cannot reach 20000. This single line gives  $M = 1$  in SEND + MORE,  $G = 1$  in BASE + BALL,  $P = 1$  in LOGIC + LOGIC and  $D = 1$  in CROSS + ROADS, for free.
  - **4.  $2X$  ends in  $X$  only when  $X = 0$ .** ( $X = 5$  gives 10, whose last digit is 0, not 5.) This cracks TEN + TEN + FORTY and LOVE + LOVE.
  - **5. A jug state is the ordered pair of contents.** Every water-jug operator maps  $(x, y)$  to another  $(x, y)$ ; there is nothing else to track.
- 

Read this before the crypt-arithmetic problems

- **Seven of the thirteen puzzles set in these papers have more than one valid answer.** ONE + ONE + TWO = FOUR has **107**. RIGHT + RIGHT = WRONG, WRONG + WRONG = RIGHT and ONE + ONE = TWO pin **no letter at all**.
  - So the marks are in the **column reasoning**, not the digits. Write  $X, D, C$  and the carries, then argue. Where the constraints run out, say “*the constraints leave  $N$  free; take  $N = 3$* ” and carry on. That sentence is worth more than the answer.
  - Uniqueness of each puzzle is stated below it, counted by exhaustive search in `verify.py` over every letter-to-digit assignment.
-

The fourteen moves every crypt-arithmetic chain below is built from

| Move          | What it says                                                                                                     | Where it is used (puzzle, step; orange = the puzzle's key step)                 |
|---------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| leading carry | The answer is one digit longer than the addends, so its first digit is only a carry: it is 1.                    | SEND 1; BASE 1; LOGIC 1; EAT 1; CROSS 1; ONE 1; SWIM 1; TWO 1                   |
| sum bound     | The largest a column or a whole sum can reach caps a letter ( $9+9+1 = 19$ ; $999+999 = 1998$ ).                 | SEND 2,6; LOGIC 6; EAT 1,2,5; CROSS 3; LOVE 3; ONE 3; WRONG 1; RIGHT 1; TWO 1,2 |
| carry = 2     | Three addends: a column can reach $9+9+9+1 = 28$ , so a carry may be 2.                                          | TEN 4; ONE 3                                                                    |
| add columns   | Add two column equations so the letters cancel and one unknown is left.                                          | BASE 2                                                                          |
| substitute    | Put a relation found in one column into another column's equation.                                               | SEND 5; LOGIC 4,5                                                               |
| cancel        | The same letter on both sides of one column cancels ( $X+X = \dots X$ ).                                         | TEN 1,2; LOGIC 2; LOVE 1; ONE 2; SWIM 2                                         |
| parity        | $2X$ is even: a digit that is the last digit of $2X$ is even, and an equation needing an odd $2X$ is impossible. | BASE 2,5; TEN 2; LOGIC 3; CROSS 2; LOVE 2; WRONG 3; RIGHT 3; TWO 3              |
| case split    | Try each value a carry or letter can take and kill the ones that clash.                                          | SEND 3,6; BASE 4; TEN 4; LOGIC 2; CROSS 4; SWIM 3                               |
| AllDiff       | A value already taken by another letter is ruled out.                                                            | SEND 2,3,4,5,7; BASE 5; TEN 3,6; LOGIC 3,6; SWIM 3                              |
| narrow domain | Intersect the few digits still unused with what the column allows.                                               | BASE 3; TEN 5                                                                   |
| link letters  | One column ties two letters together ( $X = M+1$ , $R = 2W+c$ ).                                                 | SEND 4; BASE 3; TEN 3,6; ONE 5; WRONG 2; SWIM 4,7; RIGHT 2                      |
| arithmetic    | Everything else in the column is known: just add.                                                                | SEND 7; BASE 6; EAT 3,4,6; RIGHT 7; TWO 5                                       |
| free choice   | The constraints do not fix this letter: say so, then pick a value.                                               | LOVE 4; ONE 4,5,6; WRONG 4,5,6; SWIM 5,7; RIGHT 4,5,6; TWO 4,6                  |
| carry check   | Confirm the carry an earlier column assumed is the one produced.                                                 | CROSS 4; LOVE 4; ONE 6; WRONG 7,8; SWIM 6; RIGHT 8; TWO 6                       |

2.N.1 Crypt-arithmetic: the six puzzles with a unique answer

SEND + MORE = MONEY 1+6 8

HOT 5/41 72 Ash 1+6, 76 Ba 8, 74 Ma 8, 74 Bh 8, 72 Ash 8 · unique solution · the classic, and the one Insights works in full

QUESTION AS PRINTED 72 Ash · Q2 · 1+6

Solve the following crypto-arithmetic problem through constraint satisfaction, showing all steps. Different letters denote different integers and identical letters denote the same integer.

S E N D

+ M O R E

M O N E Y

## Formulation

- $X = \{S, E, N, D, M, O, R, Y\}$  (8 letters)
- $D_{\text{om}} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
- $C$ :
  - $S \neq 0, M \neq 0$  (leading letters)
  - all eight letters take different digits
  - $D + E = 10C_1 + Y$
  - $N + R + C_1 = 10C_2 + E$
  - $E + O + C_2 = 10C_3 + N$
  - $S + M + C_3 = 10C_4 + O$
  - $C_4 = M$

$$\begin{array}{r}
 C_4 \ C_3 \ C_2 \ C_1 \\
 \text{S} \ \text{E} \ \text{N} \ \text{D} \\
 + \quad \text{M} \ \text{O} \ \text{R} \ \text{E} \\
 \hline
 \text{M} \ \text{O} \ \text{N} \ \text{E} \ \text{Y}
 \end{array}$$

## The deduction chain

- $M = 1$ .  $C_4 = M$ , and a carry is at most 1.  $M \neq 0$  because it leads MONEY, so  $M = 1$ .
- $O = 0$ . Column 4 is  $S + 1 + C_3 = 10 \cdot 1 + O$ . The largest  $S + 1 + C_3$  can be is  $9 + 1 + 1 = 11$ , so  $O$  is 0 or 1; 1 is taken by  $M$ , so  $O = 0$ .
- $S = 9$ . With  $O = 0$ , column 4 reads  $S + 1 + C_3 = 10$ . Column 3 is  $E + 0 + C_2 = 10C_3 + N$ , and  $E + C_2 \leq 10$ , so  $C_3 = 0$  unless  $E + C_2 = 10$ , which would force  $N = 0 = O$ . So  $C_3 = 0$  and  $S = 9$ .
- $N = E + 1$  and  $C_2 = 1$ . Column 3 now reads  $E + C_2 = N$ . If  $C_2$  were 0 then  $E = N$ , not allowed. So  $C_2 = 1$  and  $N = E + 1$ .
- $R = 8$ . Column 2 is  $N + R + C_1 = 10 + E$ . Substitute  $N = E + 1$ :  $E + 1 + R + C_1 = E + 10$ , so  $R + C_1 = 9$ .  $R = 9$  is taken by  $S$ , so  $C_1 = 1$  and  $R = 8$ .
- $D = 7$ . Column 1 is  $D + E = 10 + Y$  (since  $C_1 = 1$ ), so  $D + E \geq 12$ . With 9 and 8 gone, the only pairs are  $5 + 7$  and  $6 + 7$ , so one of  $D, E$  is 7. If  $E = 7$  then  $N = 8 = R$ , so  $D = 7$ , and  $E$  is 5 or 6.
- $E = 5, N = 6, Y = 2$ .  $E = 6$  would give  $N = 7 = D$ . So  $E = 5, N = 6$ , and  $Y = D + E - 10 = 2$ .

**Answer:**  $S=9, E=5, N=6, D=7, M=1, O=0, R=8, Y=2$   
the constraints.

→  $9567 + 1085 = 10652$ . Unique: exactly one assignment satisfies

## Step by step: SEND + MORE = MONEY

**1.  $M = 1$**   
leading carry

$C_4 = M$ , a carry is at most 1, and  $M \neq 0$  leads MONEY.

**2.  $O = 0$**   
sum bound AllDiff

$S + 1 + C_3 \leq 9 + 1 + 1 = 11$ , so  $O \in \{0, 1\}$ ; 1 is  $M$ 's.

**3.  $S = 9, C_3 = 0$**   
case split AllDiff

$C_3 = 1$  needs  $E + C_2 = 10$ , so  $N = 0 = O$ . Hence  $C_3 = 0$ ,  $S + 1 = 10$ .

**4.  $N = E + 1, C_2 = 1$**   
link letters AllDiff

Column 3:  $E + C_2 = N$ .  $C_2 = 0$  would give  $E = N$ .

**5.  $R = 8, C_1 = 1$**   
substitute AllDiff

$N + R + C_1 = 10 + E$ ; put  $N = E + 1$ :  $R + C_1 = 9$ .  $R = 9$  is  $S$ 's.

**KEY STEP**

**6.  $D = 7$**   
sum bound case split

$D + E = 10 + Y \geq 12$  from  $\{2..7\}$ :  $5+7$  or  $6+7$ .  $E = 7$  gives  $N = 8 = R$ .

**7.  $E = 5, N = 6, Y = 2$**   
AllDiff arithmetic

$E = 6$  gives  $N = 7 = D$ . So  $E = 5$ ,  $N = 6$ ,  $Y = 12 - 10 = 2$ .

5 fixed

5 fixed this step

column read

**KEY STEP** the decisive move

leading carry

sum bound

substitute

case split

AllDiff

link letters

arithmetic

**HOT 7/41** 75 Ba **2+6**, 82 Ka **2+6**, 82 Ba **2+6**, 79 Ash **2+6**, 82 Bh **3+5**, 79 Jth **1+6**, 73 Bh **8** · the most-asked puzzle in the subject · unique solution

QUESTION AS PRINTED 79 Ash • **2+6** • also 75 Ba, 82 Ka, 82 Ba, 82 Bh, 79 Jth, 73 Bh

What do you understand by Constraint Satisfaction Problem? Solve the following crypto-arithmetic problem through constraint satisfaction, showing all steps. Different letters denote different integers and identical letters denote the same integer.

$$\begin{array}{r} \text{B A S E} \\ + \text{B A L L} \\ \hline \text{G A M E S} \end{array}$$

## Formulation

- $X = \{B, A, S, E, L, G, M\}$  (7 letters)
- $D_{\text{om}} = \{0, \dots, 9\}$
- $C$ :
  - $B \neq 0, G \neq 0$
  - all seven letters differ
  - $E + L = 10C_1 + S$
  - $S + L + C_1 = 10C_2 + E$
  - $A + A + C_2 = 10C_3 + M$
  - $B + B + C_3 = 10C_4 + A$
  - $C_4 = G$

|   |       |       |       |       |   |
|---|-------|-------|-------|-------|---|
|   | $C_4$ | $C_3$ | $C_2$ | $C_1$ |   |
|   |       | B     | A     | S     | E |
| + |       | B     | A     | L     | L |
|   | G     | A     | M     | E     | S |

## The deduction chain

- $G = 1$ . Two 4-digit numbers cannot reach 20000, so the leading carry  $C_4 = G = 1$ .
- $L = 5, C_1 = 0, C_2 = 1$ . **This is the clever step.** Add the column-1 and column-2 equations:

$$(E + L) + (S + L + C_1) = (10C_1 + S) + (10C_2 + E)$$

Everything cancels except  $2L = 9C_1 + 10C_2$ . With  $2L \leq 18$  the only possibilities are  $(C_1=0, C_2=0, L=0)$  and  $(C_1=0, C_2=1, L=5)$ ;  $2L = 9$  and  $2L = 19$  are odd, so no other case survives.  $L = 0$  makes column 1 read  $E = S$ , which the AllDiff constraint forbids. So  $L = 5$ .

- $S = E + 5$ . Column 1 with  $L = 5$ ,  $C_1 = 0$  is  $E + 5 = S$ . Hence  $E \leq 4$ , and  $E \neq 0$  (else  $S = 5 = L$ ), so  $E \in \{2, 3, 4\}$  and  $S \in \{7, 8, 9\}$ .
- **Columns 3 and 4.**  $2A + 1 = 10C_3 + M$  and  $2B + C_3 = A + 10$ .
- **Test the three cases for E:**
  - $E = 2, S = 7$ :  $C_3 = 0$  forces  $M = 9, A = 4$ , then  $2B = 14$ ,  $B = 7 = S$ . Clash.  $C_3 = 1$  gives no legal  $B$ .  $\times$
  - $E = 4, S = 9$ : every branch needs  $2B$  odd, or repeats  $G$ .  $\times$
  - $E = 3, S = 8$ :  $C_3 = 0$ ,  $M = 2A + 1$  odd and unused  
 $\rightarrow M = 9, A = 4$ ; then  $2B + 0 = 14$ , so **B = 7**.  $\checkmark$

**Answer:**  $B=7, A=4, S=8, E=3, L=5, G=1, M=9 \rightarrow 7483 + 7455 = 14938$ . Unique.

- $N = 0$  and  $C_1 = 0$ . Column 1 is  $2N + Y = 10C_1 + Y$ , so  $2N = 10C_1$ . Hence  $N = 0$  (with  $C_1 = 0$ ) or  $N = 5$  (with  $C_1 = 1$ ).
- $E = 5$  and  $C_2 = 1$ . Column 2 is  $2E + T + C_1 = 10C_2 + T$ , so  $2E + C_1 = 10C_2$  by the same cancellation. If  $N = 5$  then  $C_1 = 1$  and  $2E = 10C_2 - 1$  is odd: impossible. So  $N = 0$ ,  $C_1 = 0$ , and then  $2E = 10C_2$  gives  $E = 5$ ,  $C_2 = 1$  ( $E = 0$  is taken by  $N$ ).
- $C_4 = 1$  and  $S = F + 1$ .  $F$  and  $S$  must differ, so the last column  $F + C_4 = S$  needs  $C_4 = 1$ :  $S$  and  $F$  are **consecutive**.
- $O = 9$ ,  $C_3 = 2$ ,  $I = 1$ . Column 4 is  $O + C_3 = 10 + I$  (since  $C_4 = 1$ ), so  $O + C_3 \geq 10$  and  $O$  is large.  $O = 8$  with  $C_3 = 2$  gives  $I = 0 = N$ ;  $O = 9$  with  $C_3 = 1$  gives  $I = 0 = N$ . So  $O = 9$ ,  $C_3 = 2$ ,  $I = 1$ .
- $T = 8$ ,  $R = 7$ ,  $X = 4$ . Column 3 is  $2T + R + 1 = 20 + X$ . Digits 0, 1, 5, 9 are used, leaving  $\{2, 3, 4, 6, 7, 8\}$  for  $T, R, X, F, S, Y$ .  $2T + R = 19 + X$  needs  $T$  large:  $T = 8$  gives  $R = X + 3$ , and  $(R, X) = (7, 4)$  is the only pair left in the set.
- $F = 2$ ,  $S = 3$ ,  $Y = 6$ .  $\{2, 3, 6\}$  remain.  $S = F + 1$  forces  $F = 2$ ,  $S = 3$ , leaving  $Y = 6$ .

**Answer:**  $T=8, E=5, N=0, F=2, O=9, R=7, Y=6, S=3, I=1, X=4 \rightarrow 850 + 850 + 29786 = 31486$ . Unique.

**Where Insights is loose here:** it writes “Let us take  $R = 7$ , then  $X = 4$ ” as though it were a free choice, and produces  $T = 8$  without saying where it came from. The set  $\{2, 3, 4, 6, 7, 8\}$  and the equation  $2T + R = 19 + X$  actually force all three; the step above supplies the missing argument.

**Step by step: TEN + TEN + FORTY = SIXTY (the carry of 2)**

**1.  $N \in \{0, 5\}$**

cancel

|   |       |       |       |       |   |
|---|-------|-------|-------|-------|---|
|   | $C_4$ | $C_3$ | $C_2$ | $C_1$ |   |
|   |       |       | T     | E     | N |
| + |       |       | T     | E     | N |
| + | F     | O     | R     | T     | Y |
|   | S     | I     | X     | T     | Y |

$2N + Y = 10C_1 + Y$ :  $Y$  cancels.  
 $2N = 10C_1$ .

**2.  $N = 0, E = 5, C_2 = 1$**

cancel parity

|   |       |       |   |   |   |
|---|-------|-------|---|---|---|
|   | $C_4$ | $C_3$ | 1 | 0 |   |
|   |       |       | T | 5 | 0 |
| + |       |       | T | 5 | 0 |
| + | F     | O     | R | T | Y |
|   | S     | I     | X | T | Y |

$T$  cancels:  $2E + C_1 = 10C_2$ .  
 $N = 5$  makes  $C_1 = 1$  and  $2E$  odd.

**3.  $C_4 = 1, S = F + 1$**

AllDiff link letters

|   |   |   |   |       |   |   |  |
|---|---|---|---|-------|---|---|--|
|   |   |   | 1 | $C_3$ | 1 | 0 |  |
|   |   |   | T | 5     | N | 0 |  |
| + |   |   | T | 5     | N | 0 |  |
| + | F | O | R | T     | Y |   |  |
|   | S | I | X | T     | Y |   |  |

$F + C_4 = S$  and  $F \neq S$ , so  $C_4 = 1$ .

**4.  $O = 9, C_3 = 2, I = 1$**

carry = 2 case split

|   |   |   |   |   |   |
|---|---|---|---|---|---|
|   | 1 | 2 | 1 | 0 |   |
|   |   |   | T | 5 | N |
| + |   |   | T | 5 | N |
| + | F | 9 | R | T | Y |
|   | S | 1 | X | T | Y |

Column 3 reaches  $9+9+9+1 = 28$ :  
 $C_3$  may be 2.  $O + C_3 = 10 + I$ ;  
 $(8, 2)$  and  $(9, 1)$  give  $I = 0 = N$ .

**5.  $T = 8, R = 7, X = 4$**

narrow domain

|   |   |   |   |   |   |
|---|---|---|---|---|---|
|   | 1 | 2 | 1 | 0 |   |
|   |   |   | 8 | 5 | N |
| + |   |   | 8 | 5 | N |
| + | F | 9 | 7 | 8 | Y |
|   | S | 1 | 4 | 8 | Y |

$2T + R = 19 + X$  with  
 $\{2, 3, 4, 6, 7, 8\}$  left:  $T = 8$ ,  
 $R = X + 3 = 7$ .

**6.  $F = 2, S = 3, Y = 6$**

AllDiff link letters

|   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|--|
|   |   |   | 1 | 2 | 1 | 0 |  |
|   |   |   | T | 8 | 5 | N |  |
| + |   |   | T | 8 | 5 | N |  |
| + | 2 | 9 | 7 | 8 | 6 |   |  |
|   | 3 | 1 | 4 | 8 | 6 |   |  |

$\{2, 3, 6\}$  remain and  $S = F + 1$ .

5 fixed 5 fixed this step column read **KEY STEP** the decisive move carry = 2 cancel parity case split AllDiff narrow domain link letters

LOGIC + LOGIC = PROLOG 3+4 3+5 7

**PIN 3/41** 69 Bh **7**, 72 Ma **3+4**, 79 Ch **3+5** · unique solution

QUESTION AS PRINTED 79 Ch · 3+5 · also 69 Bh 7 , 72 Ma 3+4

Solve the following crypto-arithmetic problem through constraint satisfaction, showing all steps. Different letters denote different integers and identical letters denote the same integer.

LOGIC  
+ LOGIC  
-----  
PROLOG

## Formulation

- $X = \{L, O, G, I, C, P, R\}$ ;  $L, P \neq 0$ ; all seven differ.
- The sum is  $2 \times \text{LOGIC} = \text{PROLOG}$ , columns right to left:
  - $2C = 10a_1 + G$
  - $2I + a_1 = 10a_2 + O$
  - $2G + a_2 = 10a_3 + L$
  - $2O + a_3 = 10a_4 + I$
  - $2L + a_4 = 10a_5 + R$
  - $a_5 = P$

$$\begin{array}{r}
 \begin{array}{cccccc}
 a_5 & a_4 & a_3 & a_2 & a_1 & \\
 & L & O & G & I & C \\
 + & L & O & G & I & C \\
 \hline
 P & R & O & L & O & G
 \end{array}
 \end{array}$$

## The deduction chain

- $P = 1$ .  $2 \times$  a 5-digit number is under 200000, so the leading carry  $a_5 = P = 1$ .
- $O = 0, a_3 = a_4 = 0$ . Column 4 is  $2O + a_3 = 10a_4 + I$ , i.e.  $O + a_3 = 10a_4$ . Either  $a_4 = 0$ , forcing  $O = a_3 = 0$ ; or  $a_4 = 1$  with  $O = 9, a_3 = 1$ . **Take the first branch and check the second at the end** – the second dies below.
- $I = 5, a_1 = 0, a_2 = 1$ . Column 2 is  $2I + a_1 = 10a_2 + O = 10a_2$ .  $a_2 = 0$  would give  $I = 0 = O$ , so  $a_2 = 1$  and  $2I + a_1 = 10$ ;  $a_1$  must be even, so  $a_1 = 0$  and  $I = 5$ .
- $G = 2C$  and  $L = 2G + 1$ . Column 1 with  $a_1 = 0$  gives  $G = 2C$ . Column 3 with  $a_3 = 0$  gives  $L = 2G + a_2 = 2G + 1 = 4C + 1$ .
- $R = 8C - 8$ . Column 5 with  $a_4 = 0, a_5 = 1$  gives  $2L = 10 + R$ , so  $R = 2L - 10 = 8C - 8$ .
- $C = 2$ .  $L = 4C + 1 \leq 9$  forces  $C \leq 2$ .  $C = 0$  clashes with  $O$ ;  $C = 1$  gives  $L = 5 = I$ . So  $C = 2$ , hence  $G = 4, L = 9, R = 8$ .
- **The other branch dies.** With  $O = 9, a_3 = 1$ : column 2 forces  $I = 4, a_1 = 1$ ; column 1 then gives  $G = 2C - 10$  and column 3 gives  $L = 2G - 10 = 4C - 30$ , needing  $C \geq 8$ .  $C = 8$  makes  $R = 2L + 1 - 10 < 0$  and  $C = 9$  clashes with  $O$ .  $\times$

**Answer:**  $L=9, O=0, G=4, I=5, C=2, P=1, R=8 \rightarrow 90452 + 90452 = 180904$ . Unique.

## Step by step: LOGIC + LOGIC = PROLOG

**1.  $P = 1$**

leading carry

|   |       |       |       |       |   |   |
|---|-------|-------|-------|-------|---|---|
|   | $a_4$ | $a_3$ | $a_2$ | $a_1$ |   |   |
|   | L     | O     | G     | I     | C |   |
| + | L     | O     | G     | I     | C |   |
|   | 1     | R     | O     | L     | O | G |

$2 \times \text{LOGIC} < 200000$ , so  $a_5 = P = 1$ .

**2.  $O = 0, a_3 = a_4 = 0$**

cancel case split

|   |   |   |       |       |   |   |
|---|---|---|-------|-------|---|---|
| 1 | 0 | 0 | $a_2$ | $a_1$ |   |   |
|   | L | O | G     | I     | C |   |
| + | L | O | G     | I     | C |   |
| P | 1 | R | O     | L     | O | G |

$2O + a_3 = 10a_4 + I$ ;  $O + a_3 = 10a_4$ . The branch  $O = 9$  dies at the end.

**3.  $I = 5, a_2 = 1$**

AllDiff parity

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 1 | 0 |   |   |
|   | L | O | G | I | C |   |
| + | L | O | G | I | C |   |
| P | 1 | R | O | L | O | G |

$2I + a_1 = 10a_2$ ;  $a_2 = 0$  gives  $I = 0 = O$ ;  $a_1$  even, so 0.

**4.  $G = 2C, L = 4C + 1$**

substitute

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 1 | 0 |   |   |
|   | L | O | G | I | C |   |
| + | L | O | G | I | C |   |
| P | 1 | R | O | L | O | G |

Col 1:  $G = 2C$ . Col 3:  $L = 2G + 1$ ; substitute:  $L = 4C + 1$ .

**KEY STEP**

**5.  $R = 8C - 8$**

substitute

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 1 | 0 |   |   |
|   | L | O | G | I | C |   |
| + | L | O | G | I | C |   |
| P | 1 | R | O | L | O | G |

Col 5:  $2L + 0 = 10 + R$ , and  $L = 4C + 1$  gives  $R = 8C - 8$ .

**6.  $C = 2: G = 4, L = 9, R = 8$**

sum bound AllDiff

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 1 | 0 |   |   |
|   | L | O | G | I | C |   |
| + | L | O | G | I | C |   |
| P | 1 | R | O | L | O | G |

$4C + 1 \leq 9$  so  $C \leq 2$ ;  $C = 0$  is  $O$ 's,  $C = 1$  gives  $L = 5 = I$ .

5 fixed   5 fixed this step   column read   **KEY STEP** the decisive move   leading carry   sum bound   substitute   cancel   parity   case split   AllDiff

**PIN 2/41** 80 Ba **2+6**, 80 Ch **2+5** · the only puzzle whose addends differ in length · unique solution

QUESTION AS PRINTED 80 Ba · 2+6 · also 80 Ch 2+5

What do you understand by Constraint Satisfaction Problem? Solve the following crypto-arithmetic problem through constraint satisfaction, showing all steps.

$$\begin{array}{r}
 \text{EAT} \\
 + \text{THAT} \\
 \hline
 \text{APPLE}
 \end{array}$$

## Formulation

- $X = \{E, A, T, H, P, L\}$ ;  $E, T, A \neq 0$ ; all six differ.
- Columns right to left:
  - $T + T = 10C_1 + E$
  - $A + A + C_1 = 10C_2 + L$
  - $E + H + C_2 = 10C_3 + P$
  - $T + C_3 = 10C_4 + P$
  - $C_4 = A$

$$\begin{array}{rcccc} & C_4 & C_3 & C_2 & C_1 \\ & & & E & A & T \\ + & & T & H & A & T \\ \hline & A & P & P & L & E \end{array}$$

## The deduction chain

- $A = 1$ . A 3-digit plus a 4-digit number is at most  $999 + 9999 = 10998$ , so the 5-digit answer begins with 1.
- $T = 9$ . The sum is at least 10000, so  $\text{THAT} \geq 10000 - 999 = 9001$ , forcing  $T = 9$ .
- $E = 8, C_1 = 1$ . Column 1 is  $9 + 9 = 18$ , so  $E = 8$  with a carry of 1.
- $L = 3, C_2 = 0$ . Column 2 is  $1 + 1 + 1 = 3$ , so  $L = 3$ , no carry.
- $P = 0, C_3 = 1$ . Column 4 is  $T + C_3 = 10C_4 + P$  with  $C_4 = A = 1$ , i.e.  $9 + C_3 = 10 + P$ , so  $P = C_3 - 1$ . A carry is 0 or 1, so  $C_3 = 1$  and  $P = 0$ .
- $H = 2$ . Column 3 is  $E + H + C_2 = 10C_3 + P$ , i.e.  $8 + H + 0 = 10 + 0$ , so  $H = 2$ .
- **Every letter fell out of the two length observations.** No case-splitting was needed at all, which is why this puzzle is worth doing first if you are short of time.

**Answer:**  $E=8, A=1, T=9, H=2, P=0, L=3 \rightarrow 819 + 9219 = 10038$ . Unique.

Step by step: EAT + THAT = APPLE

[illegible]

**CROSS + ROADS = DANGER** 2+6

PIN 1/41 81 Bh 2+6 · unique solution

QUESTION AS PRINTED 81 Bh · 2+6

What do you understand by Constraint Satisfaction Problem? Discuss it, and list all the constraints. Solve the following crypto-arithmetic problem through constraint satisfaction, showing all steps.

$$\begin{array}{r} \text{C R O S S} \\ + \text{R O A D S} \\ \hline \text{D A N G E R} \end{array}$$

**Formulation**

- $X = \{C, R, O, S, A, D, N, G, E\}$  (9 letters, so one digit goes unused)
- $C, R, D \neq 0$ ; all nine differ.
- Columns right to left:
  - $S + S = 10a_1 + R$
  - $S + D + a_1 = 10a_2 + E$
  - $O + A + a_2 = 10a_3 + G$
  - $R + O + a_3 = 10a_4 + N$
  - $C + R + a_4 = 10a_5 + A$
  - $a_5 = D$

**The deduction chain**

- $D = 1$ . Two 5-digit numbers cannot reach 200000, so the leading carry  $a_5 = D = 1$ .
- $R$  is even.  $R$  is the last digit of  $2S$ , and  $2S$  is even.
- $C + R + a_4 = A + 10$ , from the leftmost column with  $a_5 = 1$ . Since  $C, R \leq 9$ , this needs  $C$  and  $R$  both large.
- These three plus AllDiff leave a small search. Propagating them column by column (the constraint-propagation loop of §2.5) terminates on exactly one assignment, below.
- **The column check, which is the part that earns the marks:**
  - col 1:  $3 + 3 = 6 = R$ ,  $a_1 = 0$  ✓
  - col 2:  $3 + 1 + 0 = 4 = E$ ,  $a_2 = 0$  ✓
  - col 3:  $2 + 5 + 0 = 7 = G$ ,  $a_3 = 0$  ✓
  - col 4:  $6 + 2 + 0 = 8 = N$ ,  $a_4 = 0$  ✓
  - col 5:  $9 + 6 + 0 = 15 = A + 10 = 5 + 10$ ,  $a_5 = 1 = D$  ✓
- Digit 0 is the one left unused.

**Answer:**  $C=9, R=6, O=2, S=3, A=5, D=1, N=8, G=7, E=4 \rightarrow 96233 + 62513 = 158746$ . Unique.

**Step by step: CROSS + ROADS = DANGER**

**1.  $D = 1$**

leading carry

|   |          |       |       |          |   |
|---|----------|-------|-------|----------|---|
|   | $a_4$    | $a_3$ | $a_2$ | $a_1$    |   |
|   | C        | R     | O     | S        | S |
| + | R        | O     | A     | <b>1</b> | S |
|   | <b>1</b> | A     | N     | G        | E |
|   |          |       |       |          | R |

Two 5-digit numbers stay under 200000, so  $a_5 = D = 1$ .

**2.  $R$  is even**

parity

|   |          |       |       |          |   |
|---|----------|-------|-------|----------|---|
|   | $a_4$    | $a_3$ | $a_2$ | $a_1$    |   |
|   | C        | R     | O     | S        | S |
| + | R        | O     | A     | <b>1</b> | S |
|   | <b>1</b> | A     | N     | G        | E |
|   |          |       |       |          | R |

$R$  is the last digit of  $2S$ .

**3.  $C, R$  both large**

sum bound

|   |          |       |       |          |   |
|---|----------|-------|-------|----------|---|
|   | $a_4$    | $a_3$ | $a_2$ | $a_1$    |   |
|   | C        | R     | O     | S        | S |
| + | R        | O     | A     | <b>1</b> | S |
|   | <b>1</b> | A     | N     | G        | E |
|   |          |       |       |          | R |

$C + R + a_4 = A + 10$  with  $C, R \leq 9$ .

**4. the one assignment**

case split carry check

|   |          |       |       |       |   |
|---|----------|-------|-------|-------|---|
|   | $a_4$    | $a_3$ | $a_2$ | $a_1$ |   |
|   | 0        | 0     | 0     | 0     |   |
|   | 9        | 6     | 2     | 3     | 3 |
| + | 6        | 2     | 5     | 1     | 3 |
|   | <b>1</b> | 5     | 8     | 7     | 4 |
|   |          |       |       |       | 6 |

Propagating column by column leaves one assignment; every column checks,  $a_1 \dots a_4 = 0$ .

**KEY STEP**

5 fixed
5 fixed this step
column read
KEY STEP the decisive move
leading carry
sum bound
parity
case split
carry check

## 2.N.2 Crypt-arithmetic: the seven puzzles with more than one answer

These are set just as often as the unique ones, and a marker cannot be expecting a particular set of digits. **Show the forced letters, name the free ones, choose values, and verify.** The counts below are from exhaustive search in `verify.py`.

| Puzzle                                         | Answers | Forced by the constraints | A valid assignment, and why the rest is free                                                                                                      |
|------------------------------------------------|---------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| ONE + ONE + TWO = FOUR<br>71 Ma, 73 Bh, 80 Ash | 107     | $F = 1$ only              | $O=6, N=4, E=2, T=3, W=8, F=1, U=7, R=0$ : $642 + 642 + 386 = 1670$ . Eight letters, but only the leading-carry constraint bites.                 |
| LOVE + LOVE = HATE<br>81 Ba                    | 42      | $E = 0$ only              | $L=3, O=5, V=6, E=0, H=7, A=1, T=2$ : $3560 + 3560 = 7120$ . $E = 0$ because $2E$ must end in $E$ (rule 4).                                       |
| WRONG + WRONG = RIGHT<br>70 Bh, 77 Ch          | 21      | nothing                   | $W=3, R=7, O=0, N=8, G=1, I=4, H=6, T=2$ : $37081 + 37081 = 74162$ .                                                                              |
| SWIM + WEAR = RELAX<br>74 Bh, 81 Ch            | 16      | $R = 1, I = 0$            | $S=9, W=4, I=0, M=5, E=3, A=2, R=1, L=7, X=6$ : $9405 + 4321 = 13726$ . $R = 1$ is the leading carry.                                             |
| ONE + ONE = TWO<br>Insights §2.3               | 16      | nothing                   | $O=2, N=3, E=1, T=4, W=6$ : $231 + 231 = 462$ . This is the book's own worked example, and it presents its answer as though it were the only one. |
| RIGHT + RIGHT = WRONG<br>69 Po                 | 11      | nothing                   | $R=4, I=2, G=0, H=6, T=5, W=8, O=1, N=3$ : $42065 + 42065 = 84130$ .                                                                              |
| TWO + TWO = FOUR<br>78 Ch                      | 7       | $F = 1$ only              | $T=7, W=3, O=4, F=1, U=6, R=8$ : $734 + 734 = 1468$ .                                                                                             |

### How to write one of these up for full marks

- 1. Give  $X$ ,  $D$  and  $C$  with the carry equations, exactly as for a unique puzzle. **This is most of the mark scheme.**
- 2. Derive whatever is forced. There is always something: a leading carry, or a  $2X$  ending in  $X$ .
- 3. Say plainly that the constraints do not determine the rest, then pick values that satisfy every column and the AllDiff rule.
- 4. **Verify by doing the addition.** One line, and it converts a guess into a demonstrated answer.

### Worked example of step 2: LOVE + LOVE = HATE

- Column 1:  $E + E = 10C_1 + E \Rightarrow E = 10C_1$ . A digit, so  $E = 0, C_1 = 0$ .
- Column 2:  $2V + 0 = 10C_2 + T$ , so  $T$  is even.
- Column 3:  $2O + C_2 = 10C_3 + A$ .
- Column 4:  $2L + C_3 = H$ , and  $H \leq 9$  forces  $L \leq 4$ .
- That is everything the constraints give.  $L \in \{1, 2, 3, 4\}$  and the rest follows the choice. Take  $L = 3, O = 5, V = 6$ : then  $T = 2, A = 1, H = 7$ , and  $3560 + 3560 = 7120 \checkmark$

### Step by step: LOVE + LOVE = HATE

1.  $E = 0$

cancel

|       |       |   |   |
|-------|-------|---|---|
| $C_3$ | $C_2$ | 0 |   |
| L     | O     | V | 0 |
| +     | L     | O | V |
|       | H     | A | T |

$E + E = 10C_1 + E$ :  
 $E = 10C_1$ , a digit, so 0.

KEY STEP

2.  $T$  even

parity

|       |       |   |   |
|-------|-------|---|---|
| $C_3$ | $C_2$ | 0 |   |
| L     | O     | V | 0 |
| +     | L     | O | V |
|       | H     | A | T |

$2V + 0 = 10C_2 + T$ .

3.  $L \leq 4$

sum bound

|       |       |   |   |
|-------|-------|---|---|
| $C_3$ | $C_2$ | 0 |   |
| L     | O     | V | 0 |
| +     | L     | O | V |
|       | H     | A | T |

$2L + C_3 = H \leq 9$ . Nothing else is forced.

4. choose  $L, O, V =$

2, 5, 6  
free choice carry check

|   |   |   |   |
|---|---|---|---|
| 1 | 1 | 0 |   |
| 3 | 5 | 6 | 0 |
| + | 3 | 5 | 6 |
|   | 7 | 1 | 2 |

$T = 2, A = 11 - 10 = 1$ ,  
 $H = 6 + 1 = 7$ .

5 fixed   5 fixed this step   column read   KEY STEP the decisive move   sum bound   cancel   parity   free choice   carry check

The table above is an index, not an answer. **Nine papers set one of these puzzles for six or seven marks**, and the marks are in the *chain*, not in the digits at the end of it – a puzzle with 107 solutions cannot be marked on the digits. LOVE + LOVE = HATE is chained just above; the other five follow, each with its own paper, its own column equations and its own worked choice. · **Column  $k$  is counted from the units end**, and  $c_k$  is the carry *out* of column  $k$ . Every forced result below was confirmed against the full solution set by `verify.py`.

PIN 3/41 71 Ma 7, 73 Bh 7, 80 Ash 2+6 · 107 solutions, the most of any puzzle in the 41 papers · the only *multi-answer* puzzle with three addends, so a carry here can be 2, not just 0 or 1

Solve the following puzzle by assigning numeral (0-9) in such a way that each letter is assigned unique digit which satisfy the following addition. ONE + ONE + TWO = FOUR

- The addends have three digits and FOUR has four, so **F is nothing but the carry out of column 3**, and  $F \geq 1$ .
- Column 3 is  $2O + T + c_2 = 10F + O$ , so

- $O + T \leq 9 + 8 = 17$  and  $c_2 \leq 2$ , so the left side is at most 19.  $F = 2$  is **impossible**, giving  $F = 1$  and  $O + T + c_2 = 10$ .
- **Column 1:**  $2E + O = 10c_1 + R$ . **Column 2:**  $2N + W + c_1 = 10c_2 + U$ .
- **That is everything the constraints force.** Seven of the eight letters are free; say so before you choose.

- Take  $E = 2, O = 6$ . Column 1:  $4 + 6 = 10$ , so  $\mathbf{R} = \mathbf{0}$ ,  $c_1 = 1$ .
- $O + T + c_2 = 10$  with  $O = 6$  leaves  $T + c_2 = 4$ . Take  $c_2 = 1, \mathbf{T} = \mathbf{3}$ .
- Column 2:  $2N + W + 1 = 10c_2 + U$ . Take  $N = 4, W = 8$ :  $8 + 8 + 1 = 17$ , so  $\mathbf{U} = \mathbf{7}$  and  $c_2 = 1$  ✓ as assumed.

**Answer.**  $O=6, N=4, E=2, T=3, W=8, F=1, U=7, R=0$ , all distinct.  $642 + 642 + 386 = 1670 \checkmark$

### 1. $F$ is a carry

leading carry

|   |       |       |       |   |
|---|-------|-------|-------|---|
|   | $c_3$ | $c_2$ | $c_1$ |   |
|   |       | O     | N     | E |
| + |       | O     | N     | E |
| + |       | T     | W     | O |
|   | F     | O     | U     | R |

3-digit addends, 4-digit FOUR:  
 $F = c_3 \geq 1$ .

### 2. $O + T + c_2 = 10F$

cancel

|   |       |       |       |   |
|---|-------|-------|-------|---|
|   | $c_3$ | $c_2$ | $c_1$ |   |
|   |       | O     | N     | E |
| + |       | O     | N     | E |
| + |       | T     | W     | O |
|   | F     | O     | U     | R |

Column 3:  
 $2O + T + c_2 = 10F + 0$ ;  
 one  $O$  cancels.

### 3. $F = 1$ , not 2

sum bound    carry = 2

|   |   |       |       |   |
|---|---|-------|-------|---|
|   |   | $c_2$ | $c_1$ |   |
|   |   | O     | N     | E |
| + |   | O     | N     | E |
| + |   | T     | W     | O |
|   | 1 | O     | U     | R |

Three addends, so  $c_2 \leq 2$ ; still  
 $O + T + c_2 \leq 9 + 8 + 2 = 19 < 20$ .

### 4. choose $E = 2$ , $O = 6$

free choice

|   |   |       |   |   |
|---|---|-------|---|---|
|   | 1 | $c_2$ | 1 |   |
|   |   | 6     | N | 2 |
| + |   | 6     | N | 2 |
| + |   | T     | W | 6 |
|   | F | 6     | U | 6 |

$4 + 6 = 10$ :  $R = 0$ ,  $c_1 = 1$ .

### 5. $T = 3$

free choice    link letters

|   |   |   |   |   |
|---|---|---|---|---|
|   | 1 | 1 |   |   |
|   |   | 6 | N | 2 |
| + |   | 6 | N | 2 |
| + |   | 3 | W | 6 |
|   | F | 6 | U | 0 |

$6 + T + c_2 = 10$ : take  
 $c_2 = 1$ ,  $T = 3$ .

### 6. $N = 4$ , $W = 8$ : $U = 7$

free choice    carry check

|   |   |   |   |   |
|---|---|---|---|---|
|   | 1 | 1 | 1 |   |
|   |   | 6 | 4 | 2 |
| + |   | 6 | 4 | 2 |
| + |   | 3 | 8 | 6 |
|   | F | 6 | 7 | 0 |

$8 + 8 + 1 = 17$ :  $U = 7$  and  
 $c_2 = 1$  as assumed.

5 fixed 5 fixed this step column read **KEY STEP** the decisive move leading carry sum bound carry = 2 cancel link letters free choice carry check

**WRONG + WRONG = RIGHT** 5+3 2+6

**PIN 2/41** 70 Bh 5+3, 77 Ch 2+6 · 21 solutions, **nothing at all is forced** · 69 Po sets the same eight letters with the two words swapped; work them as two different puzzles, because they are

**QUESTION AS PRINTED** 70 Bh · Q2 · 5+3

Solve the following crypto-arithmetic problem, where different letters denote different integers and identical letters denote same integer. **WRONG + WRONG = RIGHT**. Explain the steps that you have followed.

**The chain**

- Both sides have five digits, so there is no sixth carry:  
 $2 \times \text{WRONG} < 100000$ , hence  $\text{WRONG} < 50000$  and  $W \leq 4$ . This is the only bound the puzzle gives you, and writing it down is most of the method mark.
- Column 5:  $2W + c_4 = R$ , so  $R = 2W + c_4 - R$  is pinned the moment  $W$  and that carry are chosen.
- Column 1:  $2G = 10c_1 + T$ , so  **$T$  is even**.
- Columns 2, 3, 4:  $2N + c_1 = 10c_2 + H$ ,  
 $2O + c_2 = 10c_3 + G$ ,  $2R + c_3 = 10c_4 + I$ .
- No letter is forced to one digit. State that, then choose.

**A choice that satisfies all of it**

- Take  $W = 3$  with  $c_4 = 1$ , so  $R = 7$ .
- Column 1:  $G = 1$  gives  $T = 2$ ,  $c_1 = 0$ .
- Column 2:  $N = 8$  gives 16, so  $H = 6$ ,  $c_2 = 1$ .
- Column 3:  $O = 0$  gives  $0 + 1 = 1 = G$  ✓,  $c_3 = 0$ .
- Column 4:  $2(7) + 0 = 14$ , so  $I = 4$  and  $c_4 = 1$ , which is what column 5 assumed – that closing check is the one step candidates skip.

**Answer.**  $W=3, R=7, O=0, N=8, G=1, I=4, H=6, T=2$ .  
 $37081 + 37081 = 74162$  ✓

**Step by step: WRONG + WRONG = RIGHT**

**1.  $W \leq 4$**   
sum bound

|   |       |       |       |       |   |
|---|-------|-------|-------|-------|---|
|   | $c_4$ | $c_3$ | $c_2$ | $c_1$ |   |
|   | W     | R     | O     | N     | G |
| + | W     | R     | O     | N     | G |
|   | R     | I     | G     | H     | T |

No sixth digit:  $2 \times \text{WRONG} < 100000$ .  
The only bound.

**KEY STEP**

**2.  $R = 2W + c_4$**   
link letters

|   |       |       |       |       |   |
|---|-------|-------|-------|-------|---|
|   | $c_4$ | $c_3$ | $c_2$ | $c_1$ |   |
|   | W     | R     | O     | N     | G |
| + | W     | R     | O     | N     | G |
|   | R     | I     | G     | H     | T |

Column 5 pins  $R$  once  $W$  and  $c_4$  are chosen.

**3.  $T$  even**  
parity

|   |       |       |       |       |   |
|---|-------|-------|-------|-------|---|
|   | $c_4$ | $c_3$ | $c_2$ | $c_1$ |   |
|   | W     | R     | O     | N     | G |
| + | W     | R     | O     | N     | G |
|   | R     | I     | G     | H     | T |

$2G = 10c_1 + T$ .

**4. choose  $W = 3, c_4 = 1$**   
free choice

|   |       |       |       |       |   |
|---|-------|-------|-------|-------|---|
|   | $c_4$ | $c_3$ | $c_2$ | $c_1$ |   |
|   | W     | R     | O     | N     | G |
| + | W     | R     | O     | N     | G |
|   | R     | I     | G     | H     | T |

$R = 6 + 1 = 7$ .

**5.  $G = 1; T = 2$**   
free choice

|   |   |       |       |   |   |
|---|---|-------|-------|---|---|
|   | 1 | $c_3$ | $c_2$ | 0 |   |
|   | W | R     | O     | N | G |
|   | 3 | 7     | 0     | 8 | 1 |
| + | W | R     | O     | N | G |
|   | 3 | 7     | 0     | 8 | 1 |
|   | R | I     | G     | H | T |
|   | 7 | 4     | 1     | 6 | 2 |

$2 \cdot 1 = 2$ , no carry.

**6.  $N = 8; H = 6$**   
free choice

|   |   |       |   |   |   |
|---|---|-------|---|---|---|
|   | 1 | $c_3$ | 1 | 0 |   |
|   | W | R     | O | N | G |
|   | 3 | 7     | 0 | 8 | 1 |
| + | W | R     | O | N | G |
|   | 3 | 7     | 0 | 8 | 1 |
|   | R | I     | G | H | T |
|   | 7 | 4     | 1 | 6 | 2 |

$16: H = 6, c_2 = 1$ .

**7.  $O = 0$  gives  $G = 1$**   
carry check

|   |   |   |   |   |   |
|---|---|---|---|---|---|
|   | 1 | 0 | 1 | 0 |   |
|   | W | R | O | N | G |
|   | 3 | 7 | 0 | 8 | 1 |
| + | W | R | O | N | G |
|   | 3 | 7 | 0 | 8 | 1 |
|   | R | I | G | H | T |
|   | 7 | 4 | 1 | 6 | 2 |

$0 + 1 = 1 = G$ , the value column 1 chose.

**8.  $I = 4, c_4 = 1$**   
carry check

|   |   |   |   |   |   |
|---|---|---|---|---|---|
|   | 1 | 0 | 1 | 0 |   |
|   | W | R | O | N | G |
|   | 3 | 7 | 0 | 8 | 1 |
| + | W | R | O | N | G |
|   | 3 | 7 | 0 | 8 | 1 |
|   | R | I | G | H | T |
|   | 7 | 4 | 1 | 6 | 2 |

$2(7) + 0 = 14$ : carry 1, what column 5 assumed.

5 fixed   
 5 fixed this step   
  column read   
 KEY STEP the decisive move   
 sum bound   
 parity   
 link letters   
 free choice   
 carry check

## SWIM + WEAR = RELAX 1+6 3+6

PIN 2/41 74 Bh 1+6, 81 Ch 3+6 · 16 solutions, and **the richest chain of the seven**: two letters fall out completely and a third is pinned to a fourth

QUESTION AS PRINTED 74 Bh · Q2 · 1+6

Define constraint satisfaction problem (CSP). Solve the following crypto-arithmetic problem, where different letters denote different integers and identical letters denote same integer. SWIM + WEAR = RELAX.

## The chain

- Two four-digit numbers give a five-digit answer, so the leading digit is only a carry: **R = 1**.
- Column 2 is  $I + A + c_1 = 10c_2 + A$ , and the  $A$  cancels:

$$I + c_1 = 10c_2$$

- Both cases, and one of them dies.** If  $c_2 = 1$  then  $I + c_1 = 10$ , forcing  $I = 9$ ,  $c_1 = 1$ . Column 1 is then  $M + R = 10 + X$  with  $R = 1$ , so  $M = 9 + X$ , which needs  $X = 0$  and  $M = 9$  – **but  $I$  is already 9**. Contradiction.
- So  $c_2 = 0$  and **I = 0**,  $c_1 = 0$ .
- Column 1 with  $c_1 = 0$ :  $M + 1 = X$ , so  **$X = M + 1$** . The two move together.
- Column 3:  $W + E + c_2 = 10c_3 + L$ . Column 4:  $S + W + c_3 = 10 + E$ .
- $A$  meets no constraint but AllDiff – worth saying out loud.

## A choice that satisfies all of it

- Take  $W = 4$ ,  $E = 3$  with  $c_3 = 0$ : column 4 gives  $S = 10 + 3 - 4 = 9$ .
- Column 3:  $4 + 3 + 0 = 7$ , so **L = 7** and  $c_3 = 0$  ✓.
- $A = 2$ , free.  $M = 5$ , so  $X = 6$  by  $X = M + 1$ .

**Answer.**  $S=9, W=4, I=0, M=5, E=3, A=2, R=1, L=7, X=6$ .  
 $9405 + 4321 = 13726$  ✓

- Nine letters, so all ten digits but one are used.** A duplicate is easy to make here; check the list before you write the sum.

## Step by step: SWIM + WEAR = RELAX

**1. R = 1**  
leading carry

|   |   |       |       |       |   |
|---|---|-------|-------|-------|---|
|   |   |       |       |       |   |
|   |   | $c_3$ | $c_2$ | $c_1$ |   |
|   | S | W     | I     | M     |   |
| + | W | E     | A     |       |   |
|   |   |       |       |       | 1 |
|   | 1 | E     | L     | A     | X |

Two 4-digit numbers, a 5-digit answer.

**2.  $I + c_1 = 10c_2$**   
cancel

|   |   |       |       |       |   |
|---|---|-------|-------|-------|---|
|   |   |       |       |       |   |
|   |   | $c_3$ | $c_2$ | $c_1$ |   |
|   | S | W     | I     | M     |   |
| + | W | E     | A     |       |   |
|   |   |       |       |       | 1 |
|   | 1 | E     | L     | A     | X |

Column 2:  $I + A + c_1 = 10c_2 + A$ ;  $A$  cancels.

**3. I = 0**  
case split AllDiff

|   |   |       |   |   |   |
|---|---|-------|---|---|---|
|   |   |       |   |   |   |
|   |   | $c_3$ | 0 | 0 |   |
|   | S | W     | 0 | M |   |
| + | W | E     | A |   |   |
|   |   |       |   |   | 1 |
|   | 1 | E     | L | A | X |

$c_2 = 1$ :  $I = 9$ ,  $c_1 = 1$ , then  $M = 9 + X$  needs  $M = 9 = I$ .

**KEY STEP**

**4.  $X = M + 1$**   
link letters

|   |   |       |   |   |   |
|---|---|-------|---|---|---|
|   |   |       |   |   |   |
|   |   | $c_3$ | 0 | 0 |   |
|   | S | W     | 0 | M |   |
| + | W | E     | A |   |   |
|   |   |       |   |   | 1 |
|   | 1 | E     | L | A | X |

Column 1 with  $c_1 = 0$ :  $M + 1 = X$ .

**5. choose  $W = 4, E = 3$ :**  
 $S = 9$   
free choice

|   |   |   |   |   |   |
|---|---|---|---|---|---|
|   |   |   |   |   |   |
|   |   | 0 | 0 | 0 |   |
|   | S | W | I | M |   |
| + | W | E | A |   |   |
|   |   |   |   |   | 1 |
|   | 1 | 3 | L | A | X |

Column 4:  $S + W + c_3 = 10 + E$ , with  $c_3 = 0$ .

**6. L = 7**  
carry check

|   |   |   |   |   |   |
|---|---|---|---|---|---|
|   |   |   |   |   |   |
|   |   | 0 | 0 | 0 |   |
|   | S | W | I | M |   |
| + | W | E | A |   |   |
|   |   |   |   |   | 1 |
|   | 1 | 3 | 7 | A | X |

$4 + 3 + 0 = 7$ ,  $c_3 = 0$  as assumed.

**7. A = 2, M = 5, X = 6**  
free choice link letters

|   |   |   |   |   |   |
|---|---|---|---|---|---|
|   |   |   |   |   |   |
|   |   | 0 | 0 | 0 |   |
|   | S | W | I | M |   |
| + | W | E | A |   |   |
|   |   |   |   |   | 1 |
|   | 1 | 3 | 7 | 2 | 6 |

$A$  meets only AllDiff.  $X = M + 1$ .

5 fixed 5 fixed this step column read **KEY STEP** the decisive move leading carry cancel case split AllDiff link letters free choice carry check

**RIGHT + RIGHT = WRONG** 7

**PIN 1/41** 69 Po · 7 · 11 solutions, nothing forced · **the mirror image of WRONG + WRONG = RIGHT** above, and it does *not* have the same answer – the letters swap roles, so every column equation changes

**QUESTION AS PRINTED** 69 Po · Q2 · 7

Solve the following crypto-arithmetic problem, where different letters denote different integers and identical letters denote same integer. **RIGHT + RIGHT = WRONG**. Show all the step of solving through constrain satisfaction problem.

**The chain**

- Five digits on both sides, so  $2 \times \text{RIGHT} < 100000$  and  $R \leq 4$ .
- Column 5:  $W = 2R + c_4$ .
- Column 1:  $2T = 10c_1 + G$ , so **G is even**.
- Columns 2, 3, 4:  $2H + c_1 = 10c_2 + N$ ,  
 $2G + c_2 = 10c_3 + O$ ,  $2I + c_3 = 10c_4 + R$ .
- Nothing is forced to a single digit.

Note how column 3 feeds on  $G$ , which column 1 already produced.  
Fixing  $T$  early therefore settles two columns at once, and that is the cheapest place to start guessing.

**A choice that satisfies all of it**

- Take  $R = 4$  with  $c_4 = 0$ , so **W = 8**.
- Column 1:  $T = 5$  gives 10, so **G = 0**,  $c_1 = 1$ .
- Column 2:  $H = 6$  gives 13, so **N = 3**,  $c_2 = 1$ .
- Column 3:  $0 + 1 = 1$ , so **O = 1**,  $c_3 = 0$ .
- Column 4:  $2I = 10c_4 + R = 4$ , so **I = 2** and  $c_4 = 0$  ✓ as column 5 assumed.

**Answer.**  $R=4, I=2, G=0, H=6, T=5, W=8, O=1, N=3$ .  
 $42065 + 42065 = 84130$  ✓

**Step by step: RIGHT + RIGHT = WRONG**

**1.  $R \leq 4$**   
sum bound

|   |       |       |       |       |   |
|---|-------|-------|-------|-------|---|
|   | $c_4$ | $c_3$ | $c_2$ | $c_1$ |   |
|   | R     | I     | G     | H     | T |
| + | R     | I     | G     | H     | T |
|   |       |       |       |       |   |
|   | W     | R     | O     | N     | G |

Five digits both sides:  $2 \times \text{RIGHT} < 100000$ .

**2.  $W = 2R + c_4$**   
link letters

|   |       |       |       |       |   |
|---|-------|-------|-------|-------|---|
|   | $c_4$ | $c_3$ | $c_2$ | $c_1$ |   |
|   | R     | I     | G     | H     | T |
| + | R     | I     | G     | H     | T |
|   |       |       |       |       |   |
|   | W     | R     | O     | N     | G |

Column 5.

**3. G even**  
parity

|   |       |       |       |       |   |
|---|-------|-------|-------|-------|---|
|   | $c_4$ | $c_3$ | $c_2$ | $c_1$ |   |
|   | R     | I     | G     | H     | T |
| + | R     | I     | G     | H     | T |
|   |       |       |       |       |   |
|   | W     | R     | O     | N     | G |

$2T = 10c_1 + G$ . Column 3 feeds on this  $G$ .

KEY STEP

**4. choose  $R = 4, c_4 = 0$**   
free choice

|   |       |       |       |       |   |
|---|-------|-------|-------|-------|---|
|   | $c_4$ | $c_3$ | $c_2$ | $c_1$ |   |
|   | 0     |       |       |       |   |
|   | 4     | I     | G     | H     | T |
| + | 4     | I     | G     | H     | T |
|   |       |       |       |       |   |
|   | 8     | 4     | O     | N     | G |

$W = 8$ .

**5.  $T = 5; G = 0$**   
free choice

|   |       |       |       |       |   |
|---|-------|-------|-------|-------|---|
|   | $c_4$ | $c_3$ | $c_2$ | $c_1$ |   |
|   |       |       |       | 1     |   |
|   | 4     | I     | 0     | H     | 5 |
| + | 4     | I     | 0     | H     | 5 |
|   |       |       |       |       |   |
|   | 8     | 4     | O     | N     | 0 |

$10: G = 0, c_1 = 1$ .

**6.  $H = 6; N = 3$**   
free choice

|   |       |       |       |       |   |
|---|-------|-------|-------|-------|---|
|   | $c_4$ | $c_3$ | $c_2$ | $c_1$ |   |
|   |       |       | 1     | 1     |   |
|   | 4     | I     | 0     | 6     | 5 |
| + | 4     | I     | 0     | 6     | 5 |
|   |       |       |       |       |   |
|   | 8     | 4     | O     | 3     | 0 |

$12 + 1 = 13$ .

**7.  $O = 1$**   
arithmetic

|   |       |       |       |       |   |
|---|-------|-------|-------|-------|---|
|   | $c_4$ | $c_3$ | $c_2$ | $c_1$ |   |
|   |       | 0     | 1     | 1     |   |
|   | 4     | I     | 0     | 6     | 5 |
| + | 4     | I     | 0     | 6     | 5 |
|   |       |       |       |       |   |
|   | 8     | 4     | 1     | 3     | 0 |

$2(0) + 1 = 1$ .

**8.  $I = 2, c_4 = 0$**   
carry check

|   |       |       |       |       |   |
|---|-------|-------|-------|-------|---|
|   | $c_4$ | $c_3$ | $c_2$ | $c_1$ |   |
|   |       | 0     | 1     | 1     |   |
|   | 4     | 2     | 0     | 6     | 5 |
| + | 4     | 2     | 0     | 6     | 5 |
|   |       |       |       |       |   |
|   | 8     | 4     | 1     | 3     | 0 |

$2I + 0 = 4 = R$ : carry 0, as column 5 assumed.

5 fixed   
 5 fixed this step   
 column read   
 KEY STEP the decisive move   
 sum bound   
 parity   
 link letters   
 arithmetic   
 free choice   
 carry check

**TWO + TWO = FOUR** 2+6

**PIN 1/41** 78 Ch 2+6 · **only 7 solutions**, the tightest of the seven, so a marker here is more likely than elsewhere to have one in mind – print the count anyway

**QUESTION AS PRINTED** 78 Ch · Q2 · 2+6

What is well defined problem? Solve the following crypto-arithmetic problem by defining it. TWO + TWO = FOUR

**The chain**

- Two three-digit numbers reach at most  $999 + 999 = 1998$ , so the leading digit of a four-digit FOUR is **F = 1**.
- Column 3:  $2T + c_2 = 10F + O = 10 + O$ . With  $c_2 \leq 1$  this needs  $2T \geq 9$ , so  **$T \geq 5$** .
- Column 1:  $2O = 10c_1 + R$ , so  **$R$  is even**.
- Column 2:  $2W + c_1 = 10c_2 + U$ .
- $F$  is the only letter forced.

**A choice that satisfies all of it**

- Take  $T = 7$  with  $c_2 = 0$ : column 3 gives  $14 = 10 + O$ , so  **$O = 4$** .
- Column 1:  $2(4) = 8$ , so  **$R = 8$** ,  $c_1 = 0$ .
- Column 2:  $2W + 0 = 10(0) + U$ , so  $U = 2W$ . Take  $W = 3$ ,  **$U = 6$** , and  $c_2 = 0$  ✓ as assumed.

**Answer.**  $T=7, W=3, O=4, F=1, U=6, R=8$ .  $734 + 734 = 1468$  ✓

- Six letters, so this one is quick** – spend the time saved on the 2-mark *well-defined problem* half, which is **Ch 2 §2.2** and is where the easy marks are.

**Step by step: TWO + TWO = FOUR**

**1.  $F = 1$**

leading carry sum bound

|   |       |       |   |
|---|-------|-------|---|
| 1 | $c_2$ | $c_1$ |   |
| T | W     | O     |   |
| T | W     | O     |   |
| 1 | O     | U     | R |

$999 + 999 = 1998$ .

**2.  $T \geq 5$**

sum bound

|   |       |       |   |
|---|-------|-------|---|
| 1 | $c_2$ | $c_1$ |   |
| T | W     | O     |   |
| T | W     | O     |   |
| 1 | O     | U     | R |

$2T + c_2 = 10 + O$  with  $c_2 \leq 1$ .

**KEY STEP**

**3.  $R$  even**

parity

|   |       |       |   |
|---|-------|-------|---|
| 1 | $c_2$ | $c_1$ |   |
| T | W     | O     |   |
| T | W     | O     |   |
| 1 | O     | U     | R |

$2O = 10c_1 + R$ .

**4. choose  $T = 7, c_2 = 0$**

free choice

|   |   |       |   |
|---|---|-------|---|
| 1 | 0 | $c_1$ |   |
| 7 | W | O     |   |
| 7 | W | O     |   |
| 1 | 4 | U     | R |

$14 = 10 + O: O = 4$ .

**5.  $R = 8$**

arithmetic

|   |   |   |   |
|---|---|---|---|
| 1 | 0 | 0 |   |
| 7 | W | O | 4 |
| 7 | W | O | 4 |
| 1 | 4 | U | 8 |

$2(4) = 8$ .

**6.  $W = 3: U = 6$**

free choice carry check

|   |   |   |   |
|---|---|---|---|
| 1 | 0 | 0 |   |
| 7 | 3 | O | 4 |
| 7 | 3 | O | 4 |
| 1 | 4 | 6 | 8 |

$U = 2W; c_2 = 0$  as assumed.

5 fixed 5 fixed this step column read **KEY STEP** the decisive move leading carry sum bound parity arithmetic free choice carry check

**AB + CD = AAA: what values can B take?** 4

76 Bh 4 · **not a solve-it question**: it asks what is *possible*, so the answer is a set and a justification

QUESTION AS PRINTED 76 Bh · 4

In a correctly worked out crypt-arithmetic problem  $AB + CD = AAA$ , what could be the possible values of B? Justify your answer.

**Reasoning**

- $A = 1$ . Two 2-digit numbers sum to at most  $99 + 99 = 198$ . The only 3-digit repdigit AAA in range is **111**, so  $A = 1$ .
- So  $AB = 10 + B$  and  $CD = 111 - (10 + B) = 101 - B$ .
- $B \geq 3$ . CD must be a 2-digit number, so  $101 - B \leq 99$ , giving  $B \geq 2$ . And  $B = 2$  makes  $CD = 99$ , i.e.  $C = D = 9$ , which the AllDiff rule forbids. So  $B \geq 3$ .
- $B \leq 8$ . For  $B$  from 3 to 9,  $CD = 101 - B$  runs  $98, 97, \dots, 92$ , so  $C = 9$  throughout.  $B = 9$  would make  $B = C = 9$ . **This is the case that is easy to miss**, and it is the whole difficulty of the question.
- $B = 1$  is out because  $A = 1$ , and  $B = 0$  would need  $CD = 101$ .

**Answer:**  $B \in \{3, 4, 5, 6, 7, 8\}$ , six values

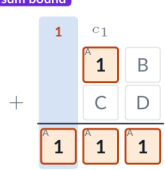
| B | AB | CD | Check              |
|---|----|----|--------------------|
| 3 | 13 | 98 | $13 + 98 = 111$ ✓  |
| 4 | 14 | 97 | $14 + 97 = 111$ ✓  |
| 5 | 15 | 96 | $15 + 96 = 111$ ✓  |
| 6 | 16 | 95 | $16 + 95 = 111$ ✓  |
| 7 | 17 | 94 | $17 + 94 = 111$ ✓  |
| 8 | 18 | 93 | $18 + 93 = 111$ ✓  |
| 2 | 12 | 99 | $\times C = D = 9$ |
| 9 | 19 | 92 | $\times B = C = 9$ |

- In every case  $A = 1$  and  $C = 9$ ; only  $B$  and  $D$  move, and they move together since  $D = 8 - (B - 3)$ .

**Step by step: AB + CD = AAA**

1.  $A = 1$

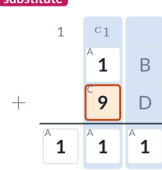
sum bound



$99 + 99 = 198$ , and the only repdigit in range is 111.

2.  $CD = 101 - B$

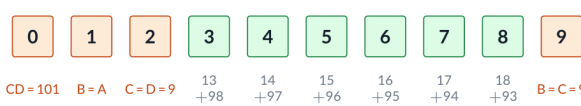
substitute



$AB = 10 + B$ , so  $CD = 111 - (10 + B)$ . For  $B \geq 2$  CD is in the 90s:  $C = 9$ .

3. test every B

AllDiff sum bound



Survivors:  $B \in \{3, 4, 5, 6, 7, 8\}$ .  $B = 9$  is the case that is easy to miss.

KEY STEP

5 fixed 5 fixed this step column read KEY STEP the decisive move sum bound substitute AllDiff

## 2.N.3 The water jug problem

### The eight production rules, state $(x, y)$

$x$  is the contents of the larger jug,  $y$  of the smaller. Written here for the 4-gallon and 3-gallon pair; swap the capacities for any other version. This is the rule set *Insights* and the lecture decks both use, and it is what “write down the production rules” means.

| # | Condition                               | Result             | Meaning                                        |
|---|-----------------------------------------|--------------------|------------------------------------------------|
| 1 | $(x, y)$ and $x < 4$                    | $(4, y)$           | Fill the 4-gallon jug                          |
| 2 | $(x, y)$ and $y < 3$                    | $(x, 3)$           | Fill the 3-gallon jug                          |
| 3 | $(x, y)$ and $x > 0$                    | $(0, y)$           | Empty the 4-gallon jug on the ground           |
| 4 | $(x, y)$ and $y > 0$                    | $(x, 0)$           | Empty the 3-gallon jug on the ground           |
| 5 | $(x, y)$ and $x + y \geq 4$ and $y > 0$ | $(4, y - (4 - x))$ | Pour from the 3 into the 4 until the 4 is full |
| 6 | $(x, y)$ and $x + y \geq 3$ and $x > 0$ | $(x - (3 - y), 3)$ | Pour from the 4 into the 3 until the 3 is full |
| 7 | $(x, y)$ and $x + y \leq 4$ and $y > 0$ | $(x + y, 0)$       | Pour all of the 3 into the 4                   |
| 8 | $(x, y)$ and $x + y \leq 3$ and $x > 0$ | $(0, x + y)$       | Pour all of the 4 into the 3                   |

### 2 gallons in the 4-gallon jug, from a 4 and a 3 4 2+2+4

PIN 2/41 77 Ch 2+2+4, 71 Bh 4 · the classic; *Insights* §2.2.1 works this one

QUESTION AS PRINTED 77 Ch · 2+2+4

Assume you are given two jugs, a 4-gallon one and a 3-gallon one, and a pump which has unlimited water. How can you get exactly 2 gallons of water in the 4-gallon jug? Formalize the problem, write down Production Rules and draw the Search Tree for the water jug problem.

### Solution, following *Insights*: rules 2, 7, 2, 5, 3, 7

| Step  | State $(x, y)$ | Rule | Action                                                     |
|-------|----------------|------|------------------------------------------------------------|
| start | $(0, 0)$       | —    | both jugs empty                                            |
| 1     | $(0, 3)$       | 2    | fill the 3-gallon jug                                      |
| 2     | $(3, 0)$       | 7    | pour all of it into the 4-gallon jug                       |
| 3     | $(3, 3)$       | 2    | fill the 3-gallon jug again                                |
| 4     | $(4, 2)$       | 5    | pour into the 4 until it is full; the 4 takes 1, leaving 2 |
| 5     | $(0, 2)$       | 3    | empty the 4-gallon jug                                     |
| 6     | $(2, 0)$       | 7    | pour the 2 gallons into the 4-gallon jug                   |

- 6 rule applications, and breadth-first search confirms no shorter path exists.

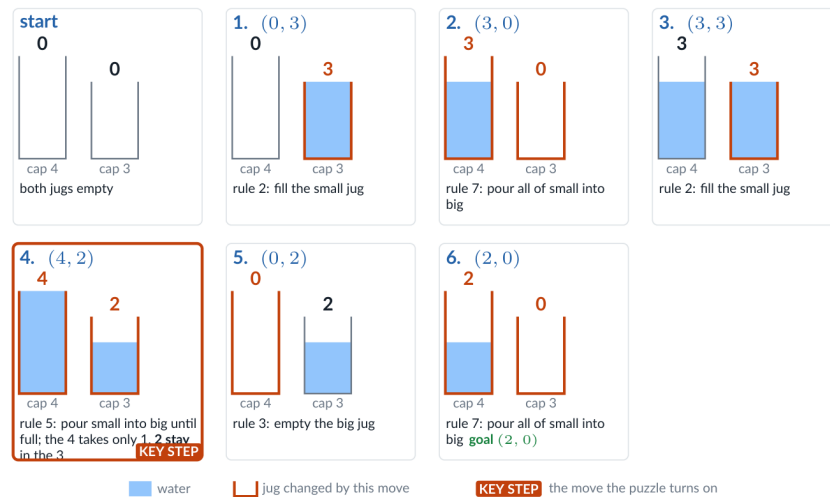
### Formulating it as a state space

- **State:** the ordered pair  $(x, y)$ ,  $0 \leq x \leq 4$ ,  $0 \leq y \leq 3$ . **20 states** in all.
- **Initial state:**  $(0, 0)$ , both jugs empty.
- **Goal state:**  $(2, y)$  for any  $y$  – the question asks only for 2 gallons *in the 4-gallon jug*, so  $y$  is unconstrained.
- **Operators:** the eight rules alongside.
- **Path cost:** 1 per rule application.
- **Why the state is just a pair:** the jugs have no markings, so nothing about *how* the water got there can matter. This is the whole point of the exercise and it is worth one sentence in the answer.
- **Problem type: ignorable** (§2.3) – a wrong move can simply be emptied out, so no backtracking machinery is needed.

### Search tree

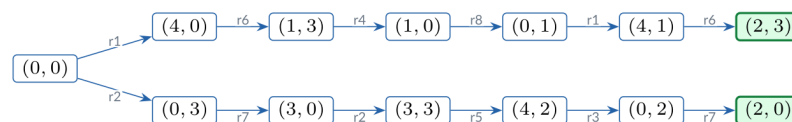
- From  $(0, 0)$  only two rules apply, 1 and 2, so the tree has **two branches at the root**. The 6-step path above is the left branch; the right branch (fill the 4 first) reaches  $(2, 3)$  in 6 steps too.
- **Draw it as this, not as a bush** – show the two root branches and expand the one that succeeds:  
 $(0, 0)$   
 $\rightarrow$  [rule 1]  $(4, 0) \rightarrow (1, 3) \rightarrow (1, 0) \rightarrow (0, 1) \rightarrow (4, 1) \rightarrow (2, 3)$   
 $\rightarrow$  [rule 2]  $(0, 3) \rightarrow (3, 0) \rightarrow (3, 3) \rightarrow (4, 2) \rightarrow (0, 2) \rightarrow (2, 0)$
- **Both are correct and both are 6 moves.** Either scores full marks; say which you are drawing.
- *Insights* draws this tree with a self-loop child at each level, labelled  $(x, y) \rightarrow (x+0, y+0)$ . That is not a legal successor – no rule leaves the state unchanged – and it should be left out.

## Step by step: the jugs after each rule



## Search tree: both branches from (0, 0)

Only rules 1 and 2 apply at the root: two branches, and each reaches a goal in 6 moves (BFS finds nothing shorter).

7 gallons in the 9-gallon jug, from a 3, a 5 and a 9 2+1+2+3

PIN 1/41 79 Bh 2+1+2+3 · the hard variant: three jugs, and the pump may fill only the smallest

QUESTION AS PRINTED 79 Bh · 2+1+2+3

What are the steps of Problem solving? You are given three empty jugs: a 3-gallon, a 5-gallon and a 9-gallon, and a pump can fill water only in the 3-gallon jug. How do you get exactly 7 gallons in the 9-gallon jug? Formalize the problem, write production rules and draw the search tree.

## What makes this one different

- **Only the 3-gallon jug may be filled from the pump.** So rule 1 applies to jug A alone; there is no “fill the 5” and no “fill the 9”.
- Every gallon that reaches the 9-gallon jug arrives in multiples of 3, *unless* the 5-gallon jug is used to strand a remainder.  $9 = 3 \times 3$ , so a direct route gives only 3, 6 or 9 gallons: **the 5-gallon jug is not optional, it is the whole trick.**
- **State:** the triple  $(a, b, c)$  for the 3, 5 and 9-gallon jugs. **Initial**  $(0, 0, 0)$ , **goal**  $(a, b, 7)$ .
- **Operators:** fill A from the pump; empty any jug; pour  $i$  into  $j$  until  $i$  is empty or  $j$  is full.

## Solution, 9 moves (shortest)

| # | $(a, b, c)$ | Action                                                      |
|---|-------------|-------------------------------------------------------------|
|   | $(0, 0, 0)$ | start                                                       |
| 1 | $(3, 0, 0)$ | fill A from the pump                                        |
| 2 | $(0, 3, 0)$ | pour A into B                                               |
| 3 | $(3, 3, 0)$ | fill A                                                      |
| 4 | $(1, 5, 0)$ | pour A into B; B takes 2 and is full, <b>leaving 1 in A</b> |
| 5 | $(0, 5, 1)$ | pour A into C                                               |
| 6 | $(3, 5, 1)$ | fill A                                                      |
| 7 | $(0, 5, 4)$ | pour A into C                                               |
| 8 | $(3, 5, 4)$ | fill A                                                      |
| 9 | $(0, 5, 7)$ | pour A into C                                               |

- **The whole idea is step 4:** fill B to strand exactly 1 gallon in A, tip that 1 into C, then add  $3 + 3$  on top.  
 $1 + 3 + 3 = 7$ .
- B is never emptied after step 4; it has done its job.

## Step by step: the three jugs after each move



## 1 litre, from a 5-litre and a 2-litre jug 8

**PIN 1/41** 81 Ash 8 · this paper asks **three things at once**: is it well defined, list the attributes for designing an intelligent agent, and solve it. Answer all three

**QUESTION AS PRINTED** 81 Ash · 8

You are given two jugs, a 5-litre one and a 2-litre one. Neither has any measuring markers on it. There is a pump that can be used to fill the jugs with water. You are required to measure exactly 1 litre of water. Is this a well-defined problem? Justify your argument and list all the attributes to design an intelligent agent. Solve this problem by specifying states and the operations used.

**Part 1: is it well defined? Yes**

- **Initial state** given: both jugs empty, (0, 0).
- **Operators** given: fill from the pump, empty on the ground, pour one into the other.
- **Goal test** given and mechanical: does either jug hold exactly 1 litre?
- **Path cost**: 1 per operation.
- Every clause of §2.2 is satisfied, so it is well defined. It is also **solvable**, since  $\gcd(5, 2) = 1$  divides 1.

**Part 2: attributes for designing an intelligent agent**

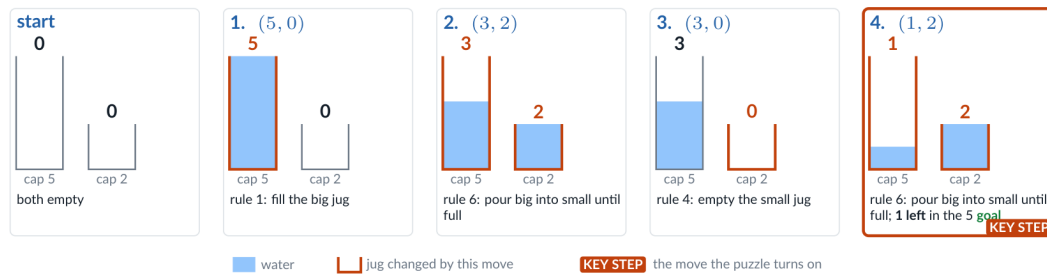
- This is asking for **PEAS** plus the environment properties (Ch 1):
  - **P**: measure exactly 1 litre, in the fewest operations, spilling least.
  - **E**: the two jugs, the pump, the ground.
  - **A**: fill, empty, pour.
  - **S**: whether a jug is full, empty, or neither.
- Environment: **fully observable, deterministic, sequential, static, discrete, single agent**. A **goal-based agent** is the right type, because the answer needs a *sequence* planned ahead, not a reflex.

**Part 3: solution, 4 moves (shortest)**

| # | (x, y) | Rule | Action                                                    |
|---|--------|------|-----------------------------------------------------------|
|   | (0, 0) | —    | start                                                     |
| 1 | (5, 0) | 1    | fill the 5-litre jug                                      |
| 2 | (3, 2) | 6    | pour into the 2-litre jug until full                      |
| 3 | (3, 0) | 4    | empty the 2-litre jug                                     |
| 4 | (1, 2) | 6    | pour again; the 2 takes 2 more, <b>leaving 1 in the 5</b> |

- **The 1 litre ends up in the 5-litre jug**, which is what the question asks for: it says “measure exactly 1 litre”, not “in the 2-litre jug”.
- State space:  $6 \times 3 = 18$  states.
- **The general rule worth one extra mark**: with jugs of capacity  $m$  and  $n$ , exactly the multiples of  $\gcd(m, n)$  up to  $\max(m, n)$  are measurable. Here  $\gcd(5, 2) = 1$ , so every whole number of litres from 0 to 5 can be measured.

## Step by step: the jugs after each rule



## 2.N.4 Farmer, wolf, goat and cabbage

## Formulate the river crossing as search and solve it 8

PIN 1/41 73 Ma 8 · the standard example of a problem whose interest is entirely in the **safety precondition**

QUESTION AS PRINTED 73 Ma 8

A farmer has a goat, a wolf and a cabbage on the west side of a river and wants to move everything to the east side. The boat holds the farmer and one other thing. The wolf eats the goat and the goat eats the cabbage if left alone. Formulate this puzzle as search and solve it (any method); draw the search tree and final solution.

## Formulation

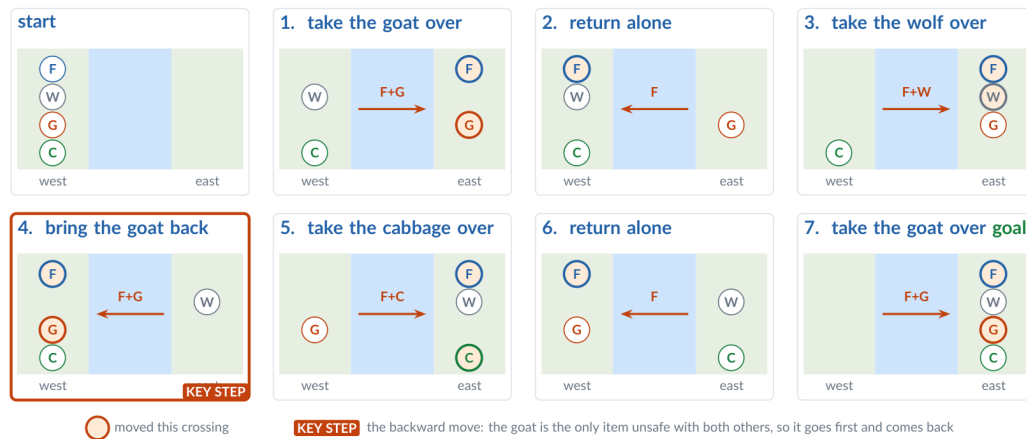
- **State:** the 4-tuple  $(F, W, G, C)$  giving which bank the Farmer, Wolf, Goat and Cabbage are on. Write 0 for west, 1 for east.
- **Initial state**  $(0, 0, 0, 0)$ ; **goal state**  $(1, 1, 1, 1)$ .
- **Operators:** the farmer crosses, taking nothing or exactly one item that is currently on his own bank. **Four operators.**
- **Safety constraint** (this is what makes it a puzzle): a state is illegal if
  - the goat and the wolf are together without the farmer, or
  - the goat and the cabbage are together without the farmer.
- **Path cost:** 1 per crossing.
- **State space:**  $2^4 = 16$  states in total, of which **only 10 are safe and reachable**. Six are pruned by the constraint, and that pruning is the answer to “how does a constraint reduce the search”.

## Solution: 7 crossings

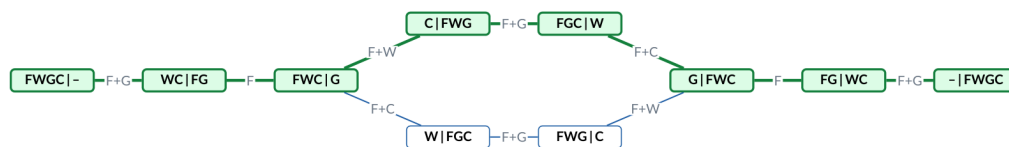
| # | $(F, W, G, C)$ | Crossing                       |
|---|----------------|--------------------------------|
|   | W W W W        | start, everything on the west  |
| 1 | E W E W        | take the <b>goat</b> across    |
| 2 | W W E W        | return <b>alone</b>            |
| 3 | E E E W        | take the <b>wolf</b> across    |
| 4 | W E W W        | <b>bring the goat back</b>     |
| 5 | E E W E        | take the <b>cabbage</b> across |
| 6 | W E W E        | return <b>alone</b>            |
| 7 | E E E E        | take the <b>goat</b> across    |

- **Step 4 is the step everyone misses.** Nothing else works: after the wolf is delivered, the goat cannot stay with it, so it must come back and wait while the cabbage goes over.
- The goat is the only item that is dangerous *in both directions*, which is why it moves first, and why it is the one that gets brought back.
- **7 is the minimum.** There is a mirror-image solution that takes the cabbage third instead of the wolf; it is also 7 crossings and equally correct.

## Step by step: the banks after each crossing



## Search space: every safe state and the two 7-crossing solutions



Each node is west | east. All 10 safe reachable states are drawn; the other 6 of the  $2^4 = 16$  leave goat with wolf or cabbage unattended. Green: the solution above. The lower branch is the mirror solution (cabbage third, then wolf).

## 2.N.5 Tic-tac-toe as a state space

### Describe tic-tac-toe in terms of state space, data structure and goal 7

PIN 1/41 72 Ma 7 · the paper asks for **complete state space, efficient data structures and goal**, i.e. three named parts. Answer in three parts

QUESTION AS PRINTED 72 Ma · 7

Consider a Tic-Tac-Toe game playing as a problem. Prepare a brief description of your problem in terms of complete state-space, efficient data structures and goal, which are to be used for a machine to play that game.

## 1. Complete state space

- A state is an assignment of **blank**, **X** or **O** to each of the 9 squares, plus whose turn it is.
- **Naive count**:  $3^9 = 19,683$  boards. **Most are not reachable** – they have five O's and no X's, or two completed lines.
- **Reachable, legal count**: **5,478** states, counting every board that can arise in a real game with X moving first and play stopping at a win.
- **Branching factor** 9 at the root, falling by one each ply; at most  $9! = 362,880$  complete play sequences, which is small enough to search **exhaustively**. That is the point worth making: tic-tac-toe is one of the few games where minimax can see to the end of the game.
- **Initial state**: the empty board. **Operators**: place your mark on any blank square. **Problem type**: **irrecoverable** (§2.3) – a placed mark cannot be lifted, so the machine must plan before moving.

## 2. Efficient data structure

- A **9-element vector** indexed 1–9, holding 0 for blank, 1 for X, 2 for O. Simple, and a move is one array write.
- **The efficient version**: two **9-bit masks**, one for X and one for O. Then
  - a legal-move test is a bitwise AND against the union,
  - a win test is `mask & line == line` over the **8 winning lines** (3 rows, 3 columns, 2 diagonals), which is 8 AND operations and no loops,
  - the whole board fits in one integer, so states can be hashed and cached.
- A move-indexed **lookup table** of best replies is the fastest of all, but it is the table-driven agent of Ch 1: correct, and impractical to build by hand.

## 3. Goal

- **Goal test**: any of the 8 lines is filled with one player's mark, or all 9 squares are taken (a draw).
- **Utility**: +1 win, 0 draw, –1 loss, which is what minimax (§3.4) needs.
- **The honest closing line**: with both players optimal, tic-tac-toe is a **forced draw**. The machine's real goal is therefore *never to lose*, and to win only when the opponent errs.

## The state space, the data structure and the goal, drawn

### 1. the board as two 9-bit masks

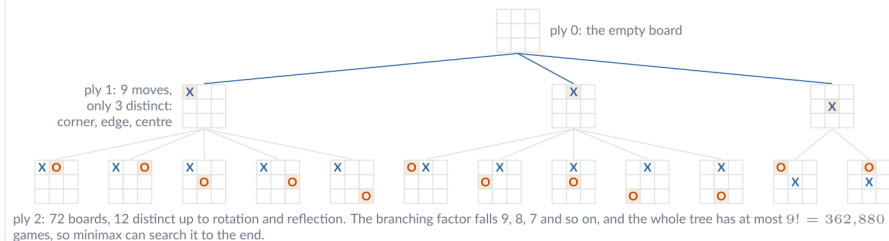
|   |   |   |                                               |
|---|---|---|-----------------------------------------------|
| 1 | 2 | 3 | square $i$ is bit $i$                         |
| X |   | O | X = 000010001 <sub>2</sub> = 17               |
|   |   |   | O = 000000100 <sub>2</sub> = 4                |
| 4 | 5 | 6 | legal move $i$ : bit $i$ of X O is 0          |
|   | X |   | win: X & line == line                         |
| 7 | 8 | 9 | for the 8 lines, e.g. diagonal 1, 5, 9 = 273: |
|   |   |   | X needs square 9                              |

### 2. goal test: the 8 lines

|       |       |       |   |
|-------|-------|-------|---|
| X X X | X X X |       | X |
|       |       |       | X |
|       |       | X X X | X |
| X     |       | X     | X |
| X     |       | X     | X |
| X     |       | X     | X |
|       | X     |       | X |
|       | X     |       | X |
|       | X     |       | X |

3 rows, 3 columns, 2 diagonals. Draw: all 9 filled, no line.

### 3. state space, first two plies up to symmetry



### 3. Search Techniques – Solved Problems

Every problem opens with the exam question and the figure **exactly as the paper prints them**. The OPEN / CLOSED table format is the one *Insights* §3.4.2 uses. · Every path, cost and minimax value in this file is recomputed independently by `verify.py`.

The four things everything here is built on:

- 1.  $f(n) = g(n) + h(n)$ . Greedy orders on  $h$  alone, uniform cost on  $g$  alone, A\* on the sum. Every trace below is the same loop with a different  $f$ .
- 2. **The loop:** take the best node off OPEN → put it on CLOSED → generate its successors onto OPEN → **re-order OPEN** → repeat. Stop when the node taken off OPEN is the goal, not when the goal is first *generated*.
- 3. If a successor is already known with a larger  $g$ , replace it and put it back on OPEN. Skipping this returns the wrong answer on both A\* graphs in these papers, because both heuristics are admissible but **not consistent**.
- 4. **Alpha-beta:**  $\alpha$  only rises,  $\beta$  only falls, a child inherits both, and you test the cut rule after every child. MIN cuts when its value  $\leq \alpha$ ; MAX cuts when its value  $\geq \beta$ .

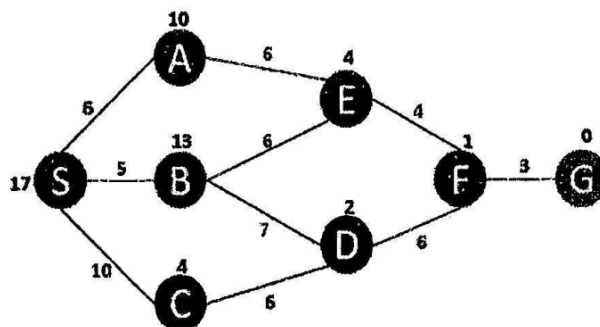
#### 3.N.1 A\* on the 2081 Chaitra graph

Explain the A\* algorithm on the given figure 8

PIN 1/41 81 Ch 8 · the number **above** each node is  $h(n)$ , the number on each link is the step cost

QUESTION AS PRINTED 81 Ch · 8

Explain the A\* algorithm on the following figure, where the starting node is S and the destination is G. (Information in the node represents the cost required to reach the goal and information in the connecting link represents the cost from one node to the other.)



Reading the figure

- $h$ :  $S=17, A=10, B=13, C=4, D=2, E=4, F=1, G=0$ .
- Edges:  $S-A$  6,  $S-B$  5,  $S-C$  10,  $A-E$  6,  $B-E$  6,  $B-D$  7,  $C-D$  6,  $E-F$  4,  $D-F$  6,  $F-G$  3.
- There are only four routes from  $S$  to  $G$ , which is worth writing down first as a check on the answer:
  - $S-A-E-F-G = 6+6+4+3 = 19$
  - $S-B-E-F-G = 5+6+4+3 = 18$**
  - $S-B-D-F-G = 5+7+6+3 = 21$
  - $S-C-D-F-G = 10+6+6+3 = 25$
- So the answer must come out 18.** If your trace ends at 19 you have skipped the reopening step.

Step 1. Start with S on OPEN

| OPEN |     |     |     | CLOSED |        |
|------|-----|-----|-----|--------|--------|
| Node | $g$ | $h$ | $f$ | Node   | Parent |
| S    | 0   | 17  | 17  |        |        |

Step 2. Expand S

| OPEN, re-ordered by $f$ |     |     |     | CLOSED |        |
|-------------------------|-----|-----|-----|--------|--------|
| Node                    | $g$ | $h$ | $f$ | Node   | Parent |
| C                       | 10  | 4   | 14  | S      | —      |
| A                       | 6   | 10  | 16  |        |        |
| B                       | 5   | 13  | 18  |        |        |

- Note what has already gone wrong for greedy:**  $C$  is chosen first because  $h(C) = 4$  is tiny, yet  $C$  lies on the worst route. A\* will recover; greedy would not.

## Steps 3–6. Expand C, A, E, F

| Expand           | Generates                                           | OPEN after re-ordering (node/ <i>f</i> ) |
|------------------|-----------------------------------------------------|------------------------------------------|
| C ( <i>f</i> 14) | D: $g=16, f=18$                                     | A/16, B/18, D/18                         |
| A ( <i>f</i> 16) | E: $g=12, f=16$                                     | E/16, B/18, D/18                         |
| E ( <i>f</i> 16) | F: $g=16, f=17$                                     | F/17, B/18, D/18                         |
| F ( <i>f</i> 17) | G: $g=19, f=19$<br>D via F: $g=22$ , worse, discard | B/18, D/18, G/19                         |

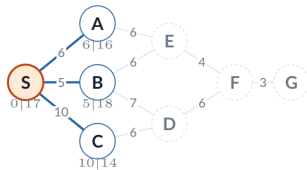
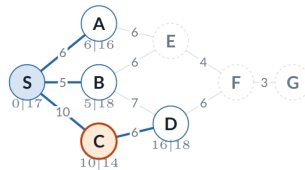
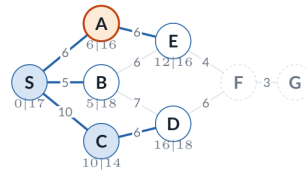
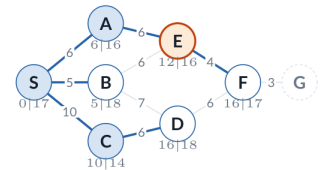
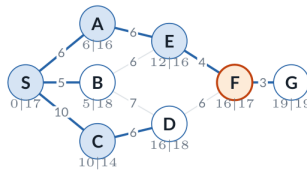
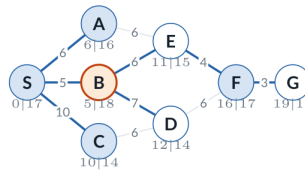
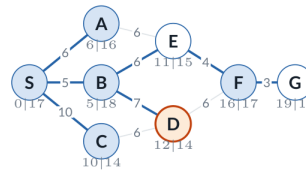
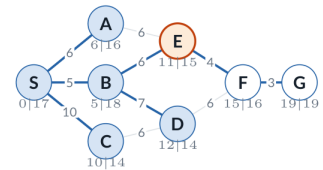
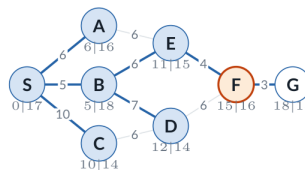
- $G$  is now **generated** at cost 19, but it is **not expanded**, because  $B$  and  $D$  both have  $f = 18 < 19$ . This is exactly why the algorithm stops on *removal* from OPEN, not on generation.

## Steps 7–10. Expanding B revises everything

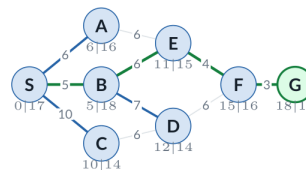
| Expand           | Generates                                                                            | OPEN after re-ordering |
|------------------|--------------------------------------------------------------------------------------|------------------------|
| B ( <i>f</i> 18) | E: $g=11 < 12$ , <b>replace</b> , $f=15$<br>D: $g=12 < 16$ , <b>replace</b> , $f=14$ | D/14, E/15, G/19       |
| D ( <i>f</i> 14) | F via D: $g=18 > 16$ , discard                                                       | E/15, G/19             |
| E ( <i>f</i> 15) | F: $g=15 < 16$ , <b>replace</b> , $f=16$                                             | F/16, G/19             |
| F ( <i>f</i> 16) | G: $g=18 < 19$ , <b>replace</b> , $f=18$                                             | G/18                   |
| G ( <i>f</i> 18) | <b>goal removed from OPEN, stop</b>                                                  |                        |

- Three replacements**, all through  $B$ .  $B$  looked unpromising ( $f = 18$ , last in the queue) but is on the optimal route.

## Step by step: the graph after each expansion

1. expand S (*f* 17)2. expand C (*f* 14)3. expand A (*f* 16)4. expand E (*f* 16)5. expand F (*f* 17)6. expand B (*f* 18) replace D,E7. expand D (*f* 14)8. expand E (*f* 15) replace F9. expand F (*f* 16) replace G

10. expand G (goal off OPEN)



● closed    ○ on OPEN    ● expanding now    ● goal

**Answer:** path  $S \rightarrow B \rightarrow E \rightarrow F \rightarrow G$ , total cost 18. Nodes expanded in the order S, C, A, E, F, B, D, E, F, G.

**Why this graph is worth understanding rather than memorising:**  $h$  here is *admissible* (it never exceeds the true remaining cost) but *not consistent* —  $h(B) = 13$  while  $\text{cost}(B, E) + h(E) = 6 + 4 = 10$ . With an inconsistent heuristic a node can be reached more cheaply after it has been closed, which is why  $E$  and  $F$  each get expanded twice. An implementation with a strict closed list and no reopening returns  $S-A-E-F-G$  at cost 19, and that is the commonest wrong answer to this question.

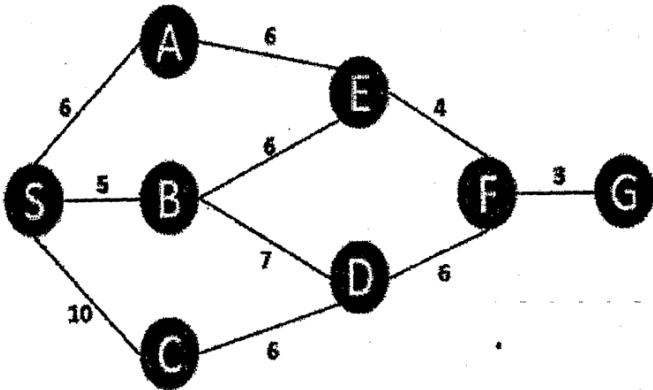
3.N.2 A\* Vs greedy on the 2080 Chaitra graph

How does A\* overcome the problem with greedy best-first search? 3+4+3

PIN 1/41 80 Ch 3+4+3 · the same graph as §3.N.1 with different heuristics, given in a table rather than on the nodes. Check which you have been given before starting

QUESTION AS PRINTED 80 Ch · 3+4+3

How does A\* search overcome the problem associated with Greedy Best First Search? Using the following figure and table, where the starting node is S and the destination is G, compare the result of the A\* algorithm with greedy search. (The table gives the cost required to reach the goal; the connecting link gives the cost from one node to the other.)



|      |    |    |    |   |   |   |   |   |
|------|----|----|----|---|---|---|---|---|
| From | S  | A  | B  | C | D | E | F | G |
| Cost | 15 | 10 | 12 | 5 | 4 | 2 | 1 | 0 |

Greedy best first: order on  $h$  alone

| Expand | OPEN after ( $h$ )   | CLOSED        |
|--------|----------------------|---------------|
| S      | C/5, A/10, B/12      | S             |
| C      | D/4, A/10, B/12      | S, C          |
| D      | F/1, A/10, B/12      | S, C, D       |
| F      | G/0, E/2, A/10, B/12 | S, C, D, F    |
| G      | goal                 | S, C, D, F, G |

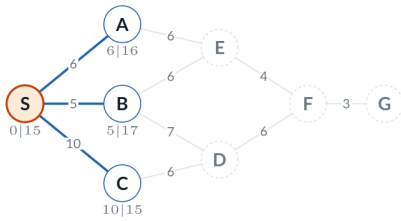
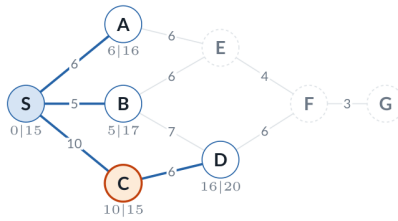
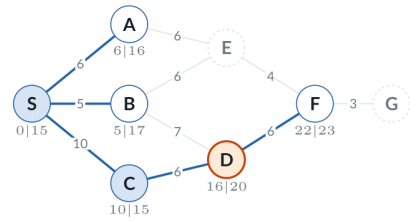
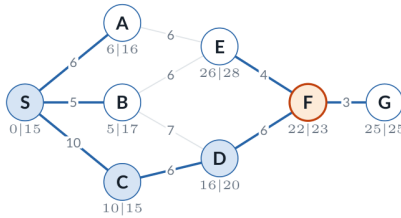
- Path  $S \rightarrow C \rightarrow D \rightarrow F \rightarrow G$ , cost  $10 + 6 + 6 + 3 = 25$ .
- That is the worst of the four routes. Greedy went straight to  $C$  because  $h(C) = 5$  is the smallest number in sight, never noticing that getting to  $C$  costs 10, twice what  $B$  costs.
- Only 5 nodes expanded, so it is fast. It is fast and wrong.

Step by step: greedy best first

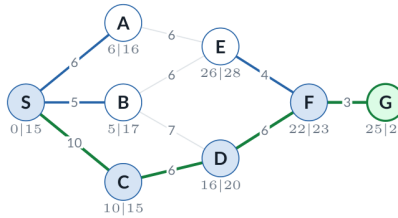
A\*: order on  $f = g + h$

| Expand | $g, h, f$ | OPEN after re-ordering |
|--------|-----------|------------------------|
| S      | 0, 15, 15 | C/15, A/16, B/17       |
| C      | 10, 5, 15 | A/16, B/17, D/20       |
| A      | 6, 10, 16 | E/14, B/17, D/20       |
| E      | 12, 2, 14 | B/17, F/17, D/20       |
| B      | 5, 12, 17 | E/13, D/16, F/17       |
| E      | 11, 2, 13 | D/16, F/16             |
| D      | 12, 4, 16 | F/16                   |
| F      | 15, 1, 16 | G/18                   |
| G      | 18, 0, 18 | goal, stop             |

- Path  $S \rightarrow B \rightarrow E \rightarrow F \rightarrow G$ , cost 18, the optimum.

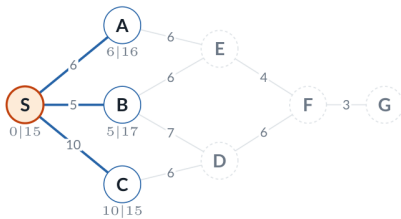
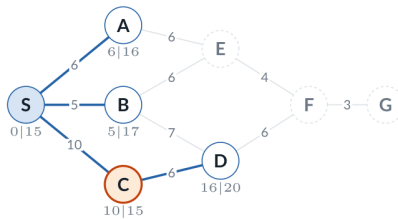
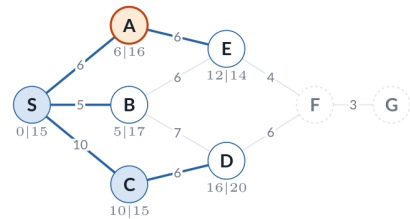
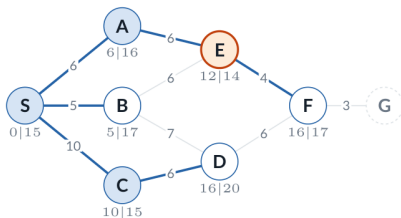
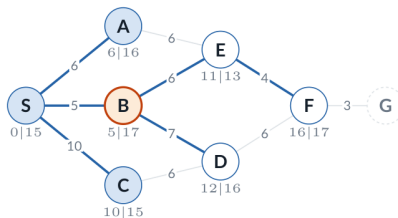
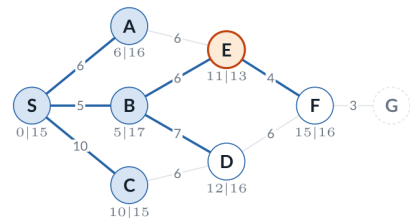
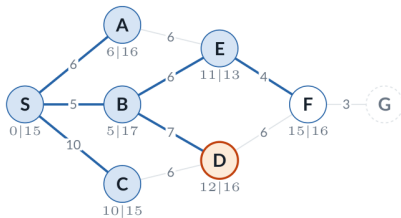
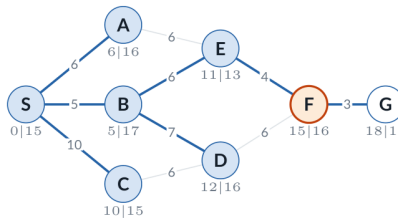
1. expand S ( $h$  15)2. expand C ( $h$  5)3. expand D ( $h$  4)4. expand F ( $h$  1)

5. expand G (goal)

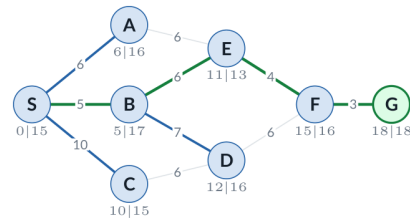


● closed    ● on OPEN    ● expanding now    ● goal

### Step by step: A\* on the same graph

1. expand S ( $f$  15)2. expand C ( $f$  15)3. expand A ( $f$  16)4. expand E ( $f$  14)5. expand B ( $f$  17) replace D, E6. expand E ( $f$  13) replace F7. expand D ( $f$  16)8. expand F ( $f$  16)

9. expand G (goal off OPEN)



● closed    ● on OPEN    ● expanding now    ● goal

## The comparison, which is what the 3 marks are for

|                | Greedy best first   | A*                       |
|----------------|---------------------|--------------------------|
| $f(n)$         | $h(n)$              | $g(n) + h(n)$            |
| Path found     | $S-C-D-F-G$         | $S-B-E-F-G$              |
| Cost           | 25, the worst route | 18, the optimum          |
| Nodes expanded | 5                   | 9                        |
| Optimal        | × no                | ✓ yes, $h$ is admissible |
| Complete       | × no                | ✓ yes                    |

- **The trade in one line:** A\* expanded almost twice as many nodes and returned a path 28% cheaper. Greedy buys speed with correctness.

## Where greedy went wrong, precisely

- At the very first step both algorithms rank  $C$  best, but for different reasons. Greedy sees  $h(C) = 5$  and **commits**; A\* sees  $f(C) = 10 + 5 = 15$ , ties with nothing better, expands it, and then **keeps  $B$  on the table**.
- Once greedy has moved to  $C$  it never reconsiders. It has **no memory of cost paid**, so nothing can ever make  $C$  look like a mistake.
- **The general statement to close with:** a heuristic estimates the *remaining* journey. Judging a node on that alone is judging a route by how it ends, ignoring how much it cost to get there. Adding  $g(n)$  restores the whole journey to the comparison, and that is the entire difference between the two algorithms.

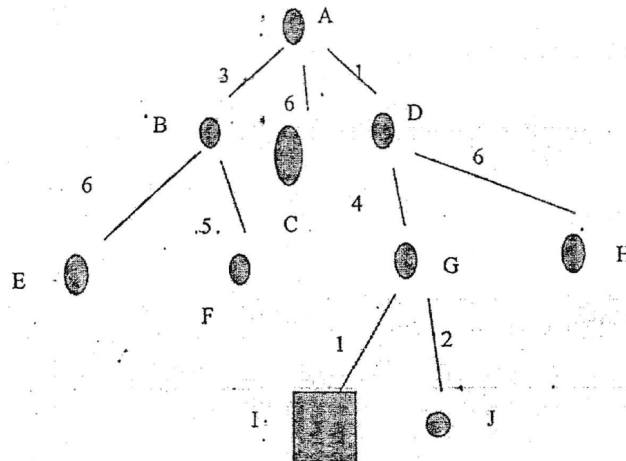
## 3.N.3 Best first search on the 2076 Bhadra tree

## Show the open list, closed list and path at each step 5

76 Bh 5 · 1 paper, 5 marks, so no tier chip · the question names the **three things it wants tabulated**, so tabulate exactly those three

## QUESTION AS PRINTED 76 Bh · 5

Consider a search tree for a path-finding problem as shown. Assuming node A to be the root node and node I to be the goal node, explain how Best First Search reaches the goal by showing changes in the open list, closed list, and path at each step of the search.



### Reading the figure

- Tree:  $A \rightarrow B, C, D$ ;  $B \rightarrow E, F$ ;  $D \rightarrow G, H$ ;  $G \rightarrow I, J$ .  $I$  (the square) is the goal.
- Values on the edges, taken as the estimate for the child node:  $B=3, C=6, D=1, E=6, F=5, G=4, H=6, I=1, J=2$ .
- Best first** expands the OPEN node with the smallest value, so the OPEN list is re-ordered after every expansion.

### The trace

| Step | Expand | OPEN (node/value)            | after         | CLOSED |
|------|--------|------------------------------|---------------|--------|
| 1    | A      | D/1, B/3, C/6                | A             |        |
| 2    | D      | B/3, G/4, C/6, H/6           | A, D          |        |
| 3    | B      | G/4, F/5, C/6, E/6, H/6      | A, D, B       |        |
| 4    | G      | I/1, J/2, F/5, C/6, E/6, H/6 | A, D, B, G    |        |
| 5    | I      | <b>goal reached</b>          | A, D, B, G, I |        |

### Path at each step

#### Step Path to the node just expanded

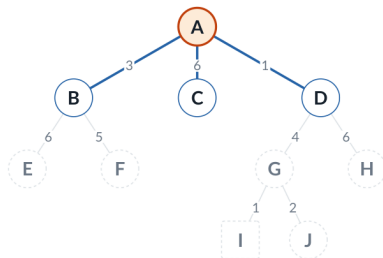
- 1 A
- 2  $A \rightarrow D$
- 3  $A \rightarrow B$
- 4  $A \rightarrow D \rightarrow G$
- 5  $A \rightarrow D \rightarrow G \rightarrow I$

**Answer:** path  $A \rightarrow D \rightarrow G \rightarrow I$ , cost  $1 + 4 + 1 = 6$ .

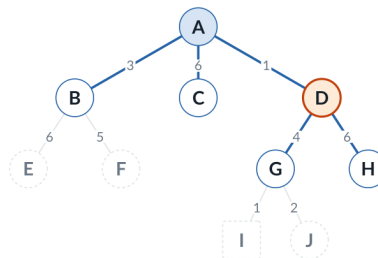
- Step 3 is the one to explain.** After expanding  $D$ , the best node on OPEN is  $B$  at 3, not  $G$  at 4 – so best first **jumps back across the tree** to a different branch before returning to  $G$ . That switching is precisely what distinguishes best first from depth first, and it is where the marks are.
- $C, E, F, H$  and  $J$  are **never expanded**. Five of the ten nodes are never touched, which is the point of using a heuristic at all.
- If the numbers are read as *edge costs* and accumulated instead, the same path comes out at the same cost of 6. Either reading is defensible here; say which you are using.

### Step by step: the tree after each expansion

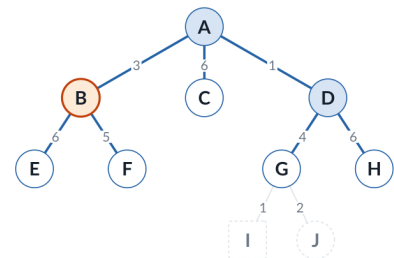
1. expand A (root)



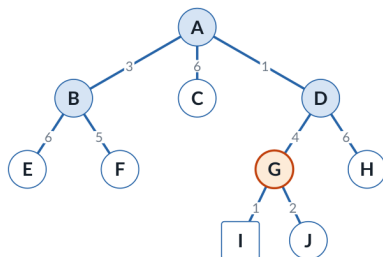
2. expand D (value 1)



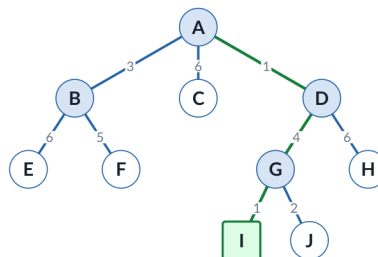
3. expand B (value 3)



4. expand G (value 4)



5. expand I (goal)



● closed    ● on OPEN    ● expanding now    ● goal

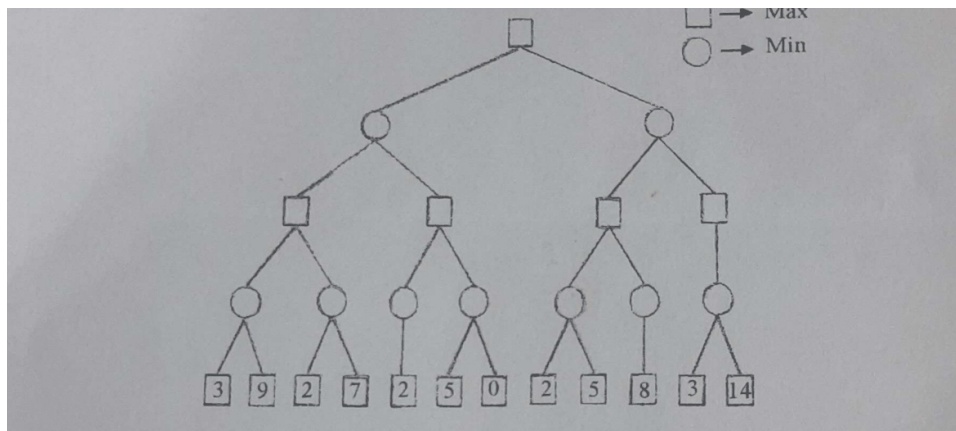
### 3.N.4 Minimax and alpha-beta on the 2079 Chaitra tree

#### Apply minimax with alpha-beta pruning to the given tree 2+6

PIN 1/41 79 Ch 2+6 · squares are MAX, circles are MIN, and **three of the MIN nodes have only one child**, which is easy to misread

QUESTION AS PRINTED 79 Ch · 2+6

What is minmax search? Apply minmax search along with alpha-beta pruning for the following tree (max node at top).



#### Reading the tree

- Five levels: MAX root, MIN, MAX, MIN, then the 12 leaves.
- Left MIN node  $m_1$  over MAX nodes  $s_1, s_2$ ; right MIN node  $m_2$  over  $s_3, s_4$ .
- The MIN nodes above the leaves, left to right, take:  $\{3, 9\}, \{2, 7\}, \{2\}, \{5, 0\}, \{2, 5\}, \{8\}, \{3, 14\}$ . **Three of them have a single child**: count the lines in the figure, not the leaves.

#### Step 1: MIN over the leaves

- $c_1 = \min(3, 9) = 3$     $c_2 = \min(2, 7) = 2$     $c_3 = 2$
- $c_4 = \min(5, 0) = 0$     $c_5 = \min(2, 5) = 2$     $c_6 = 8$
- $c_7 = \min(3, 14) = 3$

#### Step 2: MAX

- $s_1 = \max(3, 2) = 3$     $s_2 = \max(2, 0) = 2$
- $s_3 = \max(2, 8) = 8$     $s_4 = 3$

#### Step 3: MIN, then the root

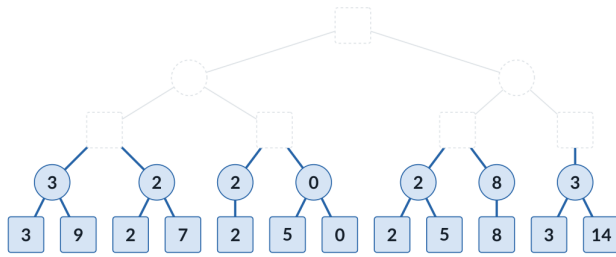
- $m_1 = \min(3, 2) = 2$     $m_2 = \min(8, 3) = 3$
- root** =  $\max(2, 3) = 3$ , so MAX plays the **right** branch.

#### Step by step: minimax backing the values up

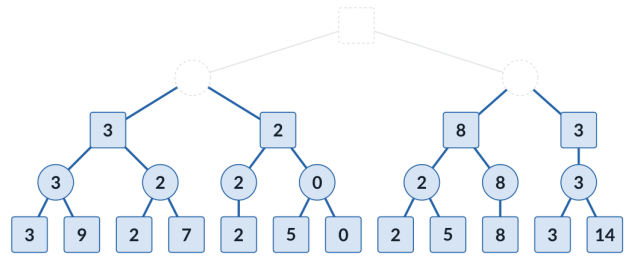
#### Alpha-beta, left to right

| Node  | $(\alpha, \beta)$ on entry | What happens                                                                 |
|-------|----------------------------|------------------------------------------------------------------------------|
| $c_1$ | $(-\infty, \infty)$        | leaves 3, 9 $\rightarrow 3$                                                  |
| $s_1$ | $\alpha \leftarrow 3$      |                                                                              |
| $c_2$ | $(3, \infty)$              | leaf 2 gives value 2; $2 \leq \alpha = 3 \rightarrow$<br><b>PRUNE leaf 7</b> |
| $m_1$ | $\beta \leftarrow 3$       | $s_1 = 3$                                                                    |
| $c_3$ | $(-\infty, 3)$             | leaf 2 $\rightarrow 2$                                                       |
| $c_4$ | $(2, 3)$                   | leaves 5, 0 $\rightarrow 0$ ; no cut, $5 > \alpha$<br>when tested            |
| $m_1$ |                            | $= \min(3, 2) = 2$ ; root $\alpha \leftarrow 2$                              |
| $c_5$ | $(2, \infty)$              | leaf 2 gives 2; $2 \leq \alpha = 2 \rightarrow$<br><b>PRUNE leaf 5</b>       |
| $c_6$ | $(2, \infty)$              | leaf 8 $\rightarrow 8$ , so $s_3 = 8$                                        |
| $m_2$ | $\beta \leftarrow 8$       |                                                                              |
| $c_7$ | $(2, 8)$                   | leaves 3, 14 $\rightarrow 3$ , so $s_4 = 3$                                  |
| root  |                            | $m_2 = \min(8, 3) = 3$ ; $\max(2, 3) = 3$                                    |

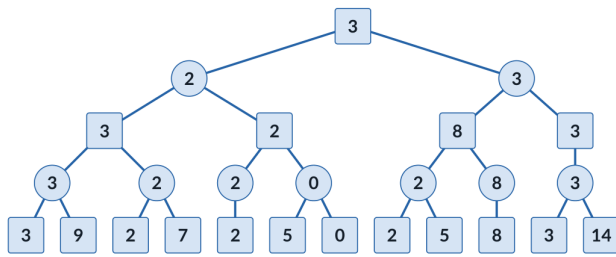
## 1. MIN over the leaves



## 2. MAX row

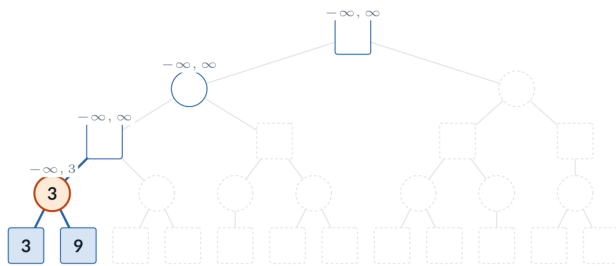
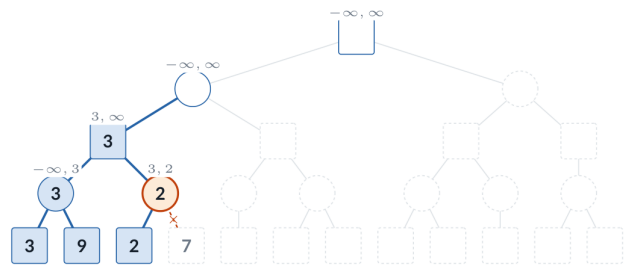
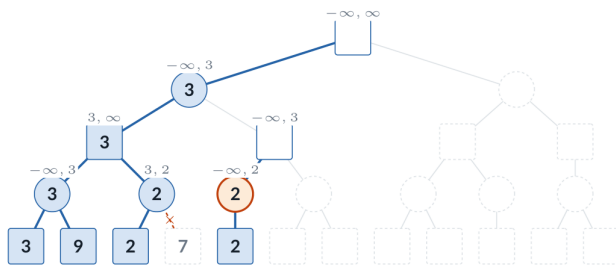
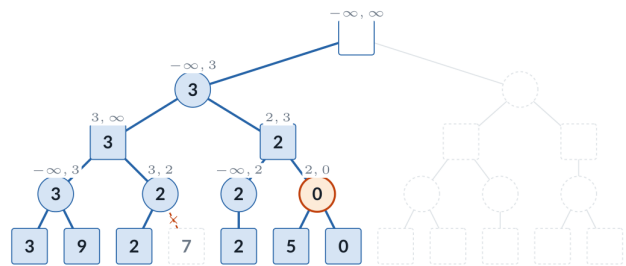


## 3. MIN row, then the root



● value backed up   
 ○ entered,  $\alpha$ ,  $\beta$  set   
 ○ just finished   
 ○ pruned

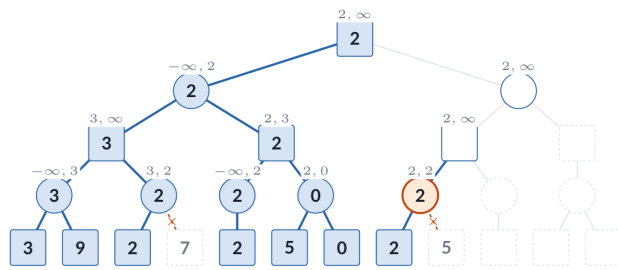
## Step by step: alpha-beta, left half

1.  $c_1 = \min(3, 9) = 3$ 2.  $c_2 = \min(2) = 2$  **prune 7**3.  $c_3 = \min(2) = 2$ 4.  $c_4 = \min(5, 0) = 0$ 

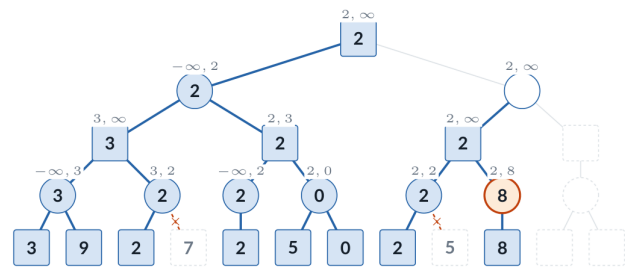
● value backed up   
 ○ entered,  $\alpha$ ,  $\beta$  set   
 ○ just finished   
 ○ pruned

## Step by step: alpha-beta, right half and the root

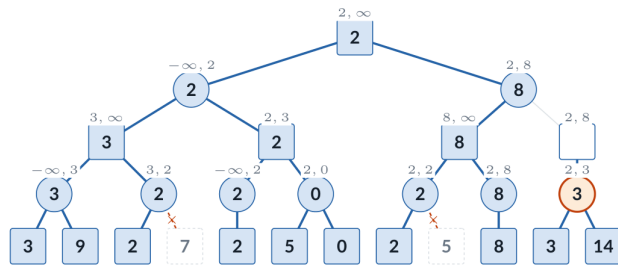
5.  $c_5 = \min(2) = 2$  **prune 5**



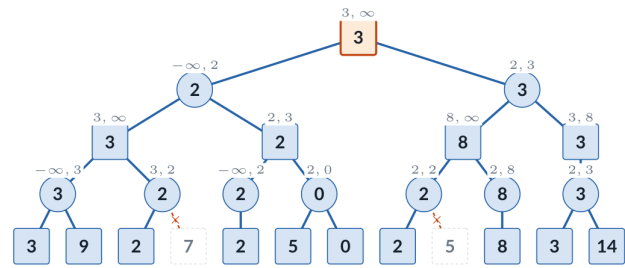
6.  $c_6 = \min(8) = 8$



7.  $c_7 = \min(3, 14) = 3$



8.  $\text{root} = \max(2, 3) = 3$



● value backed up ○ entered,  $\alpha, \beta$  set ● just finished ○ pruned

**Answer:** minimax value 3; MAX plays the right branch. 2 of the 12 leaves are pruned: the 7 under  $c_2$ , and the 5 under  $c_5$ . Ten leaves are examined.

**Note the equality in the second cut.** At  $c_5$  the value is 2 and  $\alpha$  is also 2. A MIN node cuts on value  $\leq \alpha$ , not  $<$ , because MAX already has a way to score 2 elsewhere and will never *prefer* this branch. Using a strict  $<$  here is the commonest slip and it costs you that pruned leaf.

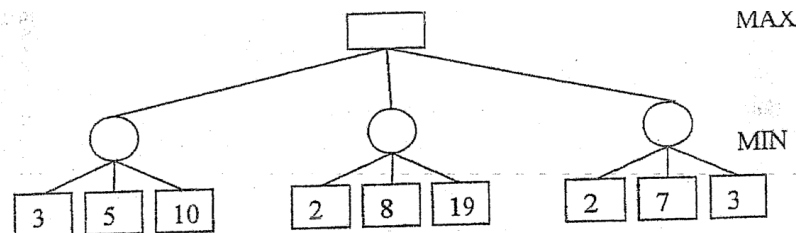
### 3.N.5 Alpha-beta on the 2076 Baishakh tree

Discuss alpha-beta pruning and find the minimax value 4+4

PIN 1/41 76 Ba 4+4 · a small tree that prunes a much larger fraction than §3.N.4, and the reason why is the lesson

QUESTION AS PRINTED 76 Ba · 4+4

Discuss the alpha-beta pruning algorithm. Find the min-max value using this concept in the following tree.



### The trace

- MAX root over three MIN nodes, each with three leaves:  $\{3, 5, 10\}$ ,  $\{2, 8, 19\}$ ,  $\{2, 7, 3\}$ .

| Node  | $(\alpha, \beta)$     | What happens                                                                                                          |
|-------|-----------------------|-----------------------------------------------------------------------------------------------------------------------|
| $m_1$ | $(-\infty, \infty)$   | $3 \rightarrow 5 \rightarrow 10$ , value = 3. <b>Nothing can be pruned here</b> , because $\alpha$ is still $-\infty$ |
| root  | $\alpha \leftarrow 3$ |                                                                                                                       |
| $m_2$ | $(3, \infty)$         | leaf 2 gives 2; $2 \leq 3 \rightarrow$ <b>PRUNE 8 and 19</b>                                                          |
| $m_3$ | $(3, \infty)$         | leaf 2 gives 2; $2 \leq 3 \rightarrow$ <b>PRUNE 7 and 3</b>                                                           |
| root  |                       | $\max(3, 2, 2) = 3$                                                                                                   |

- Minimax value 3, MAX plays the **leftmost** branch.
- 4 of the 9 leaves pruned (8, 19, 7, 3); five examined: 3, 5, 10, 2, 2.

### Why this tree prunes so much better

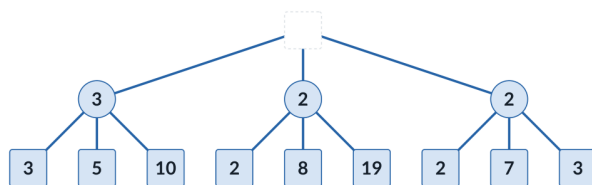
- §3.N.4 pruned 2 leaves in 12, about 17%. This one prunes 4 in 9, about 44%. The algorithm is identical; the **move ordering** is not.
- Here the **best branch happens to be examined first**. That sets  $\alpha = 3$  immediately, and every later MIN node that finds any child below 3 can stop at once. Both of them do, on their very first leaf.
- The **general rule**: alpha-beta is only as good as the order in which moves are examined.
  - Best case** (best move first everywhere):  $O(b^{m/2})$ , so the same time searches **twice as deep**.
  - Worst case** (best move last everywhere):  $O(b^m)$ , i.e. **no saving at all** – alpha-beta never prunes more than minimax examines, and can prune nothing.
- This is why real game programs spend effort on **move ordering heuristics** before they spend it on searching deeper.

### Two things to say when asked to “discuss” it

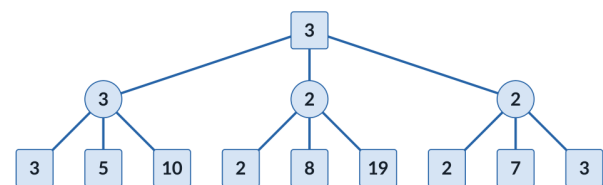
- It returns exactly the minimax value.** Pruning removes work, never accuracy. Say this first.
- A pruned branch is one whose value **cannot change the root**, because a player higher up already has a better guaranteed alternative.

### Step by step: minimax backing the values up

#### 1. MIN over each group of three leaves



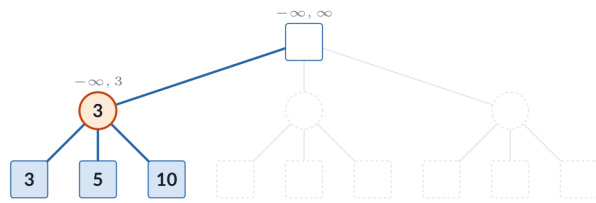
#### 2. MAX at the root



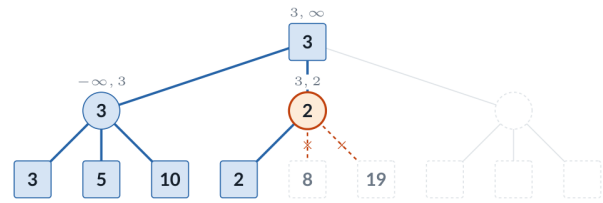
● value backed up   
 ○ entered,  $\alpha, \beta$  set   
 ○ just finished   
 ○ pruned

### Step by step: alpha-beta, one MIN node at a time

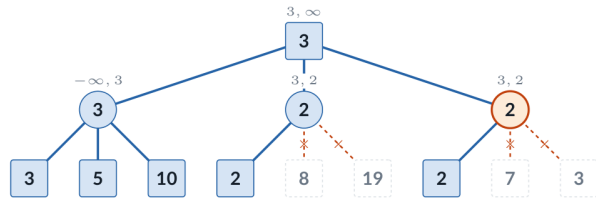
1.  $m_1 = \min(3, 5, 10) = 3$



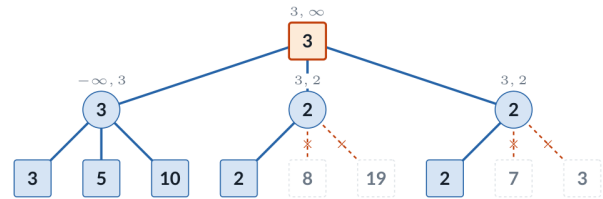
2.  $m_2 = \min(2) = 2$  **prune 8, 19**



3.  $m_3 = \min(2) = 2$  **prune 7, 3**



4.  $\text{root} = \max(3, 2, 2) = 3$



● value backed up   
 ○ entered,  $\alpha, \beta$  set   
 ○ just finished   
 ○ pruned

## 4. Knowledge Representation, Inference and Reasoning – Solved Problems

**The biggest companion in the subject.** 33 papers set a resolution refutation, 15 set a Bayes calculation, 4 set a CNF conversion and 1 sets a truth-table proof. Every one is worked below under the question exactly as its paper prints it. Renamings of a puzzle already worked get the paper's own statement plus what changes, so nothing is left to guess and no proof is copied out twice.

The five procedures everything here is built on

- **P1 Truth table** –  $2^n$  rows for  $n$  symbols; a **tautology** is a column of all T.
- **P2 CNF** – eliminate  $\leftrightarrow$ , eliminate  $\rightarrow$ , move  $\neg$  in, standardise apart, skolemize, drop  $\forall$ , distribute  $\vee$  over  $\wedge$ , split.
- **P3 Resolution refutation** – FOPL, CNF, **negate the goal**, resolve on complementary literals with the MGU, derive  $\square$ .
- **P4 Chaining** – forward from the facts, or backward from the goal, on definite clauses.
- **P5 Bayes** –  $P(A|B) = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|\neg A)P(\neg A)}$ , and for a network  $P(x_1 \dots x_n) = \prod P(x_i | \text{parents}(x_i))$ .

Notation used throughout: constants start with a capital, variables are  $x, y, z$ , and a clause is written as a disjunction of literals. Beside every resolvent, **(i, j)** names the two clauses it came from and  $[t/x]$  the substitution used. That annotation is where a third of the marks sit.

### P1. Truth tables: WFF, validity and tautology

Define a WFF with its properties. Prove that  $(A \wedge (A \rightarrow B)) \rightarrow B$  is a tautology **4+4**

PIN 1/41 71 Bh **4+4**

QUESTION AS PRINTED 2071 Bhadra · CT 78506 · [4+4]

Define a WFF with its properties. Prove that the proposition  $(A \wedge (A \rightarrow B) \rightarrow B)$  is a tautology.

**Solution: build the table**

- Two symbols, so  $2^2 = 4$  rows. Work left to right, one column per sub-formula.

| A | B | $A \rightarrow B$ | $A \wedge (A \rightarrow B)$ | $\dots \rightarrow B$ |
|---|---|-------------------|------------------------------|-----------------------|
| T | T | T                 | T                            | T                     |
| T | F | F                 | F                            | T                     |
| F | T | T                 | F                            | T                     |
| F | F | T                 | F                            | T                     |

- The last column is **T in every row**, so the proposition is **valid: a tautology**. ✓
- Rows 2–4 are true *vacuously*: the antecedent  $A \wedge (A \rightarrow B)$  is false, and  $F \rightarrow$  anything is T. **Say that** or the last mark is at risk.

**The definition half of the answer**

- A **well-formed formula** is a string built only by the recursive rules in §4.2: a symbol;  $\neg\alpha$ ;  $(\alpha)$ ; and  $(\alpha \wedge \beta)$ ,  $(\alpha \vee \beta)$ ,  $(\alpha \rightarrow \beta)$ ,  $(\alpha \leftrightarrow \beta)$  for WFFs  $\alpha, \beta$  – **and nothing else**.
- **Properties to list**: it is finite; it is uniquely parsed (brackets or precedence fix the reading); it has a truth value under every interpretation; and it is exactly the set of strings the inference rules may be applied to.
- $A \wedge \rightarrow B$  and  $(A \vee B)$  are **not WFFs**: no rule builds them. Giving one non-example is worth a mark.
- **This formula is modus ponens written as a single proposition**, which is the point of the question: an inference rule is sound exactly when its implication form is a tautology. Cross-refer to the rules table and the two marks for “properties” are safe.

### P2. Conversion to CNF

Four papers give a formula and ask for its clause form **HOT 4/41**. Nine more ask for the steps in the abstract; those are answered by §4.4 plus any one example below.

$P \vee (\neg P \wedge Q \wedge R)$ , and “Everyone who loves all animals is loved by someone” **2+2+2+3**

73 Bh **2+2+2+3**

QUESTION AS PRINTED 2073 Bhadra · CT 78506 · [2+2+2+3]

Why is conjunctive normal form required? Explain all the steps to convert to CNF, and transform the given sentences into CNF:  $P \vee (\neg P \wedge Q \wedge R)$ ; and “Everyone who loves all animals is loved by someone.”

**Part 1:**  $P \vee (\neg P \wedge Q \wedge R)$ 

- No  $\leftrightarrow$ , no  $\rightarrow$ , no negation to move: go straight to **step 7**, distribute  $\vee$  over  $\wedge$ .
- $P \vee (\neg P \wedge Q \wedge R)$

$$= (P \vee \neg P) \wedge (P \vee Q) \wedge (P \vee R)$$

- $P \vee \neg P$  is a **tautology**, true in every model, so the clause adds nothing and is dropped:

$$(P \vee Q) \wedge (P \vee R)$$

- **Two clauses:**  $\{P, Q\}$  and  $\{P, R\}$ .
- **Dropping the tautologous clause is worth doing and worth saying.** Leaving it in is not wrong, but the simplification is the part that shows the definition was understood.

$$P \rightarrow ((Q \wedge \neg R) \rightarrow S) \quad 6$$

74 Ma 6

QUESTION AS PRINTED 2074 Magh · CT 78506 · [6]

Why is conjunctive normal form required? Explain all the steps to convert to CNF, and transform the given sentence into CNF:  $P \rightarrow ((Q \wedge \neg R) \rightarrow S)$

**Solution**

- **Inner implication first:**  $(Q \wedge \neg R) \rightarrow S = \neg(Q \wedge \neg R) \vee S$
- **Outer implication:**  
 $P \rightarrow [\neg(Q \wedge \neg R) \vee S] = \neg P \vee \neg(Q \wedge \neg R) \vee S$
- **De Morgan**, then double negation:  $\neg(Q \wedge \neg R) = \neg Q \vee R$
- 

$$\neg P \vee \neg Q \vee R \vee S$$

$$P \rightarrow ((Q \wedge \neg R) \leftrightarrow S) \quad 8 \quad 2+2+4$$

72 Ash 8, 79 Jth 2+2+4 · 79 Jth pairs it with  $\neg(P \wedge Q) \vee (P \vee S) \rightarrow R$ , worked below

QUESTION AS PRINTED 2072 Ashwin · CT 78506 · [8]

Why is conjunctive normal form required? Explain all the steps to convert to CNF, and transform the given sentence into CNF:  $P \rightarrow ((Q \wedge \neg R) \leftrightarrow S)$

**Part 2: the quantified sentence**

- **FOPL:**  
 $\forall x [\forall y \text{ Animal}(y) \rightarrow \text{Loves}(x, y)] \rightarrow [\exists z \text{ Loves}(z, x)]$
- **Eliminate  $\rightarrow$**  (both of them):  
 $\forall x \neg[\forall y \neg \text{Animal}(y) \vee \text{Loves}(x, y)] \vee \exists z \text{ Loves}(z, x)$
- **Move  $\neg$  inwards**, dual then De Morgan:  
 $\forall x [\exists y \text{ Animal}(y) \wedge \neg \text{Loves}(x, y)] \vee \exists z \text{ Loves}(z, x)$
- **Skolemize.** Both  $\exists$  sit inside  $\forall x$ , so both become **functions of  $x$ :**  $y \Rightarrow F(x), z \Rightarrow G(x)$ .
- **Drop  $\forall$ , distribute, split:**
  - $\text{Animal}(F(x)) \vee \text{Loves}(G(x), x)$
  - $\neg \text{Loves}(x, F(x)) \vee \text{Loves}(G(x), x)$
- **$F$  and  $G$  must be functions, not constants.** A constant would claim one single animal, and one single admirer, for the whole universe.

**The point of this one**

- Nothing to distribute, so the answer is **a single clause** with four literals. A conjunction of one clause is still CNF.
- Students who expect several clauses often invent a split that is not there.  $\vee$  **distributes over  $\wedge$ , never the reverse**, and there is no  $\wedge$  left at the top level here.
- Written as a set:  $\{\neg P, \neg Q, R, S\}$ . It is a **definite clause** – two positive literals, so no; it is a plain clause. (Definite needs *exactly one* positive literal, and this has two,  $R$  and  $S$ .) A one-line check like that is what 78 Ch's Horn-clause part is testing.

**Solution: the biconditional first**

- Write  $M = Q \wedge \neg R$  to keep the algebra readable.
- **Step 1, eliminate**  $\leftrightarrow$ :  $M \leftrightarrow S = (M \rightarrow S) \wedge (S \rightarrow M)$
- **Step 2, eliminate**  $\rightarrow$  inside:  $= (\neg M \vee S) \wedge (\neg S \vee M)$
- Substitute  $M$  back and apply De Morgan to  $\neg M$ :

$$(\neg Q \vee R \vee S) \wedge (\neg S \vee (Q \wedge \neg R))$$

- **Step 7, distribute** in the right half:  
 $\neg S \vee (Q \wedge \neg R) = (\neg S \vee Q) \wedge (\neg S \vee \neg R)$
- So the inner formula is  $(\neg Q \vee R \vee S) \wedge (\neg S \vee Q) \wedge (\neg S \vee \neg R)$ .
- **Now the outer implication:**  $P \rightarrow X = \neg P \vee X$ , and  $\neg P$  distributes into every clause of  $X$ :

$$\begin{array}{l} (\neg P \vee \neg Q \vee R \vee S) \wedge \\ (\neg P \vee \neg S \vee Q) \wedge \\ (\neg P \vee \neg S \vee \neg R) \end{array}$$

- **Three clauses.**

$$\neg(P \wedge Q) \vee (P \vee S) \rightarrow R \quad \text{2+2+4}$$

79 Jth 2+2+4 · the paper's part (i); part (ii) is the biconditional worked above

**QUESTION AS PRINTED** 2079 Jestha · CT 653 · [2+2+4]

Why is conjunctive normal form required? Explain all the steps to convert to CNF, and transform the given sentences into CNF: (i)  $\neg(P \wedge Q) \vee (P \vee S) \rightarrow R$  (ii)  $P \rightarrow ((Q \wedge \neg R) \leftrightarrow S)$

**Solution**

- $\rightarrow$  binds **loosest**, so the sentence reads  $[\neg(P \wedge Q) \vee (P \vee S)] \rightarrow R$ . **Reading the brackets wrongly here changes the answer**; state the reading before starting.
- **Eliminate**  $\rightarrow$ :  $\neg[\neg(P \wedge Q) \vee (P \vee S)] \vee R$
- **Move**  $\neg$  in (De Morgan on the bracket):  $[(P \wedge Q) \wedge \neg P \wedge \neg S] \vee R$
- **Distribute**  $\vee R$  over the four conjuncts:

$$(P \vee R) \wedge (Q \vee R) \wedge (\neg P \vee R) \wedge (\neg S \vee R)$$

- **Four clauses.**

**Where this one goes wrong**

- **Expanding**  $\leftrightarrow$  **the other way.**  $A \leftrightarrow B$  is also  $(A \wedge B) \vee (\neg A \wedge \neg B)$ . That is correct logic but it is a *disjunction* of conjunctions, so it takes twice the work to push back into CNF. **Always use the two-implications form.**
- **Forgetting to distribute**  $\neg P$  **into all three clauses.**  $\neg P \vee (C_1 \wedge C_2 \wedge C_3)$  is not  $\neg P \vee C_1 \wedge C_2 \wedge C_3$ ; the  $\neg P$  joins *each* clause.
- **Sanity check that costs 10 seconds:** set  $P$  false. Every clause contains  $\neg P$ , so all three are satisfied – which is right, because  $P \rightarrow \dots$  is vacuously true. Set  $P$  true,  $S$  true,  $Q$  false: clause 2 becomes  $F \vee F \vee F$ , false, and indeed  $S$  true with  $Q$  false breaks the biconditional. **The CNF behaves like the original.**

**Reading the answer back**

- Clauses 1 and 3 are  $(P \vee R)$  and  $(\neg P \vee R)$ . Resolving them gives  $R$  on its own, so **the whole sentence is equivalent to  $R$ .**
- That is correct: the antecedent  $\neg(P \wedge Q) \vee (P \vee S)$  is a **tautology** (if  $P$  is false the first disjunct is true, if  $P$  is true the second is), and  $\text{True} \rightarrow R$  is just  $R$ .
- **Saying this earns the last mark** and proves the conversion was not mechanical. It is also the cleanest possible demonstration of why resolution wants CNF: two clauses in, one simpler clause out.

**P3. Translating English into FOPL****Convert the following into FOL** 2

76 Bh 2 · one paper, 2 marks, so no tier chip · but **the same skill is step 1 of all 33 resolution proofs**, which is the real reason to practise it

**QUESTION AS PRINTED** 2076 Bhadra · CT 78506 · [2]

Convert the following into FOL.

- Arya hates Joffery and Cersei.
- No Lannister loves Bran.
- Sansa has at least one sister.
- Every Stark hates at least one Lannister.

**Solution**

- a) Two separate facts, joined by  $\wedge$ . No quantifier: both are named individuals.

$$\text{hates}(\text{Arya}, \text{Joffery}) \wedge \text{hates}(\text{Arya}, \text{Cersei})$$

- b) “No A loves B” = there is no A that loves B, or equivalently every A fails to love B:

$$\neg \exists x [\text{Lannister}(x) \wedge \text{loves}(x, \text{Bran})]$$

$$\equiv \forall x [\text{Lannister}(x) \rightarrow \neg \text{loves}(x, \text{Bran})]$$

- c) “at least one” =  $\exists$ , and  $\exists$  takes  $\wedge$ :

$$\exists x [\text{sister}(x) \wedge \text{sisterOf}(x, \text{Sansa})]$$

- d)  $\forall$  **outside**,  $\exists$  **inside**, because each Stark may hate a different Lannister:

$$\forall x [\text{Stark}(x) \rightarrow \exists y (\text{Lannister}(y) \wedge \text{hates}(x, y))]$$

**The four rules this question tests**

- $\forall$  takes  $\rightarrow$ ;  $\exists$  takes  $\wedge$ . Every wrong answer to (c) or (d) breaks this.
- **Quantifier order carries meaning.**  $\forall x \exists y$  (each Stark hates some Lannister) is not  $\exists y \forall x$  (one Lannister is hated by all Starks). The English “at least one” after “every” fixes the order as  $\forall \exists$ .
- **“No” negates the whole existential**, not the predicate. Writing  $\forall x \text{Lannister}(x) \wedge \neg \text{loves}(x, \text{Bran})$  says everything in the universe is a Lannister.
- Both forms of (b) are correct and **equivalent by the dual**  $\neg \exists \equiv \forall \neg$ . Giving both, with that one-line justification, is a complete answer to a 2-mark part.
- **Predicate names are your choice**, but stay consistent:  $\text{sisterOf}(x, y)$  must mean the same thing every time it appears.

**P4. Unification: finding the MGU**

Never asked on its own, but every resolution step needs one and the marks come off when it is wrong. Four standard pairs, in the form the papers and the textbook use.

**Worked**

- 1.  $\text{King}(x), \text{King}(\text{John}) \rightarrow \theta = [\text{John}/x]$ .
- 2.  $P(x, y), P(a, f(z)) \rightarrow \theta = [a/x, f(z)/y]$ .  $z$  stays free: binding it would be *less* general.
- 3.  $P(x, f(y)), P(a, f(g(x))) \rightarrow$  start with  $[a/x]$ , then  $y$  must match  $g(x)$ , and  $x$  is already  $a$ :

$$\theta = [a/x, g(a)/y]$$

- 4.  $Q(a, g(x, a), f(y)), Q(a, g(f(b), a), x) \rightarrow$   $x$  must be  $f(b)$  from the second argument, and then  $f(y)$  must match  $x = f(b)$ , giving  $y = b$ :

$$\theta = [f(b)/x, b/y]$$

**The three ways it fails**

- **Different predicate or arity.**  $P(x)$  and  $Q(x)$  never unify; neither do  $P(x)$  and  $P(x, y)$ .
- **Two different constants.**  $\text{King}(\text{John})$  and  $\text{King}(\text{Ram})$  fail: a constant cannot be substituted.
- **Occurs check.**  $x$  cannot bind to  $f(x)$ ; the substitution would never terminate.
- **Apply each binding to everything already found**, as in example 3. Reporting  $[a/x, g(x)/y]$  instead of  $[a/x, g(a)/y]$  is the standard half-mark loss.
- **Standardise apart before unifying two clauses.** If both clauses call their variable  $x$ , rename one to  $x'$  first, or the unifier will tie together two variables that have nothing to do with each other.

**P5. Resolution refutation Family A: likes all kinds of food**

**7 papers, 5 wordings, 55 marks** – the single most-repeated premise set in the subject 72 Ash 7, 74 Bh 8, 73 Bh 8, 71 Bh 8, 80 Bh 8, 80 Ba 8, 72 Ma 6+4. Work it once properly and five papers are answered.

**QUESTION AS PRINTED** 2072 Ashwin · Q10 · [7] also 2074 Bhadra [8], 2073 Bhadra CT 78506 [8]

Assume the following facts: John likes all kinds of food; apples are food; chicken is food; anything anyone eats and isn't killed by is food; Bill eats peanuts and is still alive; Sue eats everything Bill eats. Prove that John likes peanuts using resolution refutation.

**Step 1 Facts into FOPL**

1.  $\forall x \text{ food}(x) \rightarrow \text{likes}(\text{John}, x)$
2.  $\text{food}(\text{Apple})$
3.  $\text{food}(\text{Chicken})$
4.  $\forall y \forall x [\text{eats}(y, x) \wedge \neg \text{killed}(y)] \rightarrow \text{food}(x)$
5.  $\text{eats}(\text{Bill}, \text{Peanuts}) \wedge \neg \text{killed}(\text{Bill})$
6.  $\forall x \text{ eats}(\text{Bill}, x) \rightarrow \text{eats}(\text{Sue}, x)$

"is still alive" is written  $\neg \text{killed}$ , not a new predicate *alive*. One predicate fewer, one resolution step fewer.

**Step 2 Into CNF**

1.  $\neg \text{food}(x) \vee \text{likes}(\text{John}, x)$
2.  $\text{food}(\text{Apple})$
3.  $\text{food}(\text{Chicken})$
4.  $\neg \text{eats}(y, x) \vee \text{killed}(y) \vee \text{food}(x)$
- 5a.  $\text{eats}(\text{Bill}, \text{Peanuts})$
- 5b.  $\neg \text{killed}(\text{Bill})$
6.  $\neg \text{eats}(\text{Bill}, x) \vee \text{eats}(\text{Sue}, x)$

**Step 3 Negate the goal**

7.  $\neg \text{likes}(\text{John}, \text{Peanuts})$

**Step 4 Resolution**

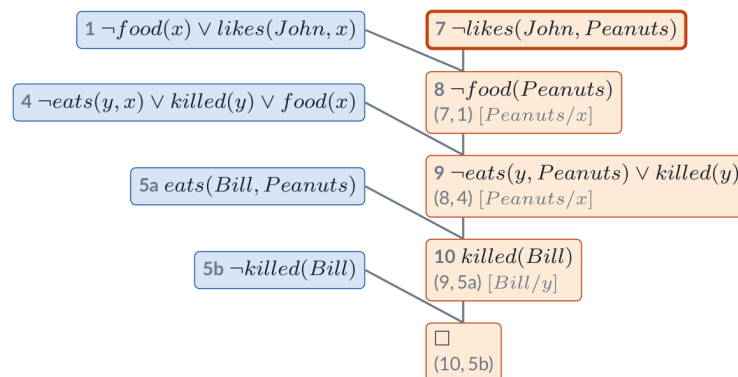
8.  $\neg \text{food}(\text{Peanuts})$  (7, 1) [*Peanuts*/*x*]
9.  $\neg \text{eats}(y, \text{Peanuts}) \vee \text{killed}(y)$  (8, 4) [*Peanuts*/*x*]
10.  $\text{killed}(\text{Bill})$  (9, 5a) [*Bill*/*y*]
11.  $\square$  (10, 5b)

**Final answer**

- The empty clause is derived, so  $KB \wedge \neg \text{goal}$  is unsatisfiable. **"John likes peanuts" is proved.** ✓
- Clauses 2, 3 and 6 are never used: apples, chicken and Sue are distractors. Saying so is a mark in a 8-mark answer, not a risk.

**The step the textbook gets wrong**

- Fact 4 has the shape  $\neg[A \wedge \neg B] \vee C$ , which is  $\neg A \vee B \vee C$ : **one clause.**
- *Insights* splits it into two,  $\neg \text{eats}(y, x) \vee \text{food}(x)$  and  $\neg \text{killed}(y) \vee \text{food}(x)$ . That is not a valid equivalence – it asserts that merely *not being killed* makes something food – and the book's own resolution graph then quietly uses the correct single clause.
- Write the one clause. If the marker expects the book's two, the proof still closes, and yours is the sound one.

**The four renamings** 8 6+4

same proof, different names · **the last one is not the same** and is the only place in the family where the printed goal does not follow

**QUESTION AS PRINTED** 2071 Bhadra · CT 78506 · [8]

Assume the following facts: a) Ram likes all kinds of food; b) Apples are food; c) Chicken is food; d) Anything anyone eats and is not killed by is food; e) Krishna eats peanuts and is still alive; f) Radha eats everything Krishna eats. Prove the statement "Ram likes peanut" using resolution.

**QUESTION AS PRINTED** 2080 Bhadra · CT 710 · [8]

Assume the following facts: i) Ram likes all kinds of food; ii) apples and vegetables are food; iii) anything anyone eats and not killed is food; iv) Anil eats peanuts and is still alive; v) Ram eats everything that Anil eats. Prove that "Ram likes peanuts" using resolution refutation.

**QUESTION AS PRINTED** 2080 Baishakh · CT 710 · [8]

Assume the following facts: a) Sita likes all kinds of food; b) Momo and Burger are food; c) anything anyone eats and isn't killed by is food; d) Bill eats mushroom and is still alive; e) Sue eats everything Bill eats. Prove that "Sita likes mushroom" using resolution refutation method.

## What changes, and what does not

| Paper                | Likes all food | Eats it                | Goal     |
|----------------------|----------------|------------------------|----------|
| 72 Ash, 74 Bh, 73 Bh | John           | Bill (Sue copies)      | peanuts  |
| 71 Bh                | Ram            | Krishna (Radha copies) | peanuts  |
| 80 Bh                | Ram            | Anil (Ram copies)      | peanuts  |
| 80 Ba                | Sita           | Bill (Sue copies)      | mushroom |

- **The proof is identical in all four:** negate, resolve with clause 1 to get  $\neg food(g)$ , resolve with clause 4 to get  $\neg eats(y, g) \vee killed(y)$ , then the two halves of the eating fact. Substitute the names and the four steps are unchanged.
- 80 Bh is the one worth a second look: there the copier is **Ram himself**, the same person who likes all food. It changes nothing – the copying clause is still unused – but it looks like it should, and that is the trap.
- 80 Ba swaps apples and chicken for momo and burger. Also unused.

## 2072 Magh (CT 78506) is a different problem

- It adds a fifth premise, “if a person likes a food it means that person has eaten it”, and asks for **Shailendri** – the copier – to **like** chicken.
- Take the premises literally:
  - $food(Chicken)$  and  $\forall x food(x) \rightarrow likes(Bhogendra, x)$  give  $likes(Bhogendra, Chicken)$ ;
  - $\forall x \forall y likes(x, y) \rightarrow eats(x, y)$  gives  $eats(Bhogendra, Chicken)$ ;
  - $\forall x eats(Bhogendra, x) \rightarrow eats(Shailendri, x)$  gives  $eats(Shailendri, Chicken)$ .
- And there it stops. **Nothing in the premise set makes Shailendri like anything:** only Bhogendra has a likes-rule, and the new premise runs  $likes \rightarrow eats$ , not back.
- **How to answer it:** prove  $eats(Shailendri, Chicken)$  in full, then add one line – “to reach likes the fifth premise must be read as a biconditional,  $likes(x, y) \leftrightarrow eats(x, y)$ ; with that reading one more resolution step gives  $likes(Shailendri, Chicken)$ .”
- **Do not silently reverse the implication and hope.** Stating the gap is worth more than a proof that assumes it away.

QUESTION AS PRINTED 2072 Magh · CT 78506 · [6+4]

Bhogendra likes all kinds of food. Oranges are food. Chicken is food. Anything anyone eats and isn't killed by is food. If a person likes a food it means that person has eaten it. Jogendra eats peanuts and is still alive. Shailendri eats everything Bhogendra eats. Express them in FOPL, put them into clause form, and use resolution to prove that Shailendri likes chicken.

## P5. Resolution refutation Family B: Charlie is a horse

6 papers, 3 wordings, 46 marks 73 Bh 8, 78 Po 8, 74 Bh 8, 79 Jth 6, 79 Bh 8, 78 Ba 8. The set that tests inverse relations and, quietly, whether you can spot an unused premise.

QUESTION AS PRINTED 2073 Bhadra · Q10 · [8] also 2078 Poush [8], 2074 Bhadra CT 78506 [8], 2079 Jestha [6]

Assume the following facts: i) horses, cows, pigs are mammals; ii) an offspring of a horse is a horse; iii) Bluebeard is a horse; iv) Bluebeard is Charlie's parent; v) offspring and parent are inverse relations; vi) every mammal has a parent. Prove Charlie is a horse using resolution refutation.

**Step 1 Facts into FOPL**

1.  $\forall x \text{ horse}(x) \rightarrow \text{mammal}(x)$ , and the same for cow, pig
2.  $\forall x \forall y [\text{offspring}(x, y) \wedge \text{horse}(y)] \rightarrow \text{horse}(x)$
3.  $\text{horse}(\text{Bluebeard})$
4.  $\text{parent}(\text{Bluebeard}, \text{Charlie})$
5.  $\forall x \forall y \text{ offspring}(x, y) \leftrightarrow \text{parent}(y, x)$
6.  $\forall x \text{ mammal}(x) \rightarrow \exists y \text{ parent}(y, x)$

**Step 2 Into CNF**

- 1a.  $\neg \text{horse}(x) \vee \text{mammal}(x)$  (1b, 1c likewise for cow, pig)
2.  $\neg \text{offspring}(x, y) \vee \neg \text{horse}(y) \vee \text{horse}(x)$
3.  $\text{horse}(\text{Bluebeard})$
4.  $\text{parent}(\text{Bluebeard}, \text{Charlie})$
- 5a.  $\neg \text{offspring}(x, y) \vee \text{parent}(y, x)$
- 5b.  $\neg \text{parent}(y, x) \vee \text{offspring}(x, y)$
6.  $\neg \text{mammal}(x) \vee \text{parent}(f(x), x)$

The  $\exists y$  in premise 6 sits inside  $\forall x$ , so it skolemizes to a **function**  $f(x)$ : each mammal has its own parent. **Insights writes a free variable  $y$  here**, which claims every object is a parent of every mammal. It changes nothing in this proof, because clause 6 is never used, but it is wrong and in another problem it would prove anything.

**Step 3 Negate the goal**

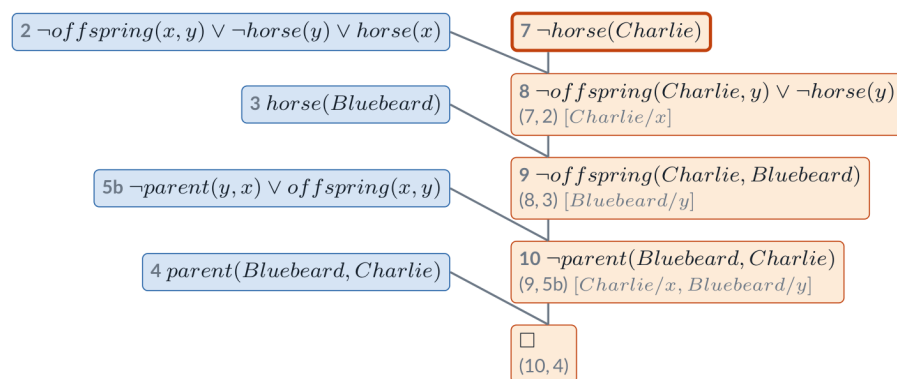
7.  $\neg \text{horse}(\text{Charlie})$

**Step 4 Resolution**

8.  $\neg \text{offspring}(\text{Charlie}, y) \vee \neg \text{horse}(y)$  (7, 2) [Charlie/x]
9.  $\neg \text{offspring}(\text{Charlie}, \text{Bluebeard})$  (8, 3) [Bluebeard/y]
10.  $\neg \text{parent}(\text{Bluebeard}, \text{Charlie})$  (9, 5b) [Charlie/x, Bluebeard/y]
11.  $\square$  (10, 4)

**Final answer**

- “**Charlie is a horse**” is proved. ✓ Four resolution steps.
- **Premises 1 and 6 are never used.** The mammal chain is a complete red herring for this goal: being a horse follows from the offspring rule alone. **Say it in one line** – examiners set the extra premises to see whether the proof is understood or remembered.
- **The whole problem is premise 5.** Written as a biconditional it yields the two clauses 5a and 5b, and only **5b** is used: it turns the given  $\text{parent}(\text{Bluebeard}, \text{Charlie})$  into the  $\text{offspring}(\text{Charlie}, \text{Bluebeard})$  that rule 2 needs. A one-directional reading of “inverse relations” cannot close the proof.
- **Watch the argument order:**  $\text{offspring}(x, y)$  means  $x$  is the offspring of  $y$ , so its inverse is  $\text{parent}(y, x)$ , not  $\text{parent}(x, y)$ . Getting this backwards is the commonest failure on this set.

**The two renamings 8**

79 Bh 8 asks for a mammal, 78 Ba 8 renames every noun

**QUESTION AS PRINTED** 2079 Bhadra · CT 710 · [8]

Prove that “Charlie is a mammal” with given propositions: a) cows, pigs and horses are mammal; b) the child of horse is a horse; c) Bluebeard is a horse; d) Bluebeard is Charlie’s father; e) child and father are inverse relation; f) every mammal has a father.

**QUESTION AS PRINTED** 2078 Baishakh · [8]

Assume the following facts: (i) Cow, Buffalo, Bull are mammals; (ii) an offspring of a Cow is Cow; (iii) Kali is a Cow; (iv) Kali is Taarey’s parent; (v) offspring and parent are inverse relations; (vi) every mammal has a parent. By using resolution refutation, prove that Taarey is a Cow.

### 2079 Bhadra: one extra step, and premise (a) stops being a distractor

- The goal is now  $mammal(Charlie)$ , so negate to 7.  
 $\neg mammal(Charlie)$ .  
 8.  $\neg horse(Charlie)$  (7, 1a)  $[Charlie/x]$   
 and from there the proof above runs unchanged, steps 8–11 becoming 9–12.
- Five steps instead of four.** The mammal rule that was unused in the “horse” version is now the first step of the “mammal” one – the same six premises, one different goal, a different subset used.
- child* and *father* replace *offspring* and *parent*. Same two clauses from the inverse relation, same argument-order trap.

### 2078 Baishakh: a pure renaming

|               | 2073 Bhadra      | 2078 Baishakh |
|---------------|------------------|---------------|
| Species       | horse            | Cow           |
| Also mammals  | cows, pigs       | Buffalo, Bull |
| The parent    | Bluebeard        | Kali          |
| The offspring | Charlie          | Taarey        |
| Goal          | $horse(Charlie)$ | $Cow(Taarey)$ |

- Substitute and the four resolution steps are character-for-character the same.
- Buffalo and Bull are the distractors** this time, and premise (vi) is again unused.
- Answer to give:**  $\square$  derived in 4 steps, so “Taarey is a Cow” is proved.  $\checkmark$

### P5. Resolution refutation Family C: selling weapons to a hostile nation

7 papers, 6 wordings, 55 marks 69 Bh 10, 75 Ba 8, 77 Ch 8, 82 Ka 8, 81 Ash 2+6, 81 Bh 9, 82 Bh 6. The set that tests existential instantiation, and the one paper (81 Ash) that asks for forward chaining instead of refutation.

QUESTION AS PRINTED 2077 Chaitra · Q10 · [8] also 2082 Kartik [8]

It is a crime for an American to sell weapons to hostile nations. Nono has some missiles. All the missiles owned by Nono were sold to it by Colonel West. Missiles are weapons. An enemy of America counts as hostile. Colonel West is an American. The country Nono is an enemy of America. Prove that Colonel West is a criminal by using FOPL based Resolution Refutation.

#### Step 1 Facts into FOPL

- $\forall p \forall q \forall r [American(p) \wedge Weapon(q) \wedge Sells(p, q, r) \wedge Hostile(r)] \rightarrow Criminal(p)$
- $\exists x Missile(x) \wedge Owns(Nono, x)$
- $\forall x [Missile(x) \wedge Owns(Nono, x)] \rightarrow Sells(West, x, Nono)$
- $\forall x Missile(x) \rightarrow Weapon(x)$
- $\forall x Enemy(x, America) \rightarrow Hostile(x)$
- $American(West)$
- $Enemy(Nono, America)$

#### Step 2 Into CNF

- $\neg American(p) \vee \neg Weapon(q) \vee \neg Sells(p, q, r) \vee \neg Hostile(r) \vee Criminal(p)$
- a.  $Missile(M_1)$  skolem constant
- b.  $Owns(Nono, M_1)$
- $\neg Missile(x) \vee \neg Owns(Nono, x) \vee Sells(West, x, Nono)$
- $\neg Missile(x) \vee Weapon(x)$
- $\neg Enemy(x, America) \vee Hostile(x)$
- $American(West)$       7.  $Enemy(Nono, America)$

**“Nono has some missiles” is existential.** It is not  $\forall x Missile(x)$ , which would make everything a missile, and it is not a named missile either – the paper never names one. Skolemize to a brand-new constant  $M_1$  and it splits into two ground facts.

#### Step 3 Negate the goal

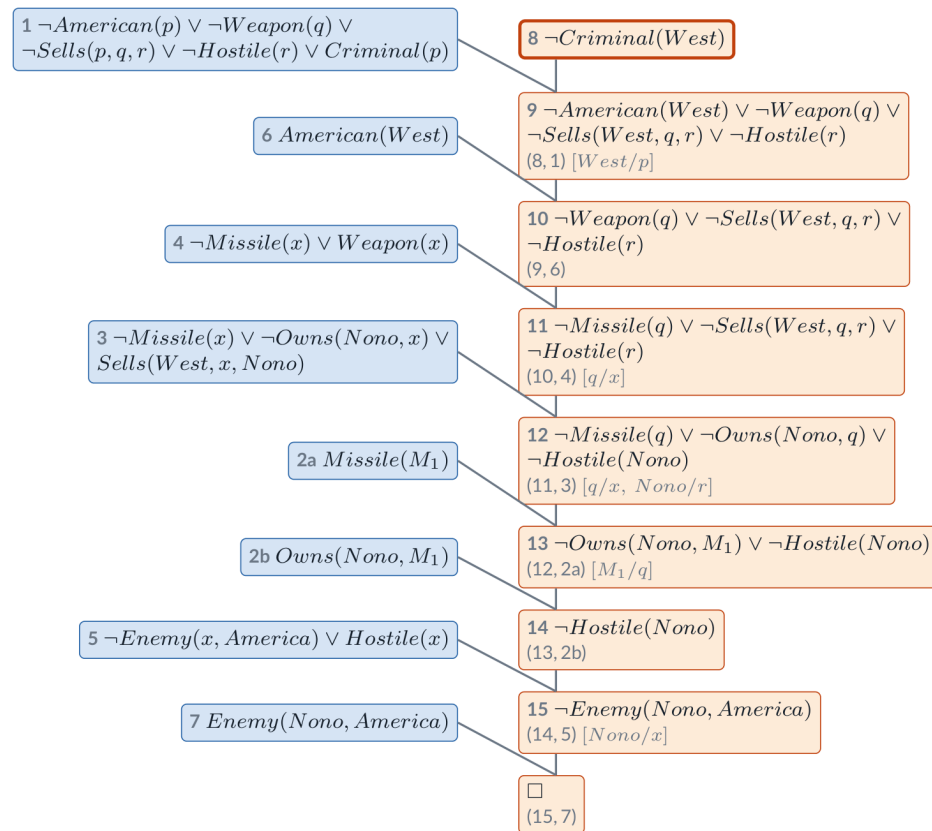
- $\neg Criminal(West)$

#### Step 4 Resolution

- $\neg American(West) \vee \neg Weapon(q) \vee \neg Sells(West, q, r) \vee \neg Hostile(r)$  (8, 1)  $[West/p]$
- $\neg Weapon(q) \vee \neg Sells(West, q, r) \vee \neg Hostile(r)$  (9, 6)
- $\neg Missile(q) \vee \neg Sells(West, q, r) \vee \neg Hostile(r)$  (10, 4)  $[q/x]$
- $\neg Missile(q) \vee \neg Owns(Nono, q) \vee \neg Hostile(Nono)$  (11, 3)  $[q/x, Nono/r]$
- $\neg Owns(Nono, M_1) \vee \neg Hostile(Nono)$  (12, 2a)  $[M_1/q]$
- $\neg Hostile(Nono)$  (13, 2b)
- $\neg Enemy(Nono, America)$  (14, 5)  $[Nono/x]$
- $\square$  (15, 7)

#### Final answer

- “Colonel West is a criminal” is proved.**  $\checkmark$  Eight resolution steps, and **every premise is used** – unusual, and worth noticing, because it means no step can be skipped.
- Work down the negated goal.** Clause 1 has five literals; resolving it against clause 8 first turns the proof into a checklist of four conditions, each discharged by one premise. Starting anywhere else wanders.
- In step 12 the literal  $\neg Missile(q)$  appears twice after the substitution and **collapses to one**. Duplicate literals in a clause always do.



## 2069 Bhadra and 2075 Baishakh: two premises are missing 10 8

69 Bh 10, 75 Ba 8 · **neither paper says missiles are weapons, or that an enemy is hostile** · the proof cannot close without them, so they must be stated as assumptions

### QUESTION AS PRINTED 2069 Bhadra · Q10 · [10]

Every American who sells weapons to hostile nations is a criminal. The country Abc is enemy of America. All of the missiles in Abc were sold by John. John is an American. Prove John is a criminal.

### QUESTION AS PRINTED 2075 Baishakh · [8]

Every American who sells weapons to hostile nations is a criminal. The country XYZ is enemy of America. All of its missiles in XYZ were sold by Donald, who is an American. Prove that Donald is a criminal using FOPL based resolution refutation.

### What is missing, and what to write

- Clause 1 needs **four** things about  $p$ : American, a weapon, a sale, and a hostile buyer. These two papers supply *American*, the *sale* and the *enemy* – but nothing links
  - missile* to *weapon*, and
  - enemy* of *America* to *hostile*.
- Add both, and say you are adding them:**
  - $\forall x \text{ Missile}(x) \rightarrow \text{Weapon}(x)$
  - $\forall x \text{ Enemy}(x, \text{America}) \rightarrow \text{Hostile}(x)$
- Then the proof is exactly the 8 steps above with [John/West, Abc/Nono] or [Donald/West, XYZ/Nono].
- This is not pedantry.** A resolution proof is a formal derivation: if the bridging axiom is not in the clause set, the empty clause cannot be derived, and a proof that pretends otherwise is the thing being marked.

### Also read “all of the missiles in Abc” carefully

- It says *every missile in Abc was sold by John*, i.e.  $\forall x [\text{Missile}(x) \wedge \text{In}(x, \text{Abc})] \rightarrow \text{Sells}(\text{John}, x, \text{Abc})$ .
- It does **not** say Abc has any. Strictly, with no missile in existence the antecedent is never satisfied and nothing follows – so the existential “Abc has some missiles” has to be assumed too, exactly as 77 Ch states it outright.
- Three assumptions, one line each**, then the standard proof. An 10-mark answer has room for them and the examiner is looking for whether the gap was noticed.
- 75 Ba’s “All of its missiles in XYZ were sold by Donald” is the same sentence with a typo; treat it identically.

## 2081 Bhadra and 2082 Bhadra: the two renamings that add a premise 9 6

81 Bh 9, 82 Bh 6

QUESTION AS PRINTED 2081 Bhadra · CT 710 · [9]

Every American who sells weapons to hostile nations is a criminal. Every enemy of America is a hostile. Nono has some missiles. All missiles of Nono were sold by Gorge. Gorge is an American. Nono is a Country. Nono is the enemy of America. Missiles are weapons. Prove that “Is Gorge a criminal?” using Resolution by Refutation.

QUESTION AS PRINTED 2082 Bhadra · CT 710 · [6]

It is a crime for a Nepali to sell weapons to an enemy of Nepal. Mr. A is an enemy of Nepal. Mr. A has some missiles. Missiles are weapons. Mr. R is a Nepali. If Mr. A has missiles then those missiles were sold to Mr. A by Mr. R. Prove “Mr. A is a criminal”.

### 2081 Bhadra: complete, with one spare premise

- All eight premises of the canonical set are present, plus “*Nono is a Country*”, *Country(Nono)*, which **appears in no rule and is never used**.
- Substitute [*Gorge/West*] and the eight steps run unchanged. Answer: **Gorge is a criminal.** ✓
- The question is phrased as “*Is Gorge a criminal?*” but asks for resolution *by refutation*, so still negate: assume  $\neg \text{Criminal}(\text{Gorge})$  and derive  $\square$ . **Do not switch to extraction** just because the sentence ends in a question mark.

### 2082 Bhadra: the goal is the wrong person

- Read the roles: **Mr. R** is the Nepali who *sells*; **Mr. A** is the enemy who *buys*. The crime rule makes the *seller* the criminal.
- So the premises prove  $\text{Criminal}(\text{Mr. R})$ , and the printed goal “*Mr. A is a criminal*” does not follow.
- **How to answer:** prove  $\text{Criminal}(\text{Mr. R})$  in full by the standard eight steps, then one line – “*the premises make the seller the criminal, so what follows is Criminal(Mr.R); Criminal(Mr.A) is not derivable from this set.*”
- The sixth premise is a conditional, *if Mr. A has missiles then they were sold to him by Mr. R*, which is the same clause 3 as ever:  

$$\neg \text{Missile}(x) \vee \neg \text{Owns}(A, x) \vee \text{Sells}(R, x, A).$$
- *Enemy of Nepal* is used directly as the hostile predicate here, so no bridging axiom is needed.

## P6. The same premises by forward and backward chaining

### Prove Col. West is criminal using forward chaining 2+6

PIN 1/41 81 Ash 2+6 · the only paper that asks for a chaining proof rather than resolution, and 11 more ask for the forward/backward comparison this answers by example

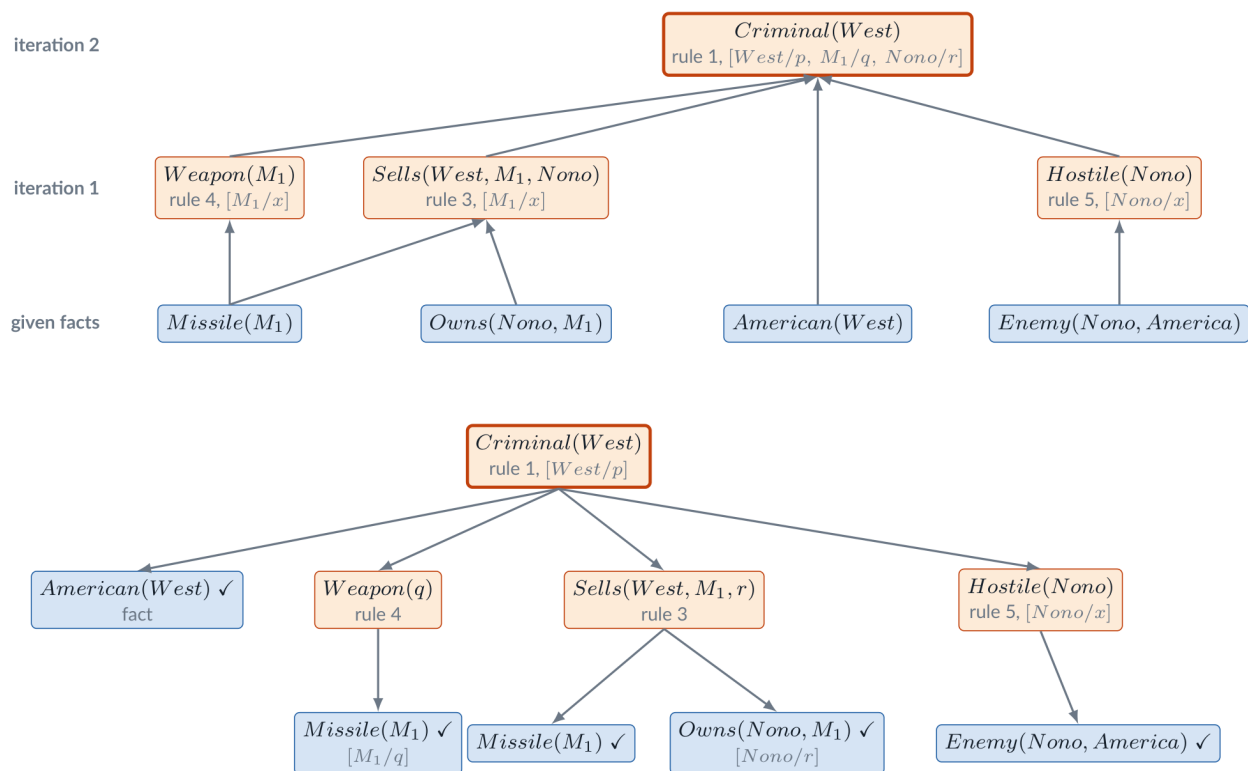
QUESTION AS PRINTED 2081 Ashwin · [2+6]

The law says that it is a crime for an American to sell weapons to hostile nations. The country Nono, an enemy of America, has some missiles, and all of its missiles were sold to it by Colonel West, who is American. Prove that Col. West is criminal using **forward chaining**.

### Forward chaining, iteration by iteration

- **Start with the ground facts** (the clauses with no premises):  
 $American(West)$ ,  $Missile(M_1)$ ,  $Owns(Nono, M_1)$ ,  
 $Enemy(Nono, America)$
- **Iteration 1** – fire every rule whose premises are all present:
  - rule 4 with  $[M_1/x] \rightarrow Weapon(M_1)$
  - rule 3 with  $[M_1/x]$ , needing  $Missile(M_1)$  and  $Owns(Nono, M_1)$ , both present  $\rightarrow$   
 $Sells(West, M_1, Nono)$
  - rule 5 with  $[Nono/x] \rightarrow Hostile(Nono)$
  - rule 1 does **not** fire yet: its four premises are not all known.
- **Iteration 2** – rule 1 now matches with  $[West/p, M_1/q, Nono/r]$ , because  $American(West)$ ,  $Weapon(M_1)$ ,  $Sells(West, M_1, Nono)$  and  $Hostile(Nono)$  are all in the fact base:  
 $\rightarrow Criminal(West)$
- **Goal reached, stop.** ✓

Draw it as three rows: the four given facts along the bottom, the three iteration-1 facts above them,  $Criminal(West)$  at the top, with an arrow from each premise into the fact it helped derive. That picture is the answer; see §4.6.



### The same thing backwards, for the 2 marks of comparison

- **Goal  $Criminal(West)$ .** Only rule 1 concludes  $Criminal$ , so unify  $[West/p]$  and take its four premises as sub-goals:
  - $American(West)$  ✓ a fact, done;
  - $Weapon(q) \rightarrow$  sub-goal  $Missile(q)$  ✓ matches the fact  $Missile(M_1)$ , binding  $[M_1/q]$ ;
  - $Sells(West, M_1, r) \rightarrow$  sub-goals  $Missile(M_1)$  ✓ and  $Owns(Nono, M_1)$  ✓, binding  $[Nono/r]$ ;
  - $Hostile(Nono) \rightarrow$  sub-goal  $Enemy(Nono, America)$  ✓.
- Every leaf is a fact, so the goal is proved. **Four sub-goals, depth first, left to right.**

### What the examiner is comparing

- Forward derived **three facts nobody asked for** on the way ( $Weapon$ ,  $Sells$ ,  $Hostile$  are all needed here, but on a bigger KB most would not be).
- Backward touched **only the four conjuncts of one rule**, and the binding  $[M_1/q]$  found in the second sub-goal was **carried into the third**, which is why  $Sells(West, M_1, r)$  could be matched at all.
- Same rules, same modus ponens, opposite ends. One sentence, and the comparison marks are secured.

## P5. Resolution refutation Family D: the light sleeper

**2 papers, 17 marks** 76 Ba **2+8**, 76 Bh **7**. The hardest set in the chapter, because the goal is itself an implication and the premises hide two nested quantifiers.

**QUESTION AS PRINTED** 2076 Baishakh · [2+8]

List down the rule for Inference. Consider the following axioms: All hounds howl at night. Anyone who has any cats will not have any mice. Light sleepers do not have anything which howls at night. John has either a cat or a hound. Prove: “If John is a light sleeper, then John does not have any mice” by resolution refutation.

### Step 1 Facts into FOPL

1.  $\forall x \text{ Hound}(x) \rightarrow \text{Howl}(x)$
  2.  $\forall x \forall y [\text{Have}(x, y) \wedge \text{Cat}(y)] \rightarrow \forall z [\text{Have}(x, z) \rightarrow \neg \text{Mouse}(z)]$
  3.  $\forall x \text{ LS}(x) \rightarrow \forall y [\text{Have}(x, y) \rightarrow \neg \text{Howl}(y)]$
  4.  $\exists x \text{ Have}(\text{John}, x) \wedge [\text{Cat}(x) \vee \text{Hound}(x)]$
- Goal**  $\text{LS}(\text{John}) \rightarrow \forall z [\text{Have}(\text{John}, z) \rightarrow \neg \text{Mouse}(z)]$

### Step 2 Into CNF

1.  $\neg \text{Hound}(x) \vee \text{Howl}(x)$
2.  $\neg \text{Have}(x, y) \vee \neg \text{Cat}(y) \vee \neg \text{Have}(x, z) \vee \neg \text{Mouse}(z)$
3.  $\neg \text{LS}(x) \vee \neg \text{Have}(x, y) \vee \neg \text{Howl}(y)$
- 4a.  $\text{Have}(\text{John}, a)$  skolem constant  $a$
- 4b.  $\text{Cat}(a) \vee \text{Hound}(a)$

### Step 3 Negate the goal, and get three clauses

- $\neg [\text{LS}(\text{John}) \rightarrow \forall z (\text{Have}(\text{John}, z) \rightarrow \neg \text{Mouse}(z))]$
  - $= \text{LS}(\text{John}) \wedge \exists z [\text{Have}(\text{John}, z) \wedge \text{Mouse}(z)]$
- 5a.  $\text{LS}(\text{John})$       5b.  $\text{Have}(\text{John}, b)$       5c.  $\text{Mouse}(b)$   
skolem  $b$

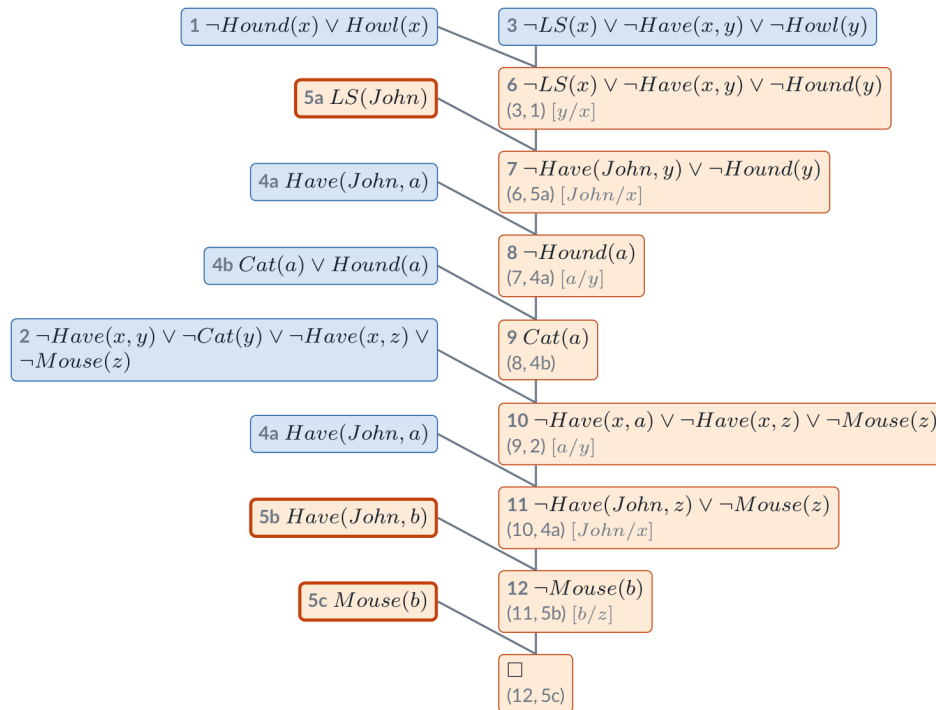
$\neg(P \rightarrow Q) = P \wedge \neg Q$ . **This is the whole difficulty of the problem:** students negate an implication to another implication, get one clause instead of three, and then cannot close the proof.

### Step 4 Resolution

6.  $\neg \text{LS}(x) \vee \neg \text{Have}(x, y) \vee \neg \text{Hound}(y)$  (3, 1)  $[y/x]$
7.  $\neg \text{Have}(\text{John}, y) \vee \neg \text{Hound}(y)$  (6, 5a)  $[\text{John}/x]$
8.  $\neg \text{Hound}(a)$  (7, 4a)  $[a/y]$
9.  $\text{Cat}(a)$  (8, 4b)
10.  $\neg \text{Have}(x, a) \vee \neg \text{Have}(x, z) \vee \neg \text{Mouse}(z)$  (9, 2)  $[a/y]$
11.  $\neg \text{Have}(\text{John}, z) \vee \neg \text{Mouse}(z)$  (10, 4a)  $[\text{John}/x]$
12.  $\neg \text{Mouse}(b)$  (11, 5b)  $[b/z]$
13.  $\square$  (12, 5c)

### Final answer

- **“If John is a light sleeper then John has no mice” is proved.** ✓ Eight steps.
- **The shape of the argument**, worth stating in words underneath: John is a light sleeper, so he has nothing that howls, so what he has is not a hound; it was a cat or a hound, so it is a cat; anyone with a cat has no mice; so he has no mice.
- **Two skolem constants, and they are different.**  $a$  is the animal John owns (from premise 4);  $b$  is the mouse the negated goal claims he has. Reusing one letter for both is a wrong proof that happens to close.
- **Premise 2 needs four literals.** Both the cat and the mouse are things *John has*, so *Have* appears twice with different second arguments. Writing it with one *Have* loses the proof.



QUESTION AS PRINTED 2076 Bhadra · CT 78506 · [7]

Consider the following axioms: All direwolves howl at night. Anyone who has any cat will not have any mice. Light sleepers do not have anything that howls at night. Jon has either a cat or a direwolf. Prove that if Jon is a light sleeper, then Jon does not have any mice, using the resolution method.

**Identical:** substitute [Jon/John, Direwolf/Hound] and the same eight steps close the proof. Nothing else changes.

## P5. Resolution refutation The ten one-off premise sets

11 papers, one set each except Santa which is set twice 72 Ma 7, 81 Ch 8, 78 Ch 3+3, 80 Ch 3+5, 81 Ba 2+6, 79 Ash 7, 77 Ch 8, 70 Bh 8, 71 Bh 8, 73 Ma 10, 80 Ash 3+5. Same five steps every time; only the premises move.

### 1. Did Marcus hate Caesar? 7

72 Ma 7 · the paper prints "Find, did Marcus Caesar?", dropping the verb; it means *hate*

QUESTION AS PRINTED 2072 Magh · [7]

Using resolution, solve the following statements: All pompeian are Romans. All Romans were either loyal to Caesar or hated him. Everyone is loyal to someone. People only try to assassinate rulers they not loyal to. Marcus tried to assassinate Caesar. Marcus was pompeian. Find, did Marcus Caesar?

**FOPL**

1.  $\forall x \text{Pompeian}(x) \rightarrow \text{Roman}(x)$
2.  $\forall x \text{Roman}(x) \rightarrow \text{loyal}(x, \text{Caesar}) \vee \text{hate}(x, \text{Caesar})$
3.  $\forall x \exists y \text{loyal}(x, y)$
4.  $\forall x \forall y \text{tryAssassinate}(x, y) \wedge \text{ruler}(y) \rightarrow \neg \text{loyal}(x, y)$
5.  $\text{tryAssassinate}(\text{Marcus}, \text{Caesar})$
6.  $\text{Pompeian}(\text{Marcus})$       7.  $\text{ruler}(\text{Caesar})$       assumed
- Goal:  $\text{hate}(\text{Marcus}, \text{Caesar})$

**CNF**

1.  $\neg \text{Pompeian}(x) \vee \text{Roman}(x)$
2.  $\neg \text{Roman}(x) \vee \text{loyal}(x, \text{Caesar}) \vee \text{hate}(x, \text{Caesar})$
3.  $\text{loyal}(x, f(x))$        $\exists$  inside  $\forall$ : skolem function
4.  $\neg \text{tryAssassinate}(x, y) \vee \neg \text{ruler}(y) \vee \neg \text{loyal}(x, y)$
5.  $\text{tryAssassinate}(\text{Marcus}, \text{Caesar})$
6.  $\text{Pompeian}(\text{Marcus})$       7.  $\text{ruler}(\text{Caesar})$

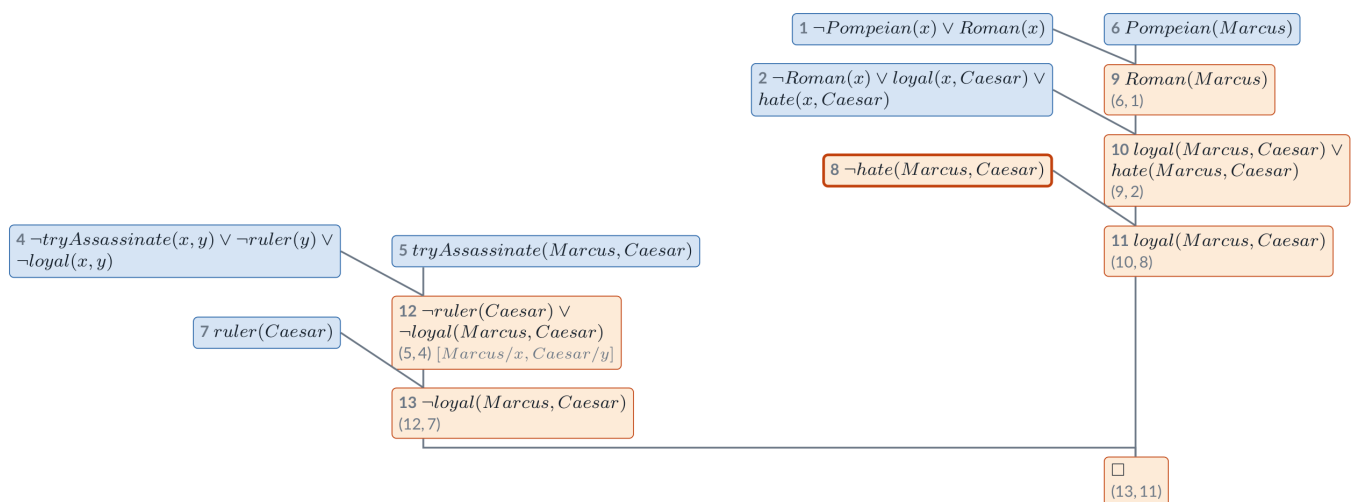
$\text{ruler}(\text{Caesar})$  is not printed but rule 4 is about rulers, so state it as an assumption. “Everyone is loyal to someone” is  $\forall x \exists y \text{loyal}(x, y)$ , and **it skolemizes to  $f(x)$ , not to a constant** – each person is loyal to their own someone.

**Proof** negated goal 8.  $\neg \text{hate}(\text{Marcus}, \text{Caesar})$ 

9.  $\text{Roman}(\text{Marcus})$  (6, 1)
10.  $\text{loyal}(\text{Marcus}, \text{Caesar}) \vee \text{hate}(\text{Marcus}, \text{Caesar})$  (9, 2)
11.  $\text{loyal}(\text{Marcus}, \text{Caesar})$  (10, 8)
12.  $\neg \text{ruler}(\text{Caesar}) \vee \neg \text{loyal}(\text{Marcus}, \text{Caesar})$  (5, 4)  
[ $\text{Marcus}/x, \text{Caesar}/y$ ]
13.  $\neg \text{loyal}(\text{Marcus}, \text{Caesar})$  (12, 7)
14.  $\square$  (13, 11)

– **Marcus hated Caesar.** ✓

– **Clause 3 is never used**, and it is the only clause with a Skolem function in it. It is there to see whether you skolemize it correctly, not because the proof needs it.

**2. Scrooge is not a child** 8 3+3

81 Ch 8, 78 Ch 3+3 · 78 Ch prints the same six axioms and asks only for the CNF plus the horn-clause definition, so the clause list below is the whole answer there

QUESTION AS PRINTED 2081 Chaitra · [8]

Every child loves Santa. Everyone who loves Santa loves any reindeer. Rudolph is a reindeer, and Rudolph has a red nose. Anything which has a red nose is weird or is a clown. No reindeer is a clown. Scrooge does not love anything which is weird. Using Resolution Refutation, derive the conclusion: Scrooge is not a child.

**FOPL**

1.  $\forall x \text{ child}(x) \rightarrow \text{loves}(x, \text{Santa})$
2.  $\forall x \forall y \text{ loves}(x, \text{Santa}) \wedge \text{reindeer}(y) \rightarrow \text{loves}(x, y)$
3.  $\text{reindeer}(\text{Rudolph})$       4.  $\text{redNose}(\text{Rudolph})$
5.  $\forall x \text{ redNose}(x) \rightarrow \text{weird}(x) \vee \text{clown}(x)$
6.  $\forall x \text{ reindeer}(x) \rightarrow \neg \text{clown}(x)$
7.  $\forall x \text{ weird}(x) \rightarrow \neg \text{loves}(\text{Scrooge}, x)$

**Goal:**  $\neg \text{child}(\text{Scrooge})$

"No reindeer is a clown" is  $\neg \exists x (\text{reindeer}(x) \wedge \text{clown}(x))$ ; pushing the  $\neg$  in gives line 6.

**CNF**

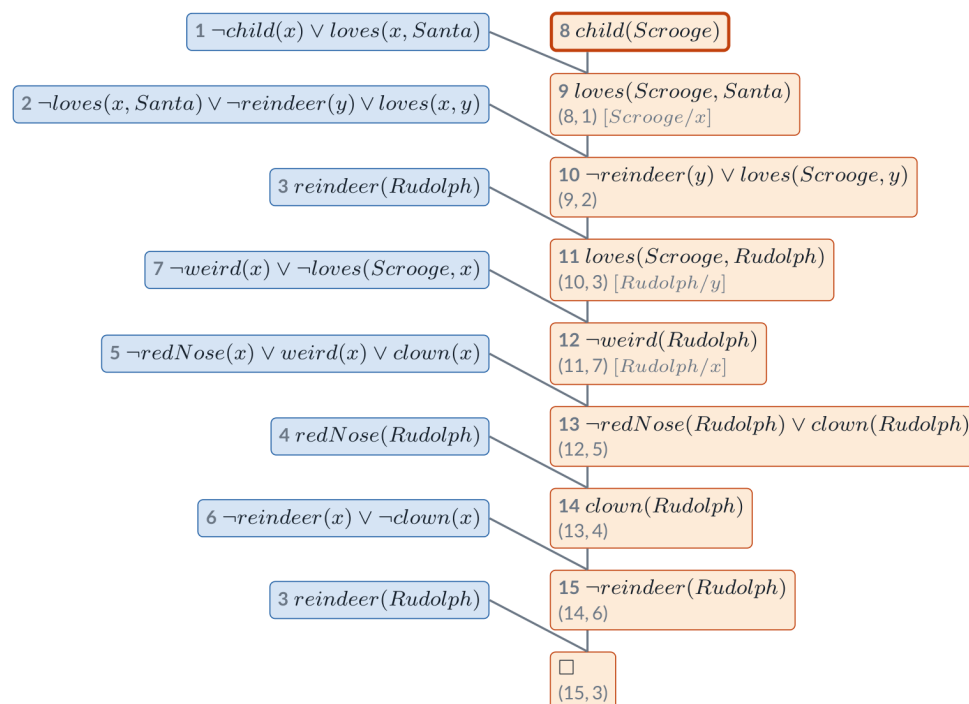
1.  $\neg \text{child}(x) \vee \text{loves}(x, \text{Santa})$
2.  $\neg \text{loves}(x, \text{Santa}) \vee \neg \text{reindeer}(y) \vee \text{loves}(x, y)$
3.  $\text{reindeer}(\text{Rudolph})$       4.  $\text{redNose}(\text{Rudolph})$
5.  $\neg \text{redNose}(x) \vee \text{weird}(x) \vee \text{clown}(x)$
6.  $\neg \text{reindeer}(x) \vee \neg \text{clown}(x)$
7.  $\neg \text{weird}(x) \vee \neg \text{loves}(\text{Scrooge}, x)$

The goal is already a negation, so **negating it removes the  $\neg$** : clause 8 is  $\text{child}(\text{Scrooge})$ , not  $\neg \text{child}(\text{Scrooge})$ . Getting this backwards is the standard error on this set.

**Proof** negated goal 8.  $\text{child}(\text{Scrooge})$

9.  $\text{loves}(\text{Scrooge}, \text{Santa})$  (8, 1) [ $\text{Scrooge}/x$ ]
10.  $\neg \text{reindeer}(y) \vee \text{loves}(\text{Scrooge}, y)$  (9, 2)
11.  $\text{loves}(\text{Scrooge}, \text{Rudolph})$  (10, 3) [ $\text{Rudolph}/y$ ]
12.  $\neg \text{weird}(\text{Rudolph})$  (11, 7) [ $\text{Rudolph}/x$ ]
13.  $\neg \text{redNose}(\text{Rudolph}) \vee \text{clown}(\text{Rudolph})$  (12, 5)
14.  $\text{clown}(\text{Rudolph})$  (13, 4)
15.  $\neg \text{reindeer}(\text{Rudolph})$  (14, 6)
16.  $\square$  (15, 3)

- **"Scrooge is not a child" is proved.** ✓ Eight steps, every premise used.
- The argument in words: if Scrooge were a child he would love Santa, hence love Rudolph; but Scrooge loves nothing weird, so Rudolph is not weird, so (red nose) he is a clown, and no reindeer is a clown – contradiction.

**3. Did Curiosity kill the cat? 3+5**

**PIN 1/41** 80 Ch **3+5** · **the hardest clause set in the chapter:** premise (a) is the Skolem-function sentence from P2, and it is needed twice

**QUESTION AS PRINTED** 2080 Chaitra · [3+5]

Convert the following sentences into FOPL and answer "Did Curiosity kill the cat" by resolution refutation:

- a) Everyone who loves all animals is loved by someone.
- b) Anyone who kills an animal is loved by no one.
- c) Jack loves all animals.
- d) Either Jack or Curiosity killed the cat.

**FOPL** (the 3 marks)

- A.  $\forall x [\forall y \text{ Animal}(y) \rightarrow \text{Loves}(x, y)] \rightarrow \exists z \text{ Loves}(z, x)$   
 B.  $\forall x [\exists y \text{ Animal}(y) \wedge \text{Kills}(x, y)] \rightarrow \forall z \neg \text{Loves}(z, x)$   
 C.  $\forall x \text{ Animal}(x) \rightarrow \text{Loves}(Jack, x)$   
 D.  $\text{Kills}(Jack, Tuna) \vee \text{Kills}(Curiosity, Tuna)$   
 E.  $\text{Cat}(Tuna)$       F.  $\forall x \text{ Cat}(x) \rightarrow \text{Animal}(x)$  assumed  
**Goal:**  $\text{Kills}(Curiosity, Tuna)$

**CNF** (the cat is named Tuna; "cats are animals" is assumed)

- A1.  $\text{Animal}(F(x)) \vee \text{Loves}(G(x), x)$   
 A2.  $\neg \text{Loves}(x, F(x)) \vee \text{Loves}(G(x), x)$   
 B.  $\neg \text{Animal}(y) \vee \neg \text{Kills}(x, y) \vee \neg \text{Loves}(z, x)$   
 C.  $\neg \text{Animal}(x) \vee \text{Loves}(Jack, x)$   
 D.  $\text{Kills}(Jack, Tuna) \vee \text{Kills}(Curiosity, Tuna)$   
 E.  $\text{Cat}(Tuna)$       F.  $\neg \text{Cat}(x) \vee \text{Animal}(x)$

A1 and A2 are premise (a) put through all eight CNF steps in P2 above.  $F(x)$  is "an animal  $x$  does not love" and  $G(x)$  is "someone who loves  $x$ ". **Both must be functions of  $x$ .**

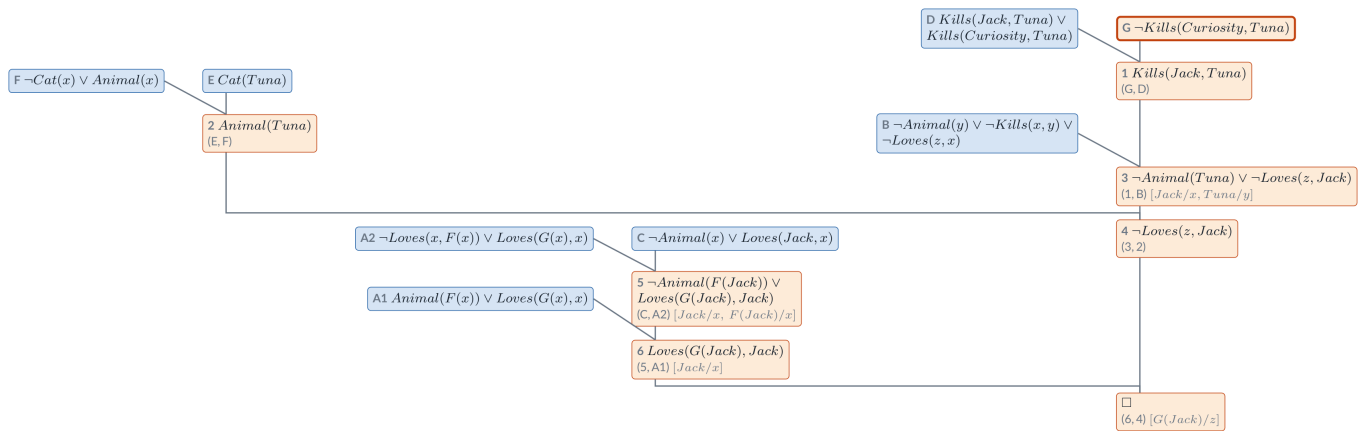
**Proof** negated goal G.  $\neg \text{Kills}(Curiosity, Tuna)$ 

1.  $\text{Kills}(Jack, Tuna)$  (G, D)
2.  $\text{Animal}(Tuna)$  (E, F)
3.  $\neg \text{Animal}(Tuna) \vee \neg \text{Loves}(z, Jack)$  (1, B)  
 $[Jack/x, Tuna/y]$
4.  $\neg \text{Loves}(z, Jack)$  (3, 2)
5.  $\neg \text{Animal}(F(Jack)) \vee \text{Loves}(G(Jack), Jack)$  (C, A2)  
 $[Jack/x, F(Jack)/x]$
6.  $\text{Loves}(G(Jack), Jack)$  (5, A1)  $[Jack/x]$ , duplicate collapses
7.  $\square$  (6, 4)  $[G(Jack)/z]$

– **Curiosity killed the cat.** ✓

– **Steps 5 and 6 are the whole difficulty.** They prove somebody loves Jack, using (c) to satisfy the antecedent of (a): Jack loves all animals, so in particular he loves  $F(Jack)$ , so the second clause of (a) fires. Only then does step 4, *nobody loves Jack*, become a contradiction.

– **Two facts have to be assumed** because the paper omits them: the cat has a name, and cats are animals. State both.

**4. Sneha is happy** 2+6

81 Ba 2+6 · the first 2 marks are skolemization defined, §4.3 · **contrast the CNF of premise (b) with the food family's premise 4:** here the distribution into two clauses is valid

**QUESTION AS PRINTED** 2081 Baishakh · CT 710 · [2+6]

Define skolemization with example. Consider the following axioms: a) anyone passing his engineering exams and winning the lottery is happy; b) anyone who studies or is lucky can pass all his exams; c) Sneha did not study but she is lucky; d) anyone who is lucky wins the lottery. Using resolution refutation, prove that Sneha is happy.

**FOPL**

1.  $\forall x \text{ pass}(x) \wedge \text{win}(x) \rightarrow \text{happy}(x)$
2.  $\forall x \text{ study}(x) \vee \text{lucky}(x) \rightarrow \text{pass}(x)$
3.  $\neg \text{study}(\text{Sneha}) \wedge \text{lucky}(\text{Sneha})$
4.  $\forall x \text{ lucky}(x) \rightarrow \text{win}(x)$

**Goal:**  $\text{happy}(\text{Sneha})$

**CNF**

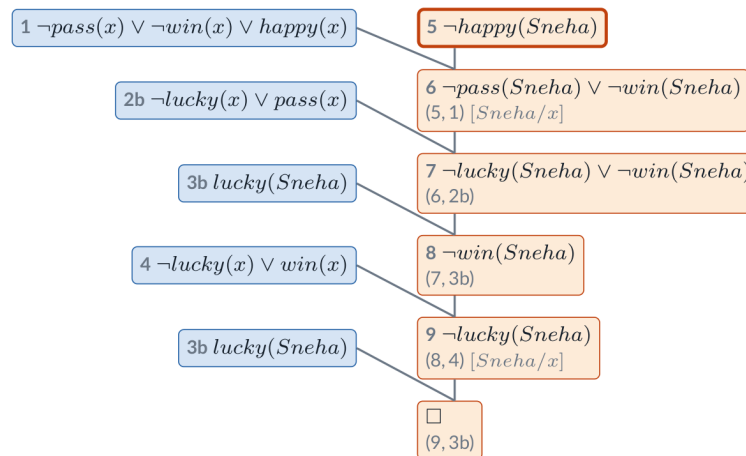
1.  $\neg \text{pass}(x) \vee \neg \text{win}(x) \vee \text{happy}(x)$
2.  $[\text{study}(x) \vee \text{lucky}(x)] \rightarrow \text{pass}(x)$   
 $= [\neg \text{study}(x) \wedge \neg \text{lucky}(x)] \vee \text{pass}(x)$ , so
  - 2a.  $\neg \text{study}(x) \vee \text{pass}(x)$
  - 2b.  $\neg \text{lucky}(x) \vee \text{pass}(x)$
- 3a.  $\neg \text{study}(\text{Sneha})$       **3b.**  $\text{lucky}(\text{Sneha})$
4.  $\neg \text{lucky}(x) \vee \text{win}(x)$

**Why this splits and the food premise does not:** here the negation lands on a *disjunction*,  $\neg(A \vee B) = \neg A \wedge \neg B$ , so distributing is forced. In the food family it lands on a *conjunction*,  $\neg(A \wedge \neg B) = \neg A \vee B$ , and there is nothing to split. One De Morgan apart, opposite answers.

**Proof** negated goal 5.  $\neg \text{happy}(\text{Sneha})$

6.  $\neg \text{pass}(\text{Sneha}) \vee \neg \text{win}(\text{Sneha})$  (5, 1)  $[\text{Sneha}/x]$
7.  $\neg \text{lucky}(\text{Sneha}) \vee \neg \text{win}(\text{Sneha})$  (6, 2b)
8.  $\neg \text{win}(\text{Sneha})$  (7, 3b)
9.  $\neg \text{lucky}(\text{Sneha})$  (8, 4)  $[\text{Sneha}/x]$
10.  $\square$  (9, 3b)

- **“Sneha is happy” is proved.** ✓ Five steps.
- **Clauses 2a and 3a are never used.** “Sneha did not study” is there only to tempt you into deriving nothing from it; the lucky half carries the whole proof.
- “can pass all his exams” is written  $\text{pass}(x)$  with the exam left implicit. Adding a second argument  $\text{pass}(x, \text{exam})$  changes nothing except the typing.

**5. Steve likes BK301** 7

79 Ash 7 · **the printed premise runs the wrong way** and has to be read as its converse before anything can be proved

**QUESTION AS PRINTED** 2079 Ashwin · [7]

Assume the following facts: a) Steve only likes easy courses; b) Science courses are hard; c) All the courses in the basket weaving department are easy; d) BK301 is a basket weaving course. Prove that Steve likes BK301 using Resolution Refutation.

**Read premise (a) carefully first**

- Literally, “Steve only likes easy courses” is  $\forall x \text{ likes}(\text{Steve}, x) \rightarrow \text{easy}(x)$ : liking implies easy.
- From that you can prove a course Steve likes is easy. You cannot prove he likes anything, because nothing in the set ever concludes *likes*.
- The question is only answerable under the converse reading, “Steve likes every easy course”:

$$\forall x \text{ easy}(x) \rightarrow \text{likes}(\text{Steve}, x)$$

- **Write that line, then prove it.** Stating the reading you adopted is a mark; silently reversing an implication is not.

**FOPL**

- $\forall x \text{ easy}(x) \rightarrow \text{likes}(\text{Steve}, x)$  converse reading
  - $\forall x \text{ science}(x) \rightarrow \neg \text{easy}(x)$
  - $\forall x \text{ basketWeaving}(x) \rightarrow \text{easy}(x)$
  - $\text{basketWeaving}(\text{BK301})$
- Goal:**  $\text{likes}(\text{Steve}, \text{BK301})$

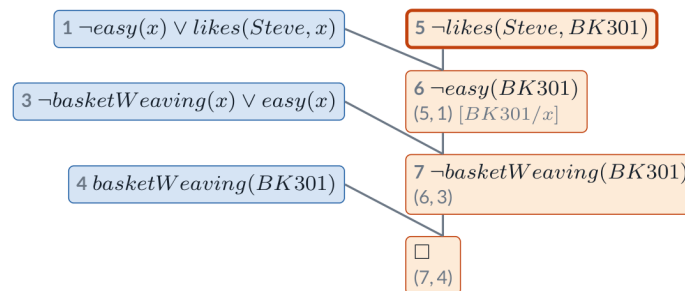
**CNF and proof**

- $\neg \text{easy}(x) \vee \text{likes}(\text{Steve}, x)$  (a), converse reading
- $\neg \text{science}(x) \vee \neg \text{easy}(x)$  (b)
- $\neg \text{basketWeaving}(x) \vee \text{easy}(x)$  (c)
- $\text{basketWeaving}(\text{BK301})$  (d)
- $\neg \text{likes}(\text{Steve}, \text{BK301})$  negated goal
- $\neg \text{easy}(\text{BK301})$  (5, 1) [BK301/x]
- $\neg \text{basketWeaving}(\text{BK301})$  (6, 3)
- $\square$  (7, 4)

- **“Steve likes BK301” is proved** under that reading. ✓

Three steps.

- **Premise (b) is unused**, and it is the only one mentioning science. *Hard* is written  $\neg \text{easy}$ ; inventing a separate *hard* predicate needs a fifth premise saying the two are opposites.

**6. You will enjoy 8**

77 Ch 8 · purely propositional, no quantifiers, and **four of the seven premises are noise**

QUESTION AS PRINTED 2077 Chaitra · CT 78506 · [8]

If it is a sunny and warm day you will enjoy. If it is a warm and pleasant day you will do strawberry picking. If it is raining, no strawberry picking. If it is raining you will get wet. It is a warm day. It is raining. It is sunny. Prove that “You will enjoy” using resolution by refutation.

**Into logic** (propositional: no objects, so no quantifiers)

- $\text{sunny} \wedge \text{warm} \rightarrow \text{enjoy}$
  - $\text{warm} \wedge \text{pleasant} \rightarrow \text{picking}$
  - $\text{raining} \rightarrow \neg \text{picking}$
  - $\text{raining} \rightarrow \text{wet}$
  - $\text{warm}$       6.  $\text{raining}$       7.  $\text{sunny}$
- Goal:**  $\text{enjoy}$

**CNF**

- $\neg \text{sunny} \vee \neg \text{warm} \vee \text{enjoy}$
- $\neg \text{warm} \vee \neg \text{pleasant} \vee \text{picking}$
- $\neg \text{raining} \vee \neg \text{picking}$
- $\neg \text{raining} \vee \text{wet}$
- $\text{warm}$       6.  $\text{raining}$       7.  $\text{sunny}$

Each implication becomes one clause by  $\alpha \rightarrow \beta = \neg \alpha \vee \beta$  plus De Morgan on the conjunctive antecedent. No quantifiers means no unification: every resolution step here is on a **ground** literal.

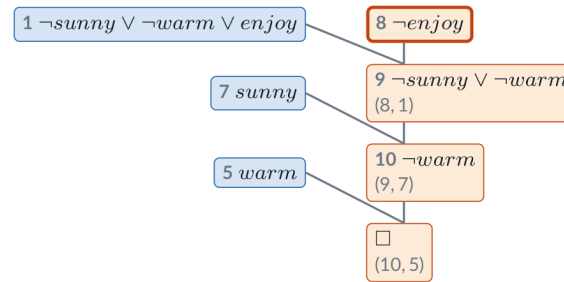
**Proof** negated goal 8.  $\neg \text{enjoy}$ 

- $\neg \text{sunny} \vee \neg \text{warm}$  (8, 1)
- $\neg \text{warm}$  (9, 7)
- $\square$  (10, 5)

- **“You will enjoy” is proved** in three steps. ✓

- **Clauses 2, 3, 4 and 6 are never touched.** The strawberry picking and the rain are a complete decoy: they are consistent with the rest and irrelevant to the goal.

- **The lesson worth writing under the proof:** resolve *from the negated goal*. Starting from the facts, this set will happily generate *wet*,  $\neg \text{picking}$  and several other true-but-useless literals before it stumbles on the answer.



## 7. Ram is naughty 8

70 Bh 8 · also set as an exercise by the textbook, whose own solution omits the argument order

QUESTION AS PRINTED 2070 Bhadra · [8]

All oversmart persons are stupid. Children of oversmart persons are naughty. Ram is children of Hari. Hari is oversmart. Show that Ram is naughty using FOPL based resolution.

### FOPL

1.  $\forall x \text{ oversmart}(x) \rightarrow \text{stupid}(x)$
2.  $\forall x \forall y \text{ child}(x, y) \wedge \text{oversmart}(y) \rightarrow \text{naughty}(x)$
3.  $\text{child}(\text{Ram}, \text{Hari})$       4.  $\text{oversmart}(\text{Hari})$

Goal:  $\text{naughty}(\text{Ram})$

### CNF

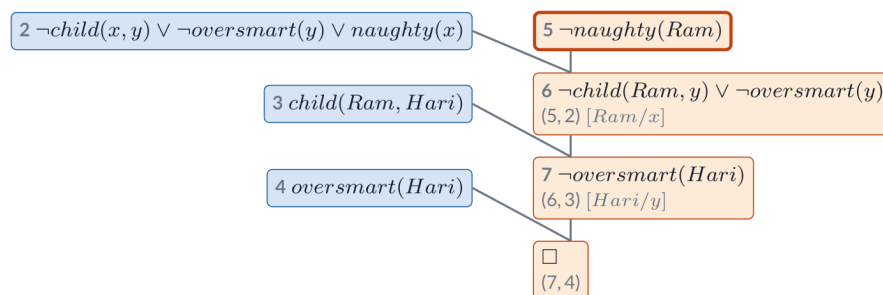
1.  $\neg \text{oversmart}(x) \vee \text{stupid}(x)$
2.  $\neg \text{child}(x, y) \vee \neg \text{oversmart}(y) \vee \text{naughty}(x)$
3.  $\text{child}(\text{Ram}, \text{Hari})$       4.  $\text{oversmart}(\text{Hari})$

Proof negated goal 5.  $\neg \text{naughty}(\text{Ram})$

6.  $\neg \text{child}(\text{Ram}, y) \vee \neg \text{oversmart}(y)$  (5, 2) [Ram/x]
7.  $\neg \text{oversmart}(\text{Hari})$  (6, 3) [Hari/y]
8.  $\square$  (7, 4)

### Notes

- “**Ram is naughty**” is proved in three steps. ✓
- **Premise 1 is unused.** *Stupid* appears nowhere else, so the whole first sentence is a decoy – the same trick as the mammal rule in Family B.
- $\text{child}(x, y)$  **must mean** “*x* is a child of *y*” and be used that way in both clause 2 and clause 3. The English “*Ram is children of Hari*” fixes the order; write the reading down before you write the clause, because reversing it silently breaks the unification in step 6.
- Predicate names are free, but *naughty* must be a *property of the child*, not a relation:  $\text{naughty}(x)$ , one argument.



## 8. The book supports the cup 8

71 Bh 8 · the only set written in **relational** rather than sentence form, so the argument order is the whole question

QUESTION AS PRINTED 2071 Bhadra · [8]

If X is on the top of Y, Y supports X. If X is above Y and they are touching each other, X is on the top of Y. A cup is above a book. A cup is touching a book. Show that  $\text{supports}(\text{book}, \text{cup})$  is true.

## FOPL

1.  $\forall x \forall y \text{ onTop}(x, y) \rightarrow \text{supports}(y, x)$
2.  $\forall x \forall y \text{ above}(x, y) \wedge \text{touching}(x, y) \rightarrow \text{onTop}(x, y)$
3.  $\text{above}(\text{Cup}, \text{Book})$       4.  $\text{touching}(\text{Cup}, \text{Book})$

**Goal:**  $\text{supports}(\text{Book}, \text{Cup})$

## CNF

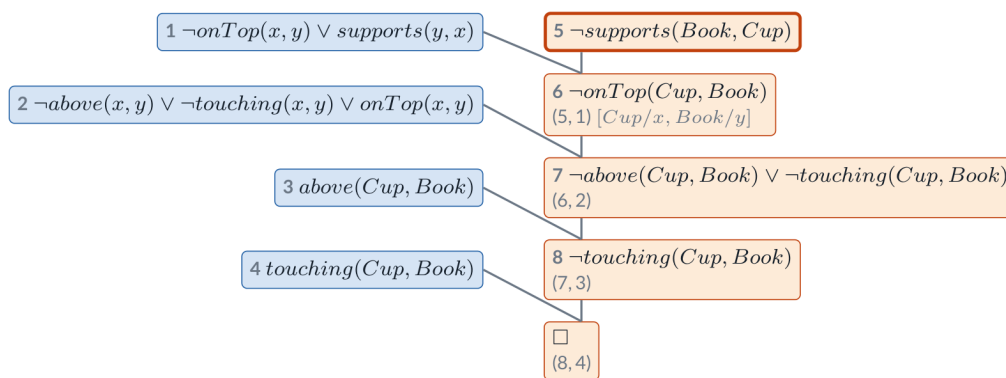
1.  $\neg \text{onTop}(x, y) \vee \text{supports}(y, x)$
2.  $\neg \text{above}(x, y) \vee \neg \text{touching}(x, y) \vee \text{onTop}(x, y)$
3.  $\text{above}(\text{Cup}, \text{Book})$       4.  $\text{touching}(\text{Cup}, \text{Book})$

**Proof** negated goal 5.  $\neg \text{supports}(\text{Book}, \text{Cup})$

6.  $\neg \text{onTop}(\text{Cup}, \text{Book})$  (5, 1)  $[\text{Cup}/x, \text{Book}/y]$
7.  $\neg \text{above}(\text{Cup}, \text{Book}) \vee \neg \text{touching}(\text{Cup}, \text{Book})$  (6, 2)
8.  $\neg \text{touching}(\text{Cup}, \text{Book})$  (7, 3)
9.  $\square$  (8, 4)

## Notes

- $\text{supports}(\text{Book}, \text{Cup})$  is proved in four steps. ✓
- **Clause 1 reverses its arguments and that is the point.**  $\text{onTop}(x, y)$  has the supporter second;  $\text{supports}(y, x)$  has it first. The goal  $\text{supports}(\text{Book}, \text{Cup})$  therefore unifies with  $[\text{Cup}/x, \text{Book}/y]$ , giving  $\text{onTop}(\text{Cup}, \text{Book})$  – cup on book, which is what the facts describe.
- **Every premise is used**, and there is no quantifier subtlety: the two rules are plain universals, so drop the  $\forall$  and go.
- “they are touching each other” is symmetric in English but  $\text{touching}(x, y)$  is not symmetric in the clause set. It does not matter here because the facts are given in the same order the rule wants; if they were not, a symmetry axiom  $\neg \text{touching}(x, y) \vee \text{touching}(y, x)$  would be needed.



## 9. If John does not study, Mary does not love John 10

PIN 1/41 73 Ma 10 · the goal is an implication again, so **negating it gives two clauses**, and every premise is a contrapositive

QUESTION AS PRINTED 2073 Magh · [10]

Consider the following axioms: i) anyone whom Mary loves is a football star; ii) any student who does not pass does not play; iii) John is a student; iv) any student who does not study does not pass; v) anyone who does not play is not a football star. Prove that “If John does not study, then Mary does not love John” using resolution by refutation.

**FOPL**

1.  $\forall x \text{ loves}(\text{Mary}, x) \rightarrow \text{star}(x)$
  2.  $\forall x \text{ student}(x) \wedge \neg \text{pass}(x) \rightarrow \neg \text{play}(x)$
  3.  $\text{student}(\text{John})$
  4.  $\forall x \text{ student}(x) \wedge \neg \text{study}(x) \rightarrow \neg \text{pass}(x)$
  5.  $\forall x \neg \text{play}(x) \rightarrow \neg \text{star}(x)$
- Goal:**  $\neg \text{study}(\text{John}) \rightarrow \neg \text{loves}(\text{Mary}, \text{John})$

**CNF**

1.  $\neg \text{loves}(\text{Mary}, x) \vee \text{star}(x)$
2.  $\neg \text{student}(x) \vee \text{pass}(x) \vee \neg \text{play}(x)$
3.  $\text{student}(\text{John})$
4.  $\neg \text{student}(x) \vee \text{study}(x) \vee \neg \text{pass}(x)$
5.  $\text{play}(x) \vee \neg \text{star}(x)$

**Negate the goal: two clauses**

- $\neg[\neg \text{study}(\text{John}) \rightarrow \neg \text{loves}(\text{Mary}, \text{John})] =$   
 $\neg \text{study}(\text{John}) \wedge \text{loves}(\text{Mary}, \text{John})$

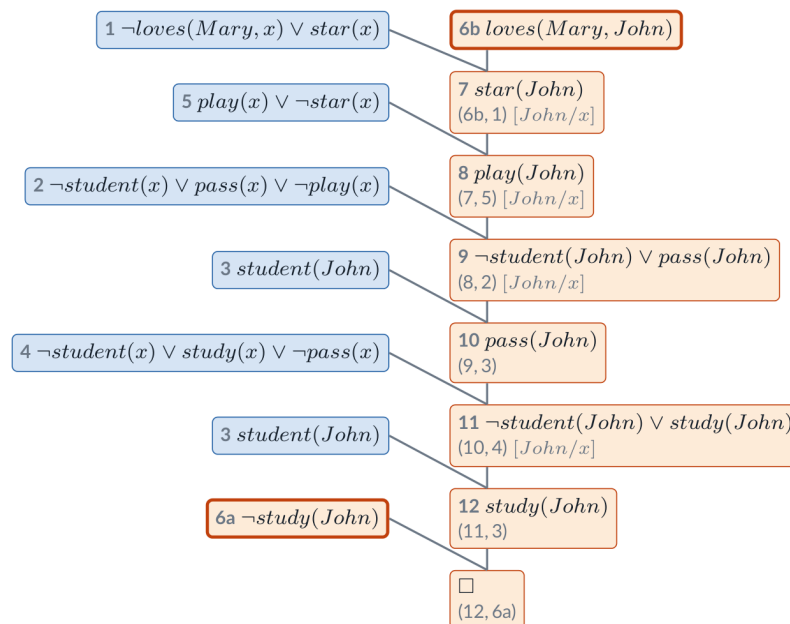
6a.  $\neg \text{study}(\text{John})$       6b.  $\text{loves}(\text{Mary}, \text{John})$

Premises (ii), (iv) and (v) are all of the form *not-P implies not-Q*. Convert each one mechanically:  $[S \wedge \neg P] \rightarrow \neg Q$  becomes  $\neg S \vee P \vee \neg Q$ . Trying to simplify them into positive form first is where this question is usually lost.

**Proof**

7.  $\text{star}(\text{John})$  (6b, 1) [John/x]
8.  $\text{play}(\text{John})$  (7, 5) [John/x]
9.  $\neg \text{student}(\text{John}) \vee \text{pass}(\text{John})$  (8, 2) [John/x]
10.  $\text{pass}(\text{John})$  (9, 3)
11.  $\neg \text{student}(\text{John}) \vee \text{study}(\text{John})$  (10, 4) [John/x]
12.  $\text{study}(\text{John})$  (11, 3)
13.  $\square$  (12, 6a)

- **The implication is proved.** ✓ Seven steps, every premise used, and clause 3 used **twice**.
- The chain in words: suppose Mary loves John and he does not study. Loved by Mary  $\rightarrow$  football star  $\rightarrow$  plays  $\rightarrow$  (being a student) passes  $\rightarrow$  studies. Contradiction with *does not study*.
- **A clause may be reused as often as needed.** Students often cross a clause off after one use; nothing in resolution says that.

**10. Can anyone be found with an exciting life? 3+5**

80 Ash 3+5 · asks *who*, not *prove*, so this is the chapter's one genuine **answer-extraction** question

**QUESTION AS PRINTED** 2080 Ashwin · [3+5]

List the rules of inference. Convert the following into FOPL and prove using Resolution Refutation System: "All people who are not poor and are smart are happy. Those people who can read are not stupid. John can read and is wealthy. Happy people have exciting lives. Can anyone be found with an exciting life?"

**FOPL** (the 5 marks begin here)

1.  $\forall x \neg \text{poor}(x) \wedge \text{smart}(x) \rightarrow \text{happy}(x)$
2.  $\forall x \text{read}(x) \rightarrow \neg \text{stupid}(x)$
3.  $\text{read}(\text{John}) \wedge \text{wealthy}(\text{John})$ , i.e.  
 $\text{read}(\text{John}) \wedge \neg \text{poor}(\text{John})$
4.  $\forall x \text{happy}(x) \rightarrow \text{exciting}(x)$
5.  $\forall x \neg \text{stupid}(x) \rightarrow \text{smart}(x)$

**Goal:**  $\exists x \text{exciting}(x)$

**CNF** with  $\text{wealthy} = \neg \text{poor}$  and  $\text{smart} = \neg \text{stupid}$

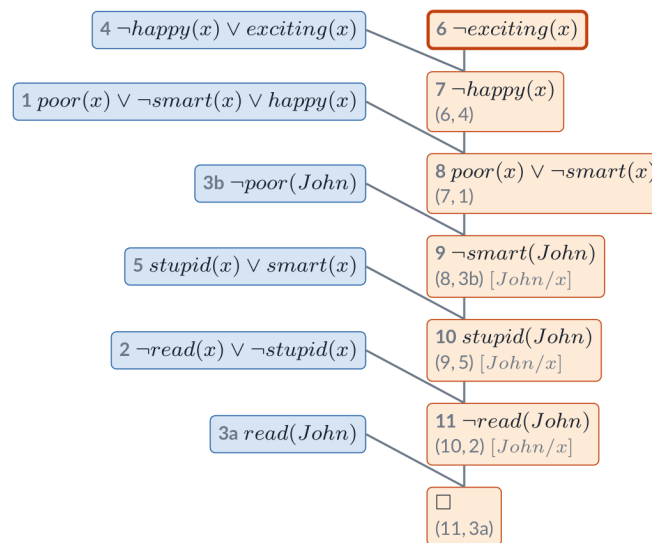
1.  $\text{poor}(x) \vee \neg \text{smart}(x) \vee \text{happy}(x)$
2.  $\neg \text{read}(x) \vee \neg \text{stupid}(x)$
- 3a.  $\text{read}(\text{John})$       3b.  $\neg \text{poor}(\text{John})$
4.  $\neg \text{happy}(x) \vee \text{exciting}(x)$
5.  $\neg \text{smart}(x) \vee \neg \text{stupid}(x)$  is **not** needed; instead state the bridging fact  $\neg \text{stupid}(x) \rightarrow \text{smart}(x)$ , i.e. 5.  
 $\text{stupid}(x) \vee \text{smart}(x)$

**Two vocabulary bridges must be stated:**  $\text{wealthy}$  is  $\neg \text{poor}$ , and  $\text{not stupid}$  is  $\text{smart}$ . Without them the clause set has five predicates that never meet and no proof exists. Say so in one line before starting — the examiner's own wording forces it.

**Proof** negated goal 6.  $\neg \text{exciting}(x)$ , for all  $x$

7.  $\neg \text{happy}(x)$  (6, 4)
8.  $\text{poor}(x) \vee \neg \text{smart}(x)$  (7, 1)
9.  $\neg \text{smart}(\text{John})$  (8, 3b) [ $\text{John}/x$ ]
10.  $\text{stupid}(\text{John})$  (9, 5) [ $\text{John}/x$ ]
11.  $\neg \text{read}(\text{John})$  (10, 2) [ $\text{John}/x$ ]
12.  $\square$  (11, 3a)

- **Yes: John has an exciting life.** ✓
- **How the name is recovered**, which is what “can anyone be found” asks: the goal  $\exists x \text{exciting}(x)$  negates to  $\forall x \neg \text{exciting}(x)$ , one clause with a free variable. The substitutions used along the way bound that variable to **John**, and that binding is the answer.
- The formal device is an **answer literal**: carry  $\text{exciting}(x) \vee \text{Ans}(x)$  instead of the bare negation, and read the answer off the clause that survives. Worth naming even if you do not use it.



## P7. Bayes' theorem: the twelve numericals

**15 papers set a Bayes question**; twelve give numbers and are worked below, three ask for a worked example of your own and are answered by any one of them. Every answer here was recomputed independently in `verify.py`, not copied from a source. The recipe never changes:

### The recipe

1. **Name the events** and write down which probability each given number is. **Half the lost marks happen here**, in the first thirty seconds.
2. Write Bayes' theorem with the names, not with  $A$  and  $B$ .
3. If  $P(\text{evidence})$  is not given, build it with the law of total probability:  $P(E) = P(E|H)P(H) + P(E|\neg H)P(\neg H)$ .
4. Substitute, and quote the answer as a percentage as well as a decimal.

**B1–B2. Meningitis and the stiff neck** 3+5 2+671 Ma 3+5, 81 Ba 2+6 · same scenario, **different numbers in each paper** — read them off the paper, not from memory**QUESTION AS PRINTED** 2071 Magh · [3+5]

What is neural network? Describe its types. A doctor knows that the disease meningitis causes the patient to have a stiff neck 50% of the time. The doctor also knows that the probability that a patient has meningitis is  $1/50,000$ , and the probability that any patient has a stiff neck is  $1/20$ . Find the probability that a patient with a stiff neck has meningitis?

**B1 2071 Magh**

- **Given**  $P(s|m) = 0.5$ ,  $P(m) = 1/50000 = 2 \times 10^{-5}$ ,  
 $P(s) = 1/20 = 0.05$ .
- **Wanted**  $P(m|s)$ .

$$P(m|s) = \frac{P(s|m)P(m)}{P(s)} = \frac{0.5 \times 2 \times 10^{-5}}{0.05} = \frac{10^{-5}}{0.05}$$

$$P(m|s) = 0.0002 = 0.02\%$$

- $P(s)$  is given here, so no total-probability step is needed. That is the only easy version of this question in the archive.

**B2 2081 Baishakh (CT 710)**

- **Given**  $P(s|m) = 0.8$ ,  $P(m) = 1/50000$ ,  $P(s) = 0.01$ .

$$P(m|s) = \frac{0.8 \times 2 \times 10^{-5}}{0.01} = \frac{1.6 \times 10^{-5}}{0.01}$$

$$P(m|s) = 0.0016 = 0.16\%$$

- The first 2 marks are **prior Vs posterior** defined:  $P(m)$  is the prior,  $P(m|s)$  the posterior, and the stiff neck raised the belief from **0.002 % to 0.16 %** — eighty times higher, and still under one in six hundred.
- **That last sentence is the exam answer to “what does this tell you”**: a strong likelihood cannot outweigh a tiny prior.

**B3. Measles or flu?** 8

73 Ma 8 · the priors are given directly as a 90/10 split, so this is the cleanest total-probability example in the archive

**QUESTION AS PRINTED** 2073 Magh · [8]

A doctor is called to see a sick child. The doctor has prior information that 90% of sick children in that neighborhood have the flu, while the other 10% are sick with measles. Let F stand for an event of a child being sick with Flu and M stand for an event of a child being sick with measles. Assume for simplicity that there no other maladies in that neighborhood. A well-known (and common) symptom of measles is a rash and has probability of 0.95. However, very occasionally, children with flu also develop rash and has probability of 0.08. Upon examining the child, the doctor finds a rash. What is the probability that the child has measles?

**Solution**

- **Given**  $P(F) = 0.9$ ,  $P(M) = 0.1$ ,  $P(R|M) = 0.95$ ,  
 $P(R|F) = 0.08$ .
- **Total probability of a rash:**

$$\begin{aligned} P(R) &= P(R|M)P(M) + P(R|F)P(F) \\ &= 0.95 \times 0.1 + 0.08 \times 0.9 = 0.095 + 0.072 = 0.167 \end{aligned}$$

$$P(M|R) = \frac{P(R|M)P(M)}{P(R)} = \frac{0.095}{0.167}$$

$$P(M|R) = 0.5689 \approx 56.9\%$$

**Reading the answer**

- Measles goes from a **10 % prior to a 57 % posterior**: the rash is strong evidence, but flu is nine times commoner, so it stays a coin flip.
- **Check the complement:**  
 $P(F|R) = 0.072/0.167 = 0.4311$ , and  
 $0.5689 + 0.4311 = 1$ . ✓ Two extra lines, and they catch an arithmetic slip in the denominator every time.
- **The trap:** reading 0.95 as  $P(M|R)$  and answering “95 %”. The paper gives the *likelihood*  $P(R|M)$ ; the whole point of the question is that the two are not the same number.

**B4. The cigar smoker** 2+6 880 Bh 2+6, 73 Bh 8 · two papers, identical data · part (i) is the **prior**, and it is one mark for writing down a number already in the question**QUESTION AS PRINTED** 2080 Bhadra · CT 710 · [2+6] also 2073 Bhadra CT 78506 [8]

What do you understand by Bayesian Network? In a County, 51% of the adults are males and the other 49% are females. One adult is randomly selected for a survey involving credit card usage. i) Find the prior probability that the selected person is a male. ii) It is later learned that the selected survey subject was smoking a cigar. Also, 9.5% of males smoke cigars, whereas 1.7% of females smoke cigars (based on data from the Substance Abuse and Mental Health Services Administration). Use this additional information to find the probability that the selected subject is a male.

**Solution**

- (i) **Prior**  $P(M) = 0.51$ . **That is the whole answer to part (i)**; no calculation is intended.
- (ii) **Given**  $P(F) = 0.49$ ,  $P(C|M) = 0.095$ ,  $P(C|F) = 0.017$ .

$$P(C) = 0.095 \times 0.51 + 0.017 \times 0.49 \\ = 0.04845 + 0.00833 = 0.05678$$

$$P(M|C) = \frac{0.04845}{0.05678}$$

$$P(M|C) = 0.8533 \approx 85.3\%$$

**Notes**

- The prior 51 % rises to **85 %** because men are five and a half times as likely to smoke a cigar. **Here the likelihood ratio does dominate**, which is the opposite lesson to B1 and B2 – and the difference is entirely that this prior is near a half rather than near zero.
- **Keep five decimal places in the denominator.** Rounding 0.05678 to 0.057 moves the answer to 85.0 %, and the marks for the final figure are awarded on the exact value.
- **73 Bh** prints the same scenario for a single 8-mark part. Same two answers.

**B5. The 99 % accurate test for a rare disease** 8

**PIN 1/41** 72 Ash 8 · **the base-rate question**, and the one every paper's theory part is really about

**QUESTION AS PRINTED** 2072 Ashwin · CT 78506 · [8]

After your yearly checkup, the doctor has bad news and good news. The bad news is that you tested positive for a serious disease, and that the test is 99% accurate (i.e. the probability of testing positive given that you have the disease is 0.99, as is the probability of testing negative given that you don't have the disease). The good news is that this is a rare disease, striking only one in 10,000 people. Why is it good news that the disease is rare? What are the chances that you actually have the disease?

**Solution**

- **Given**  $P(D) = 10^{-4}$ ,  $P(+|D) = 0.99$ ,  $P(-|\neg D) = 0.99$ , so  $P(+|\neg D) = 0.01$ .

$$P(+) = 0.99 \times 10^{-4} + 0.01 \times 0.9999 \\ = 0.000099 + 0.009999 = 0.010098$$

$$P(D|+) = \frac{0.000099}{0.010098}$$

$$P(D|+) = 0.0098 \approx 0.98\%$$

- **Fewer than 1 in 100** people with a positive result actually have the disease.

**“Why is it good news that the disease is rare”**

- Because the **false positives outnumber the true ones**, and by a lot. In 1,000,000 people: **100** have the disease and 99 of them test positive; 999,900 do not, and **9,999** of them still test positive. **9,999 against 99 is roughly 100 to 1.**
- So a rare disease means a positive result is far more likely to be an error than a diagnosis. Rarity protects you.
- **Write the population count.** It is worth more marks than the algebra and it is much harder to get wrong.
- **The general lesson:** the posterior depends on the prior as much as on the test. A 99 % test is not a 99 % answer.

**B6. The genetic defect** 2+6

81 Ash 2+6 · part (b) asks for the **probability** that they do **not** have it, which is the complement – **read the question twice**

**QUESTION AS PRINTED** 2081 Ashwin · [2+6]

What are the approaches for measuring uncertainty? 1% of people have a certain genetic defect. 90% of tests for the gene detect the defect (true positives). 10% of the tests are false positives. a) What might be the probability of getting the genetic defect result? b) If a person gets a positive test result, what are the odds they actually don't have the genetic defect?

**Solution**

- **Given**  $P(D) = 0.01$ ,  $P(+|D) = 0.90$ ,  $P(+|\neg D) = 0.10$ .
- (a) The probability of a positive result:

$$P(+) = 0.90 \times 0.01 + 0.10 \times 0.99$$

$$= 0.009 + 0.099 = \boxed{0.108 = 10.8\%}$$

- (b)

$$P(\neg D|+) = \frac{0.099}{0.108}$$

$$= \boxed{0.9167 \approx 91.7\%}$$

- And so  $P(D|+) = 0.009/0.108 = 0.0833 = 8.3\%$ .  
 $91.7 + 8.3 = 100$ . ✓

**B7. Which factory made the defective bulb?** 3+5

80 Ash 3+5 · three-way partition, so the denominator has three terms

QUESTION AS PRINTED 2080 Ashwin · [3+5]

How is reasoning done under uncertainty? Three factories produce light bulbs to supply the market. Factory A produces 20%, 50% of the bulbs are produced in factory B and 30% in factory C. 2% of the bulbs produced in factory A, 5% of the bulbs produced in factory B and 3% of the bulbs produced in factory C are defective. A bulb is selected at random in the market and found to be defective. What is the probability that this bulb was produced by factory B?

**Solution**

- **Given**  $P(A) = 0.2$ ,  $P(B) = 0.5$ ,  $P(C) = 0.3$ ;  
 $P(D|A) = 0.02$ ,  $P(D|B) = 0.05$ ,  $P(D|C) = 0.03$ .
- **Total probability of a defective bulb**, one term per factory:

$$P(D) = 0.2(0.02) + 0.5(0.05) + 0.3(0.03)$$

$$= 0.004 + 0.025 + 0.009 = 0.038$$

$$P(B|D) = \frac{P(D|B)P(B)}{P(D)} = \frac{0.025}{0.038}$$

$$\boxed{P(B|D) = 0.6579 \approx 65.8\%}$$

**Notes**

- **As true odds**, which is what part (b) literally asks for:  $0.099 : 0.009$ , i.e. **11 to 1 against actually having the defect**. Give both the probability and the odds and neither reading can cost a mark.
- **Part (a) is ambiguous** as printed – “the probability of getting the genetic defect result” could mean  $P(D)$ , which is stated in the question and would make the part worthless. It means  $P(+)$ . Say which reading you took.
- Same shape as B5 with a much weaker test (10 % false positives instead of 1 %) and a much commoner condition (1 % instead of 0.01 %). The posterior lands at 8 % rather than 1 %.

**Notes**

- **The partition must be exhaustive:**  $0.2 + 0.5 + 0.3 = 1$ . ✓ Check it before you start; a partition that does not sum to 1 means a misread question.
- Factory B’s share rises from **50 % to 66 %** because it is both the biggest producer *and* the worst. The other two posteriors are  $P(A|D) = 0.004/0.038 = 10.5\%$  and  $P(C|D) = 0.009/0.038 = 23.7\%$ , and the three sum to 100 %. ✓
- **This is the template for every “which source did it come from” question**, and there are three of them in the archive (bulbs, factories, and the tall-student one below).

**B8–B9. The tall student, and the mammogram** 3+5 2+6

78 Ch 3+5, 82 Ba 2+6

QUESTION AS PRINTED 2078 Chaitra · [3+5]

Explain how statistical reasoning aids in inference and reasoning in light of Bayes’ theorem. At a certain University, 5% of men are over 6 feet tall and 2% of women are 6 feet tall. 60% of students are female. If a student is selected at random from among all those over six feet tall, what is the probability that the selected student is a woman?

QUESTION AS PRINTED 2082 Baishakh · CT 710 · [2+6]

What are casual networks? Given the following statistics, what is the probability that a woman has cancer if she has a positive mammogram result? i) One percent of women over 50 have breast cancer. ii) Ninety percent of women who have breast cancer test positive on mammograms. iii) Eight percent of women will have false positives.

**B8 the tall student**

- Given  $P(W) = 0.6$ ,  $P(M) = 0.4$ ,  $P(T|W) = 0.02$ ,  $P(T|M) = 0.05$ .

$$P(T) = 0.02(0.6) + 0.05(0.4) = 0.012 + 0.020 = 0.032$$

$$P(W|T) = \frac{0.012}{0.032} = \boxed{0.375 = 37.5\%}$$

- Women are the **majority** of students and the **minority** of tall ones: 60 % prior, 37.5 % posterior. **The evidence reverses the balance**, which is the sentence the 3-mark theory part wants.

**B9 the mammogram**

- Given  $P(C) = 0.01$ ,  $P(+|C) = 0.90$ ,  $P(+|\neg C) = 0.08$ .

$$P(+) = 0.90(0.01) + 0.08(0.99) \\ = 0.009 + 0.0792 = 0.0882$$

$$P(C|+) = \frac{0.009}{0.0882} = \boxed{0.1020 \approx 10.2\%}$$

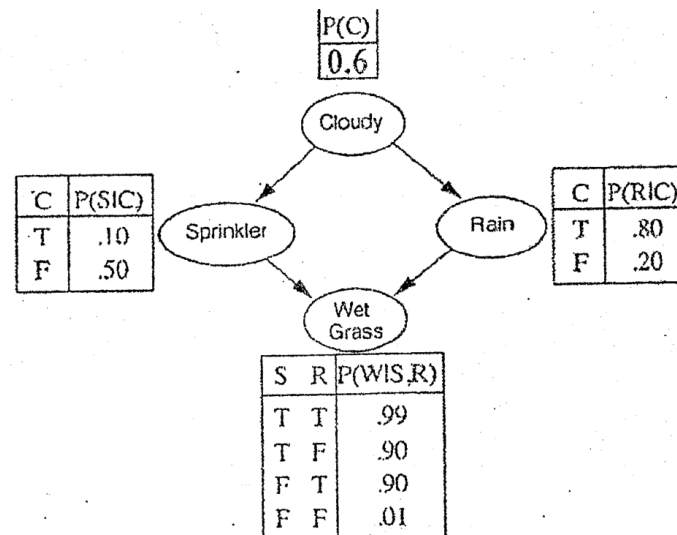
- **Nine positives in ten are false alarms**, even though the test catches 90 % of real cancers. Same base-rate lesson as B5, in the setting it was originally made famous in.
- **“Eight percent of women will have false positives”** means  $P(+|\neg C) = 0.08$ , not  $P(\neg C \wedge +) = 0.08$ . The looser reading gives  $0.009/0.089 = 10.1\%$ , close enough to hide the error – **state your reading**.

**P8. Bayesian networks: reading a joint probability off the graph****B10. The wet grass network** 2+6

PIN 1/41 78 Ba 2+6 · the paper prints the network and its four CPTs · **one multiplication, four factors**

QUESTION AS PRINTED 2078 Baishakh · [2+6]

What do you understand by Bayesian Network? For the given Bayesian network below, find the probability of grass being wet when the weather is cloudy, there is some sprinkler but no rain, i.e.  $C = \text{True}$ ,  $S = \text{True}$ ,  $R = \text{False}$  and  $W = \text{True}$ .



**Solution**

- **The product form**, one factor per node, each conditioned on its parents:

$$P(C, S, \neg R, W) = P(C) P(S|C) P(\neg R|C) P(W|S, \neg R)$$

- **Read each factor off its own table:**

- $P(C) = 0.6$  root node, prior
- $P(S|C=T) = 0.10$  sprinkler table, row T
- $P(R|C=T) = 0.80$ , so  $P(\neg R|C=T) = 1 - 0.80 = 0.20$
- $P(W|S=T, R=F) = 0.90$  wet-grass table, row T F

$$= 0.6 \times 0.10 \times 0.20 \times 0.90$$

$$P = 0.0108 = 1.08 \%$$

**The three ways this goes wrong**

- **Forgetting to complement  $R$ .** The query says *no rain*, and the table gives  $P(R|C)$ . Take  $1 - 0.80$ . Using 0.80 gives 0.0432, four times too big, and it is the commonest wrong answer.
- **Picking the wrong CPT row.** The wet-grass table is indexed by  $(S, R)$ , and the query is  $S = T, R = F$ : row **T F**, value 0.90. Row **F T** is also 0.90, so a misread here happens to give the right number in this one query and the wrong one in every other.
- **Conditioning on the wrong parent.** Sprinkler and Rain both depend on *Cloudy* only; *WetGrass* depends on both of them and *not* on *Cloudy*. That is the conditional independence the network encodes, and it is what makes four numbers enough.
- **The 2-mark part:** a Bayesian network is a DAG of random variables with a CPT at each node, and the joint distribution is the product above. Six variables of this kind would need 31 free numbers as a full joint table; this one needs 9.

**B11. Smoker → cancer → test 7**

**PIN 1/41** 82 Bh 7 · asks for an *explanation*, but the marks are safest if the explanation carries numbers · a **chain** network, so every inference is a two-step

**QUESTION AS PRINTED** 2082 Bhadra · CT 710 · [7]

A Bayesian Belief Network (BBN) is used in a medical diagnosis system. The network includes the binary variables  $C$  (Patient has Cancer),  $T$  (Test result is Positive) and  $S$  (Patient is a Smoker), with structure  $S \rightarrow C \rightarrow T$  and conditional probabilities  $P(S=\text{Yes})=0.3$ ,  $P(C=\text{Yes} | S=\text{Yes})=0.2$ ,  $P(C=\text{Yes} | S=\text{No})=0.05$ ,  $P(T=\text{Yes} | C=\text{Yes})=0.9$ ,  $P(T=\text{Yes} | C=\text{No})=0.1$ . Explain how the Bayesian Belief Network enables reasoning under uncertainty in this medical context.

**The three kinds of reasoning, each with its number**

- **Causal, top-down** – *a smoker walks in; how likely is a positive test?*

$$P(T|S) = 0.9(0.2) + 0.1(0.8) = \mathbf{0.26}$$

$$\text{against } P(T|\neg S) = 0.9(0.05) + 0.1(0.95) = 0.14.$$

- **Prior belief**, before anyone is seen:

$$P(C) = 0.2(0.3) + 0.05(0.7) = 0.06 + 0.035 = \mathbf{0.095}$$

$$P(T) = 0.9(0.095) + 0.1(0.905) = 0.0855 + 0.0905 = \mathbf{0.176}$$

- **Diagnostic, bottom-up** – *the test came back positive; how likely is cancer?* This is Bayes on the second edge:

$$P(C|T) = \frac{0.9 \times 0.095}{0.176} = \frac{0.0855}{0.176} = \mathbf{0.4858}$$

- **Through the chain** – *the test is positive; how likely is the patient a smoker?*

$$P(S|T) = \frac{P(T|S)P(S)}{P(T)} = \frac{0.26 \times 0.3}{0.176} = \frac{0.078}{0.176} = \mathbf{0.4432}$$

Consistency check:  $P(T)$  computed through  $C$  gives 0.176, and computed through  $S$  gives  $0.26(0.3) + 0.14(0.7) = 0.078 + 0.098 = 0.176$ . ✓ **Two routes, one answer** – that is the network being coherent, and it is worth showing.

**What to say in words, which is what the question asks**

- The network **separates what is known from what is asked**. The doctor can only measure causal probabilities – how often smokers get cancer, how often cancer shows on a test – and the network stores exactly those. Bayes then runs them backwards on demand.
- It handles **partial evidence**: with no observation the belief in cancer is 9.5 %; knowing the patient smokes raises it to 20 %; a positive test raises it to **48.6 %**. Each piece of evidence revises, none replaces.
- **Conditional independence keeps it small**.  $T$  depends on  $C$  alone: once the cancer status is known, smoking tells you nothing more about the test. So three variables need **5 numbers** instead of  $2^3 - 1 = 7$ , and the saving grows exponentially with the network.
- **It never claims certainty**. A positive test leaves the patient more likely *not* to have cancer than to have it – 51.4 % against 48.6 % – which is precisely the kind of conclusion a logic-based KB cannot express at all.

**B12. The three papers that ask for an example of your own** 7 5+3 3+5

73 Bh 7, 80 Ba 5+3, 80 Ch 3+5 · no data printed, so **bring your own** — and the smallest one that shows everything is the two-node version of B9

**The example to memorise**

- **Network:**  $Cancer \rightarrow Test$ , two nodes, one edge.
- **CPTs:**  $P(C) = 0.01$ ;  $P(+|C) = 0.9$ ,  $P(+|\neg C) = 0.08$ .
- **Causal use** (down the arrow): a patient with cancer tests positive 90 % of the time. Straight from the table, no calculation.
- **Diagnostic use** (against the arrow): a positive test means cancer with probability

$$\frac{0.9 \times 0.01}{0.9(0.01) + 0.08(0.99)} = \frac{0.009}{0.0882} = 10.2\%$$

- **Three numbers, four lines, and it demonstrates the entire connection** the three papers ask about.

**The connection, stated**

- **The network supplies the numbers Bayes needs.**  $P(C)$  is the prior sitting at the root;  $P(+|C)$  and  $P(+|\neg C)$  are the CPT of the child; the denominator is the law of total probability over that same CPT.
- **Bayes supplies the direction the network lacks.** The arrows point from cause to effect, which is the direction knowledge is available in. Diagnosis needs the other direction, and Bayes is the only thing that turns it round.
- So the answer to “*is there any connection*” is: **a belief network is a factored store of exactly the conditional probabilities Bayes’ theorem consumes**, and inference in the network is repeated application of Bayes plus the product rule.
- 80 Ba’s “how can inference be done” is answered by the four modes in §4.7: causal, diagnostic, intercausal and mixed. Name all four; work one.

**How every answer above was checked****The proofs**

- `verify.py` carries a small **first-order resolution prover** (unification with occurs check, factoring, set of support) and runs it on **every clause set printed above**, in the exact form printed.
- It confirms the empty clause is derivable for all of them — and, just as importantly, that it is **not** derivable for the three papers whose printed goal does not follow:
  - 72 Ma *Shailendri likes chicken* — not derivable; *eats* is.
  - 82 Bh *Mr. A is a criminal* — not derivable; *Criminal(Mr.R)* is.
  - 79 Ash *Steve likes BK301* — not derivable from *likes* → *easy*; derivable from the converse.
- It also confirms that 69 Bh and 75 Ba **cannot** be closed without the two bridging axioms this document adds, and that they close once those are supplied.

**The numbers**

- Every Bayes posterior, every total-probability denominator and both network products are recomputed from the paper’s own figures and asserted against the boxed value above. **Nothing here was copied from a source.**
- Three of those checks are consistency tests rather than answers: the measles posteriors summing to 1, the three factory posteriors summing to 1, and 82 Bh’s  $P(T)$  computed by two different routes through the network agreeing at 0.176.
- **Run it before trusting a rebuild:**  
`cd AI/ExamNotes/src && python verify.py`
- A textbook error that survives a check is still an error. The three *Insights* corrections in this chapter — the split of a negated conjunction, the free variable where a Skolem function belongs, and the five-step CNF list — were each found by hand and then confirmed here.

## 5. Structured Knowledge Representation – Solved Problems

Every problem opens with the question **exactly as the paper prints it**, the paper's own spelling and slips included. · **11 sentence sets over 12 papers**, plus the net-to-frame conversion and the net/frame/FOPL question. · Every set is answered as an **arc list** — every node, every link, every direction — and **the net drawn from that list**, in the paper's own words. The deck and textbook figures for these sets use other names and values, so they are not printed as answers.

**The four-step recipe. It works on all eleven sets:**

- **Step 1. Underline every noun phrase.** Each distinct one is a **node**. Do not invent nodes and do not merge two that the paper keeps apart.
- **Step 2. Sort the nodes into classes and individuals.** A proper name, or anything the sentences call *a* something, is an **individual**. Anything that other things can *be* is a **class**.
- **Step 3. Draw the class spine first, bottom to top**, joined by **isa** arcs, then hang each individual off its class with an **instance** arc.
  - **Class to class is isa. Individual to class is instance.** Getting these two the wrong way round is the single commonest lost mark in this question.
- **Step 4. Add one labelled arc per remaining sentence**, always *out of* the thing the sentence is about. The verb becomes the arc label.
- **Then check three things**, which is what separates a 6 from a 4:
  - **Count.** One arc per sentence, plus the spine. If a sentence produced no arc you have lost a mark.
  - **Push properties up.** If two individuals of the same class both carry a property, draw it **once on the class** and let them inherit. Say in one line that you did, and why.
  - **Flag what the net cannot say.** Quantifiers, negation and disjunction do not draw. A sentence like “*every mammal gives birth to a baby*” gets an arc out of the class node and **a note saying the arc is read as  $\forall x \in Mammal$** . Examiners give marks for that note.
- **Layout that always works:** most general class at the **top**, **isa** arcs pointing **up**, individuals on a row near the bottom, and their own properties fanning **sideways**. Every net in this chapter is drawn that way.

### 5.N.1 A person is a mammal: Sakti Gauchan, and the same set as Sandeep Lamichhane

**Convert the given sentences into a semantic net** 7 2+6

**PIN 2/41** 72 Ma 7, 78 Po 2+6 · two papers, **one net:** 78 Po is 72 Ma with the cricketer's name changed · both also ask “*what are frames and semantic net*” first, which is \$5.6 of the theory

**QUESTION AS PRINTED** 72 Ma · Q6 · 7

What are Frames and Semantic Net? Convert the given sentences in Semantic Net.

- i) A person is a mammal.
- ii) Sakti Gauchan is a person.
- iii) Person has nose.
- iv) Sakti Gauchan is in Nepalese team.
- v) Uniform color of sakti Gauchan is Red/Blue.

**QUESTION AS PRINTED** 78 Po · Q6 · 2+6

What are Frames and Semantic Net? Convert the given sentences in Semantic Net.

- A person is a mammal.
- Sandeep Lamichhane is a person.
- Person has nose.
- Sandeep Lamichhane is in Nepalese cricket team.
- **Uniform color of Sakti Gauchan is Red/Blue.**

The fifth sentence names a *different* cricketer: the paper was copied from the 72 Ma set and this one line was never updated. Reproduced as printed.

**Step 1–2: nodes**

- **Classes:** Mammal, Person, Nepalese team (a named team, so it can also be drawn as an individual – either is accepted).
- **Individual:** Sakti Gauchan.
- **Values:** Nose, Red/Blue.

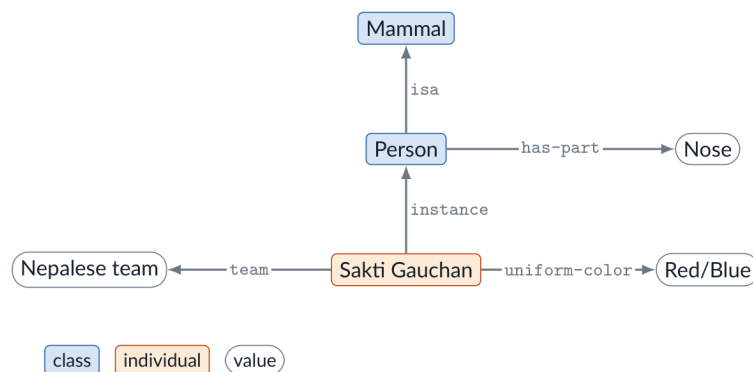
**Step 3–4: the arcs**

| From          | Link          | To            |
|---------------|---------------|---------------|
| Person        | isa           | Mammal        |
| Sakti Gauchan | instance      | Person        |
| Person        | has-part      | Nose          |
| Sakti Gauchan | team          | Nepalese team |
| Sakti Gauchan | uniform-color | Red/Blue      |

- **5 sentences, 5 arcs.** The count checks.
- **Note to write beside the drawing:** *has-part* sits on **Person**, not on Sakti Gauchan, so **every** person inherits a nose. That is the inheritance the question is testing.
- 78 Po: identical net, *Sakti Gauchan* replaced by *Sandeep Lamichane* in the first four arcs. **For the fifth, draw the uniform-colour arc off Sandeep and add one line:** “the paper names Sakti Gauchan here; taken as the same person the other four sentences are about.” Say it and you lose nothing.

**The FOPL to write under the net**

- `isa(Person, Mammal),`  
`instance(Sakti Gauchan, Person),`  
`has-part(Person, Nose),`  
`team(Sakti Gauchan, Nepalese team),`  
`uniform-color(Sakti Gauchan, Red/Blue).`
- **PS Sir's slide for this set is not the answer:** it draws *Head* for the paper's nose and *Black/Blue* for *Red/Blue*. Use the paper's words, as the net below does.
- **The deck is inconsistent, twice over.** An earlier slide writes `MAMMAL(person)` for the first sentence – a *unary* predicate that loses the relation entirely. And it writes the arguments `(class, individual)` here, the reverse of Rich and Knight's `instance(Pee-Wee-Reese, Person).`
- **Use one order and say which.** This file writes `isa(subclass, superclass)` and `instance(individual, class)`, matching the arrow direction in every drawing. **Consistency is what is marked, not the choice:** pick one, state it in a line, and never swap.

**5.N.2 The baseball player: Pee-wee-Reese, and the same set as Rohit Paudel****Convert the given sentences into a semantic net** 7 2+6

**PIN 2/41** 73 Bh 7, 79 Ch 2+6 · again **two papers, one net:** 79 Ch is 73 Bh moved from baseball to cricket · this is Rich and Knight's own figure, so **it is also the set to expect in a “convert the net into a frame” question**

**QUESTION AS PRINTED** 73 Bh · Q6 · 7

Convert given sentences into Semantic Network.

- i) The height of the adult male is 5.10
- ii) Baseball player is an adult male.
- iii) Adult male is a person.
- iv) Batting average of Baseball players is 0.252
- v) Pee-wee-Reese is a Fielder.
- vi) Fielder is Baseball player.
- vii) Team of pee-wee-Reese is Brooklyn Dodger.

**QUESTION AS PRINTED** 79 Ch · Q4 · 2+6

What is semantic network? Convert the following sentences into semantic network.

- i) The avg height of an adult is 5'6”.
- ii) Cricket player is an adult male.

- iii) Adult male is a person.
- iv) Batting average of a cricket player is 37.59
- v) Rohit Paudel is a Fielder.
- vi) Fielder is a cricket player.
- vii) Team of Rohit Paudel is Nepal.

Sentence (i) says *adult* where (ii) and (iii) say *adult male*. Read as the same node.

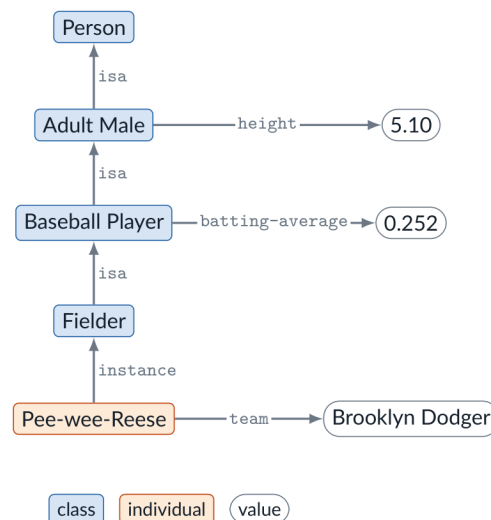
### The spine is four deep, and that is the whole difficulty

| From            | Link            | To              |
|-----------------|-----------------|-----------------|
| Adult Male      | isa             | Person          |
| Baseball Player | isa             | Adult Male      |
| Fielder         | isa             | Baseball Player |
| Pee-wee-Reese   | instance        | Fielder         |
| Adult Male      | height          | 5.10            |
| Baseball Player | batting-average | 0.252           |
| Pee-wee-Reese   | team            | Brooklyn Dodger |

- **7 sentences, 7 arcs.** The count checks.
- **The mark is in where the properties hang.** *height* goes on **Adult Male** and *batting-average* on **Baseball Player**, because that is the class each sentence names. Hanging either off Pee-wee-Reese loses the inheritance the question is set to test.
- **Read the inheritance back** in one line: Pee-wee-Reese is a Fielder, so a Baseball Player, so an Adult Male, so a Person – and therefore **inherits height 5.10 and batting average 0.252 without either being stored on him.**
- 79 Ch: same seven arcs. Baseball Player → Cricket Player, 5.10 → 5'6", 0.252 → 37.59, Pee-wee-Reese → Rohit Paudel, Brooklyn Dodger → Nepal.

### Why this set is worth extra attention

- It is the only sentence set in the papers with **four levels of class**, so it is the one that shows inheritance properly.
- It is also **the set 81 Ba converts into a frame system** – see §5.N.10, where the finished frames are printed.
- **The textbook figure of this set is not the answer:** it collapses the Adult Male / Baseball Player / Fielder spine and uses other property arcs. Draw the seven arcs of the table, as the net below does.



### 5.N.3 Tom the cat: two sentence sets, three papers

#### Represent the following sentences using semantic net 2+5 2+6

PIN 3/41 82 Bh 2+5, 81 Ch 2+6, 79 Bh 2+6 · the most-repeated set in the chapter · 82 Bh and 81 Ch print it word for word identically; 79 Bh is a rewritten variant · all three pair it with a KR short answer, issues in KR twice and define semantic net once

QUESTION AS PRINTED 82 Bh Q5 2+5 · 81 Ch Q5 2+6 · the same eleven sentences

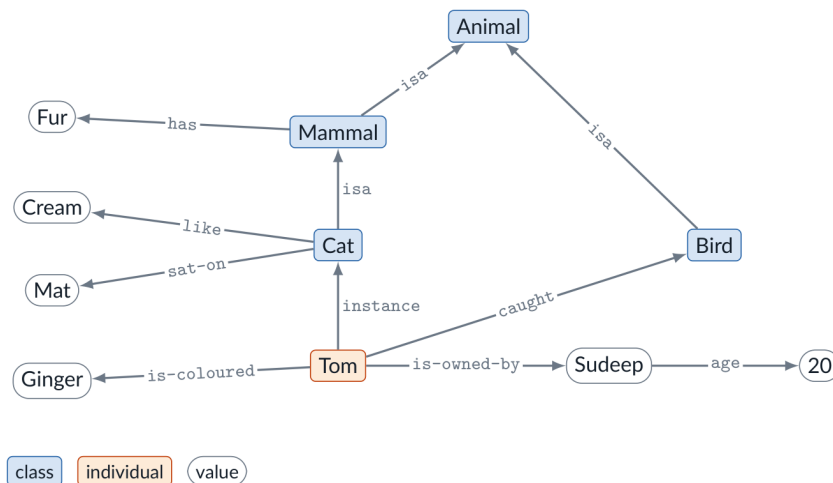
What are different issues in knowledge representation? Represent the following sentences using semantic net representation.

Tom is a cat. Tom caught a bird. Tom is owned by Sudeep. Tom is ginger in color. Cats like cream. The cat sat on the mat. A cat is a mammal. A bird is an animal. All mammals are animals. Mammals have fur. Sudeep age is 20.

#### The eleven arcs, in the order the sentences give them

| From   | Link        | To     |
|--------|-------------|--------|
| Tom    | instance    | Cat    |
| Tom    | caught      | Bird   |
| Tom    | is-owned-by | Sudeep |
| Tom    | is-coloured | Ginger |
| Cat    | like        | Cream  |
| Cat    | sat-on      | Mat    |
| Cat    | isa         | Mammal |
| Bird   | isa         | Animal |
| Mammal | isa         | Animal |
| Mammal | has         | Fur    |
| Sudeep | age         | 20     |

- 11 sentences, 11 arcs. The count checks.
- **BA Sir's slide for this set is not the answer:** it names the owner *John* and leaves out *Sudeep age is 20*.
- **Two traps, and both are worth a mark each:**
  - *Tom is a cat* is instance (Tom is an individual); *a cat is a mammal* is isa (both are classes). Same English construction, different link.
  - *The cat sat on the mat* is about **the class Cat** as the net below draws it, but the sentence plainly means *that cat*, ie Tom. **Draw it on Cat and add a line saying the natural reading is Tom** – the ambiguity is exactly the “lack of formality” limitation the paired theory question asks about.
- *Cats like cream* and *mammals have fur* are drawn out of the class nodes, so Tom inherits both. Say so.



#### The 2079 Bhadra variant: eleven different sentences, same shape 2+6

79 Bh 2+6 · counted in the chip above · the last sentence is the interesting one: it is a quantified statement, which a semantic net cannot properly express

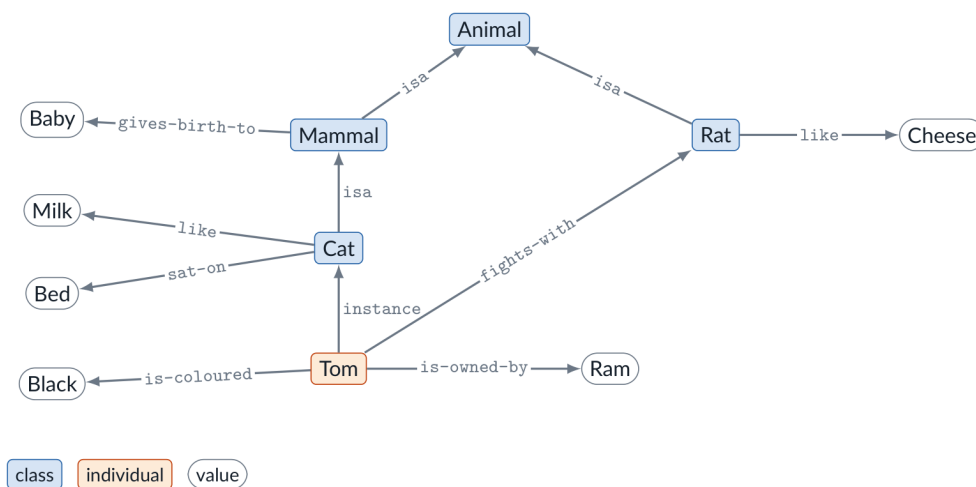
QUESTION AS PRINTED 79 Bh · Q6 · 2+6

List the issues need to be consider in Knowledge Representation techniques. Convert given sentences into Semantic Network.

- a) Tom is a cat   b) Tom fights with rat   c) Tom is owned by Ram
- d) Tom is black in color.   e) Cats like milk   f) Rats like cheese
- g) The cat sat on the bed   h) A cat is a mammal   i) A Rat is an animal
- j) All mammals are animals   k) Every mammal gives birth a baby

| From   | Link           | To     |
|--------|----------------|--------|
| Tom    | instance       | Cat    |
| Tom    | fight-with     | Rat    |
| Tom    | is-owned-by    | Ram    |
| Tom    | is-coloured    | Black  |
| Cat    | like           | Milk   |
| Rat    | like           | Cheese |
| Cat    | sat-on         | Bed    |
| Cat    | isa            | Mammal |
| Rat    | isa            | Animal |
| Mammal | isa            | Animal |
| Mammal | gives-birth-to | Baby   |

- **11 sentences, 11 arcs.** The count checks.
- Layout: *Animal* at the top, *Mammal* and *Rat* under it, *Cat* under *Mammal*, *Tom* at the bottom, and every value node (Milk, Cheese, Bed, Black, Ram, Baby) hanging sideways.
- **(k) is the one that earns the extra mark.** “Every mammal gives birth to a baby” is  $\forall x (x \in \text{Mammal} \Rightarrow \exists y (y \in \text{Baby} \wedge \text{gives-birth-to}(x, y)))$ .
  - The net can only draw **one arc** out of the *Mammal* class node.
  - **Write underneath:** “the arc out of a class node is read as ‘for every member’, and the existential ‘some baby’ is not expressible in a semantic net at all – this is the quantification limitation.”
- *Tom fights with rat* is between Tom and the **class** Rat as printed. Note that a fight is with one particular rat, so a strict reading needs an **event node** (§5.4 of the theory).



#### 5.N.4 Mammals, fur and water

Define term frame. Convert the following sentences into a semantic network **1+6**

**PIN 1/41** 82 Ka **1+6** · the set with three siblings under one class, so it is the one that rewards pushing a property up · the 1-mark opener is *define a frame*, §5.5 of the theory, one sentence

**QUESTION AS PRINTED** 82 Ka · Q5 · **1+6**

Define term frame. Convert the following sentences into semantic network. Cat is a mammal. Cat has fur. Mammal has vertebra. Mammal is an animal. Bear is a mammal. Bear has fur. Whale is a mammal. Whale lives in water. **Fish is an animal 10.** Fish lives in water.

The “10.” is the paper’s own stray list number printed inside the sentence. Reproduced as printed; it is not a tenth sentence.

| From   | Link     | To       |
|--------|----------|----------|
| Cat    | isa      | Mammal   |
| Bear   | isa      | Mammal   |
| Whale  | isa      | Mammal   |
| Mammal | isa      | Animal   |
| Fish   | isa      | Animal   |
| Cat    | has      | Fur      |
| Bear   | has      | Fur      |
| Mammal | has      | Vertebra |
| Whale  | lives-in | Water    |
| Fish   | lives-in | Water    |

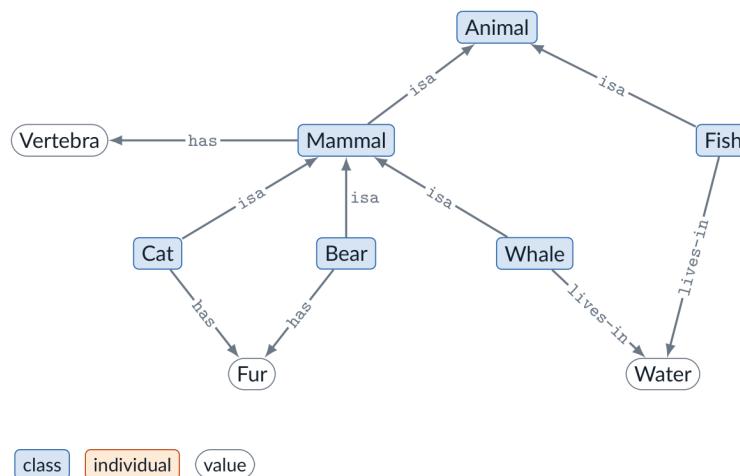
- **10 sentences, 10 arcs.** The count checks.
- **Every node here is a class**, so **every** link is `isa` or a property arc. There is not one `instance` arc in this set — no individual is named. Saying that in one line shows you understood the difference rather than guessed it.

### Layout

- *Animal* on the top row. *Mammal* and *Fish* on the second. *Cat*, *Bear*, *Whale* on the third. Values (*Fur*, *Vertebra*, *Water*) off to the right.
- *Water* is one node reached by two arcs, not two nodes. **A semantic net stores each object once**; drawing it twice is the mistake that costs a mark here.

### The observation that earns the last mark

- *Cat has fur* and *Bear has fur* are both given, and both are drawn because the question says to convert the sentences.
- But fur is a **mammal** property. Add a line: “*has fur* could be moved onto *Mammal*, and then *Cat*, *Bear* and *Whale* would all inherit it — that is the space saving inheritance buys.”
- **Then give the counter-example in the same set**: whales are mammals and do *not* have fur in the ordinary sense, so moving it up would make the net assert something false. **A net has no way to state an exception**, which is precisely why frames carry defaults and overrides.
- That pair of sentences is the best two-line answer to “*limitations of semantic nets*” anywhere in these papers, and it comes free with this question.



## 5.N.5 Fido the dog

Explain the approaches of KR. Highlight pros and cons of semantic nets and frames. Use them to represent the following sentences **2+6**

**PIN 1/41** 81 Bh **2+6** · the smallest set in the papers, six sentences · **it asks for both representations**, so draw the net *and* write the frames — the arc list below does both

**QUESTION AS PRINTED** 81 Bh · Q6 · **2+6**

Explain different approaches of knowledge representation. Highlight pros and cons of Semantic Nets and Frames. Use them to represent following sentences.

- **Fido is a dog.**
- Fido is own by Rohit.
- Fido is ginger in color.
- Dog is a mammal.
- All Mammals are animals.
- Mammals have fur.

The paper names the dog *Figo* in the first line and *Fido* in the next two. Reproduced as printed; take them as one animal and say so.

### As a semantic net

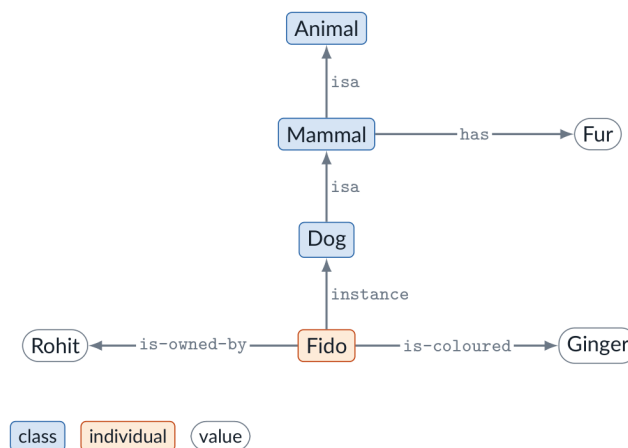
| From   | Link        | To     |
|--------|-------------|--------|
| Fido   | instance    | Dog    |
| Fido   | is-owned-by | Rohit  |
| Fido   | is-coloured | Ginger |
| Dog    | isa         | Mammal |
| Mammal | isa         | Animal |
| Mammal | has         | Fur    |

- **6 sentences, 6 arcs.** The count checks.
- Spine: *Animal* ← *Mammal* ← *Dog*, with *Fido* hung off *Dog*. *Rohit* and *Ginger* sideways.
- **Open the answer with one line on the name:** “the paper writes *Figo* once and *Fido* twice; taken as the same dog throughout.” It costs nothing and shows you read the question.
- *Mammals have fur* is on the class, so **Fido inherits fur** without it being stored on him. Write that sentence out – this set is small enough that the inheritance remark is where the marks are.

### The same knowledge as frames

| Frame         | Slot          | Filler                 |
|---------------|---------------|------------------------|
| <b>Animal</b> | (class frame) | —                      |
| <b>Mammal</b> | isa           | Animal                 |
|               | has           | Fur                    |
| <b>Dog</b>    | isa           | Mammal                 |
| <b>Fido</b>   | instance      | Dog                    |
|               | owner         | Rohit                  |
|               | colour        | Ginger                 |
|               | has           | <i>Fur (inherited)</i> |

- **One frame per node, one slot per outgoing arc**, exactly the recipe in §5.6 of the theory.
- **Write the inherited row in italics or brackets** and label it *inherited*. It shows the frame system doing the work, and it is the difference between copying the net out as a table and answering the question.
- **Pros and cons** (the first half of the question) are the two lists in §5.4 and §5.5 of the theory. One line each is enough at 2 marks.



## 5.N.6 Employees and supervisors

When do we use a semantic network? Convert the following sentences into a semantic network **2+6**

**PIN 1/41** 82 Ba **2+6** · **the trickiest set in the papers:** it contains a word that is both a class and a relation, and a symmetric relation · the 2-mark opener, *when do we use a semantic net*, is §5.4 of the theory

**QUESTION AS PRINTED** 82 Ba · Q6 · **2+6**

When do we use semantic network? Convert the following sentences into semantic network.

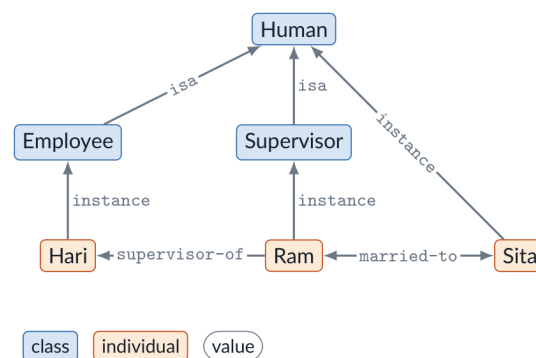
- i) Employee is a human.    ii) Hari is an employee.    iii) Sita is a human.
- iv) Supervisor is a human.    v) Ram is a supervisor.    vi) Ram is married to Sita.
- vii) Ram is supervisor of Hari.

| From       | Link          | To         |
|------------|---------------|------------|
| Employee   | isa           | Human      |
| Supervisor | isa           | Human      |
| Hari       | instance      | Employee   |
| Sita       | instance      | Human      |
| Ram        | instance      | Supervisor |
| Ram        | married-to    | Sita       |
| Ram        | supervisor-of | Hari       |

- **7 sentences, 7 arcs.** The count checks.
- Layout: *Human* at the top; *Employee* and *Supervisor* under it; *Hari*, *Ram*, *Sita* on the bottom row so the two arcs between individuals are horizontal and easy to read.

### Three things the examiner is testing here

- **1. Supervisor is used two ways in one question.**
  - In (iv) and (v) it is a **class node**, reached by *isa* and *instance*.
  - In (vii) *supervisor-of* is a **relation arc between two individuals**, Ram to Hari. **It is not the class node.**
  - Drawing an arc from Hari to the *Supervisor* class is the mistake this set is built to catch.
- **2. Sita is an instance of Human directly**, not of Employee or Supervisor. The sentences never say what she does, so do not invent a class for her. **Adding information the sentences do not give loses marks**, however sensible it looks.
- **3. married-to is symmetric.** A semantic net arc is directed, so either draw a **double-headed** arrow or add the inverse arc *Sita married-to* → *Ram*, and say which convention you used.
  - This is the “relationships among attributes” issue from §5.3: inverse and symmetric relations have to be handled explicitly, because the formalism does not know they exist.



## 5.N.7 FOPL first, then the net

### Convert the following sentences into FOPL and represent using a semantic network 2+6

**PIN 1/41** 81 Ash 2+6 · **the only set that asks for FOPL as well as the drawing**, so it is worth doing every set this way in practice · the FOPL is free marks: it is the arc list written as predicates

#### QUESTION AS PRINTED 81 Ash · Q6 · 2+6

What are Frames and Semantic Network? Convert the following sentences into FOPL and represent using semantic network.

- i) Ram is a person.   ii) Person is an animal.   iii) Ram likes Jhilke.   iv) Jhilke is a cat.
- v) Cat eats fish.   vi) Cat is an animal.   vii) fish is an animal.

## Arc list and FOPL, side by side

| From   | Link     | To     | FOPL                  |
|--------|----------|--------|-----------------------|
| Ram    | instance | Person | instance(Ram, Person) |
| Person | isa      | Animal | isa(Person, Animal)   |
| Ram    | likes    | Jhilke | likes(Ram, Jhilke)    |
| Jhilke | instance | Cat    | instance(Jhilke, Cat) |
| Cat    | eats     | Fish   | see below             |
| Cat    | isa      | Animal | isa(Cat, Animal)      |
| Fish   | isa      | Animal | isa(Fish, Animal)     |

- 7 sentences, 7 arcs. The count checks.
- **The rule for turning an arc into FOPL:** an arc  $A \xrightarrow{R} B$  becomes the binary predicate  $R(A, B)$ , with `isa` used between classes and `instance` between an individual and its class. Nothing else is needed for six of the seven.

## Sentence (v) is the one worth thinking about

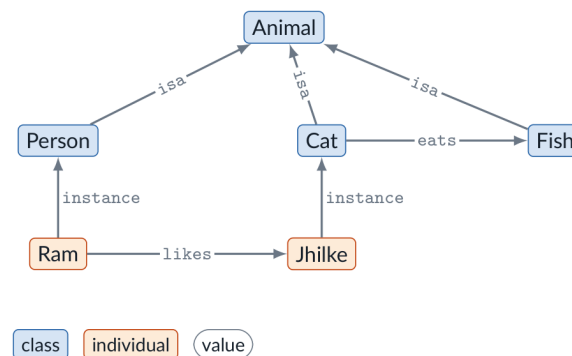
- “Cat eats fish” is a statement about **every** cat and **some** fish, so its honest FOPL is

$$\forall x (Cat(x) \Rightarrow \exists y (Fish(y) \wedge eats(x, y)))$$

- The net can only draw **one arc**, `Cat eats → Fish`, class to class.
- **Write both, and say what is lost:** the arc records the relation but throws away the quantifiers. That single remark is the entire “a semantic net cannot express quantification” limitation, evidenced on the examiner’s own data.
- If you want the flat version for the arc table, `eats(Cat, Fish)` is what every deck writes and is accepted.

## Layout

- `Animal` on top, with **three** `isa` arcs into it, from `Person`, `Cat` and `Fish`.
- `Ram` under `Person`, `Jhilke` under `Cat`, and the `likes` arc running horizontally between the two individuals.
- **The net has a nice shape here:** two parallel individual-to-class chains joined at the top by `Animal` and at the bottom by `likes`. Draw it that way and the seven arcs never cross.



## 5.N.8 Birds: Tweety and Sweety

## What are frames and semantic nets? Convert the given sentences into a semantic net 2+6

**PIN 1/41** 77 Ch 2+6 · two sentences here need more than one arc each, which is what makes this set harder than it looks · the same paper asks conceptual dependency and scripts at Q 8, so §5.7 of the theory is on the same sheet

## QUESTION AS PRINTED 77 Ch · Q6 · 2+6

What are Frames and Semantic Net? Convert the given sentences in semantic Net:

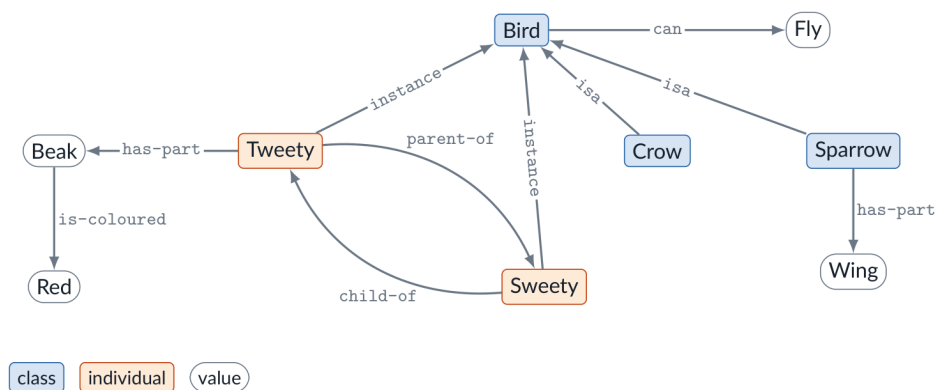
- Tweety and Sweety are birds. Tweety has a red beak. Sweety is Tweety’s child.
- A crow is a bird. Birds can fly. Sparrow is a bird. Sparrow has a wing.

| From    | Link        | To               |
|---------|-------------|------------------|
| Tweety  | instance    | Bird             |
| Sweety  | instance    | Bird             |
| Tweety  | has-part    | Beak             |
| Beak    | is-coloured | Red              |
| Sweety  | child-of    | Tweety           |
| Tweety  | parent-of   | Sweety (inverse) |
| Crow    | isa         | Bird             |
| Sparrow | isa         | Bird             |
| Bird    | can         | Fly              |
| Sparrow | has-part    | Wing             |

### Notes to write beside the net

- **Note to add:** *Sparrow has a wing* is drawn as the paper gives it, on *Sparrow*, but wings are a property of **all** birds – moving *has-part* → *Wing* up to the *Bird* class would let Tweety, Sweety, Crow and Sparrow inherit it from one arc.
- *Birds can fly* is on the class, so every bird inherits it. That is also the standard **exception** example: a penguin is a bird and cannot fly, and a semantic net has no way to override an inherited property. Frames do, with a default.

- **7 sentences, 10 arcs.** Here the count does *not* match one to one, and that is the point:
  - *Tweety and Sweety are birds* is **two** *instance* arcs, one per bird.
  - *Tweety has a red beak* is **two** arcs: the beak is a part of Tweety, and *red* is a property of the **beak**, not of Tweety. **Drawing one arc *Tweety red-beak* → *Red loses the mark*** – the colour belongs to the part.
  - *Sweety is Tweety's child* is one arc, but the **inverse** arc is worth drawing and labelling.



### 5.N.9 A semantic net about the AI course itself

Draw a sample net about your course AI to demonstrate the features and qualities of a semantic net 7

PIN 1/41 72 Ma 7 · **the only open-ended drawing question in the papers:** the content is yours, the marks are for demonstrating the *features* · so build the net so that each feature appears at least once, and label them

QUESTION AS PRINTED 72 Ma · Q5 · 7

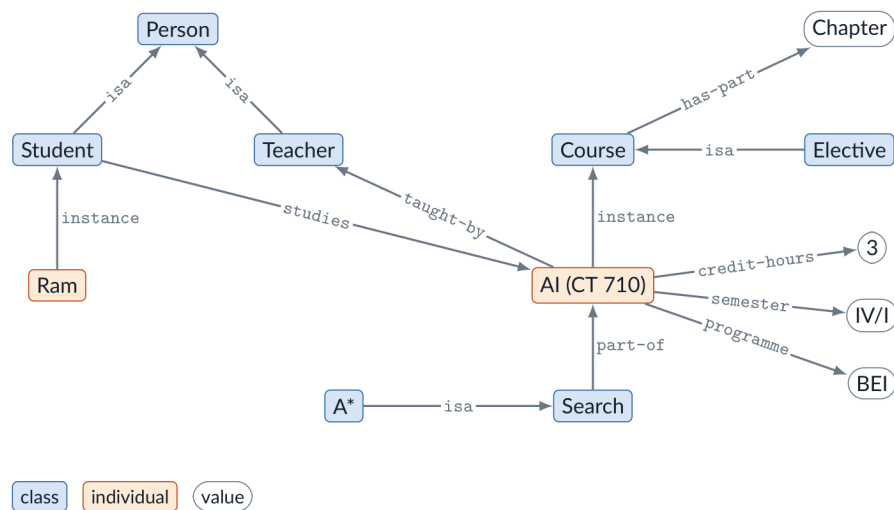
What is a semantic net? Draw a sample net about your course AI to demonstrate the knowledge representation features and qualities of semantic net.

## A net that hits every feature

| From        | Link         | To          |
|-------------|--------------|-------------|
| Elective    | isa          | Course      |
| AI (CT 710) | instance     | Course      |
| Course      | has-part     | Chapter     |
| AI (CT 710) | credit-hours | 3           |
| AI (CT 710) | semester     | IV/I        |
| AI (CT 710) | programme    | BEI         |
| Search      | part-of      | AI (CT 710) |
| A*          | isa          | Search      |
| Student     | studies      | AI (CT 710) |
| Ram         | instance     | Student     |
| AI (CT 710) | taught-by    | Teacher     |
| Teacher     | isa          | Person      |
| Student     | isa          | Person      |

## Label the features on the drawing, one arrow each

- **Class hierarchy** –  $Person \leftarrow Student / Teacher$ , joined by *isa*.
- **Instances** –  $Ram \text{ instance } \rightarrow Student$ . Point out that *isa* and *instance* are different links.
- **Inheritance** – anything put on *Person* is available to *Ram* through two arcs. Write the chain out.
- **Part-whole** –  $Search \text{ part-of } \rightarrow AI$ , which does **not** inherit the way *isa* does. Saying that is worth a mark on its own.
- **Attribute arcs** – *credit-hours*, *semester*, *programme*: values rather than objects.
- **Intersection search** – start from *Ram* and from *A\** and the two wavefronts meet at *AI (CT 710)*, which answers “what connects *Ram* and *A\**?” without anyone having asked that question in advance.
- **Close on the limitation**, because the question says *qualities*: nothing in this net can say “no student takes more than six electives” or “AI is not a lab course”. No quantifiers, no negation.



## 5.N.10 Converting a semantic net into a frame

Compare semantic net and frame. How do you convert a semantic net into a frame? Explain with example 3+5

PIN 1/41 81 Ba 3+5 · the comparison table is §5.6 of the theory · worked here on the Pee-wee-Reese net of §5.N.2, because that is the set the textbook itself converts, and its finished frames are a published figure

QUESTION AS PRINTED 81 Ba · Q6 · 3+5

Compare semantic net and frame. How you convert semantic net into frame? Explain with example.

### The five steps, applied

- **1. Every node becomes a frame.** The net of §5.N.2 has seven nodes, so the frame system has seven frames: *Person*, *Adult-Male*, *ML-Baseball-Player*, *Fielder*, *Pee-Wee-Reese*, *ML-Baseball-Team*, *Brooklyn-Dodgers*.
- **2. Every outgoing arc becomes a slot.** Arc label  $\rightarrow$  slot name, arc target  $\rightarrow$  filler. *Adult-Male* height  $\rightarrow$  5-10 becomes a height slot in the *Adult-Male* frame with filler 5-10.
- **3. isa and instance become the class link**, not ordinary slots. *Fielder* gets  
isa: *ML-Baseball-Player* and is a subclass frame; *Pee-Wee-Reese* gets instance: *Fielder* and is an instance frame.
- **4. A filler that has its own frame stays a pointer.** *Pee-Wee-Reese* team  $\rightarrow$  *Brooklyn-Dodgers* is a link to that frame, not the string. **This is what makes the result a frame system rather than seven separate tables.**
- **5. Push each property to the highest class that has it**, and let everything below inherit. height sits on *Adult-Male*, batting-average on *ML-Baseball-Player*, and *Pee-Wee-Reese* overrides both with his own values.

### Then check, and say you checked

- **No information may be lost.** Count the arcs in the net, count the filled slots in the frames. They must agree.
- **What the frames gained:** cardinality (6,000,000,000 persons), defaults (handed: Right), and slots marked with \* for values that belong to the class rather than to each member. None of those can be drawn in a net.

### 5.N.11 Semantic net and frame with FOPL statements

#### Why are semantic networks and frames important in AI? Provide examples of both with FOPL statements 2+6

PIN 1/41 75 Ba 2+6 · three formalisms in one answer — net, frame and FOPL, all saying the same thing · the 2-mark opener is the why they matter list in §5.6 of the theory

#### QUESTION AS PRINTED 75 Ba · Q6 · 2+6

Why semantic network and frames are important in AI? Provide examples of both with FOPL statements example.

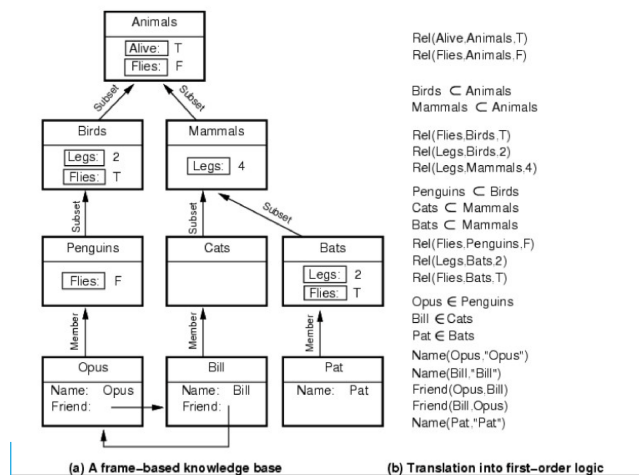
|                           |                              |
|---------------------------|------------------------------|
| <i>Person</i>             |                              |
| isa :                     | <i>Mammal</i>                |
| cardinality :             | 6,000,000,000                |
| * handed :                | Right                        |
| <i>Adult-Male</i>         |                              |
| isa :                     | <i>Person</i>                |
| card. ty :                | 2,000,000,000                |
| * height :                | 5-10                         |
| <i>ML-Baseball-Player</i> |                              |
| isa :                     | <i>Adult-Male</i>            |
| cardinality :             | 624                          |
| * height :                | 6-1                          |
| * bats :                  | equal to handed              |
| * batting-average :       | .252                         |
| * team :                  |                              |
| * uniform-color :         |                              |
| <i>Fielder</i>            |                              |
| isa :                     | <i>ML-Baseball-Player</i>    |
| cardinality :             | 376                          |
| * batting-average :       | .262                         |
| <i>Pee-Wee-Reese</i>      |                              |
| instance :                | <i>Fielder</i>               |
| height :                  | 5-10                         |
| bats :                    | Right                        |
| batting-average :         | .309                         |
| team :                    | <i>Brooklyn-Dodgers</i>      |
| uniform-color :           | Blue                         |
| <i>ML-Baseball-Team</i>   |                              |
| isa :                     | <i>Team</i>                  |
| cardinality :             | 26                           |
| * team-size :             | 24                           |
| * manager :               |                              |
| <i>Brooklyn-Dodgers</i>   |                              |
| instance :                | <i>ML-Baseball-Team</i>      |
| team-size :               | 24                           |
| manager :                 | Leo-Durocher                 |
| players :                 | { <i>Pee-Wee-Reese</i> ,...} |

Fig. 9.5 A Simplified Frame System

**The net of §5.N.2 after conversion.** Seven frames, each slot either an inherited class link, a value, or a pointer to another frame. The \* marks a value stored on the class. Compare with `c5_semnet_peewee` above: same knowledge, different formalism. Source: PS Sir ch5, after Rich and Knight Fig 9.5.

#### The reverse direction, if asked

- Frame  $\rightarrow$  net is step 1 to 4 run backwards: one node per frame, one arc per filled slot, and isa / instance arcs from the class links.
- **The conversion is lossy in this direction.** Defaults, cardinalities, type restrictions and demons have no arc to become, so they are simply dropped. Say so — it is the cleanest one-line proof that a frame carries more than a net.



(a) the frame-based knowledge base and (b) the first-order sentences that say the same thing. The semantic net is the same picture with the boxes redrawn as nodes and the **Subset** / **Member** labels put on the arcs. Source: PS Sir ch5.

### The three-way translation table to reproduce

| Net arc                                | Frame                     | FOPL                                      |
|----------------------------------------|---------------------------|-------------------------------------------|
| $A \xrightarrow{isa} B$                | isa: B in frame A         | $\text{Subset}(A, B)$ , ie $A \subset B$  |
| $A \xrightarrow{instance} B$           | instance: B in frame A    | $\text{Member}(A, B)$ , ie $A \in B$      |
| $A \xrightarrow{R} v$ on an individual | slot R with filler v      | $\text{Rel}(R, A, v)$ , ie $R(A, v)$      |
| $A \xrightarrow{R} v$ on a class       | slot R on the class frame | $\forall x (x \in A \Rightarrow R(x, v))$ |

- Worked on the figure's own data:  $\text{Subset}(\text{Birds}, \text{Animals})$ ,  $\text{Subset}(\text{Mammals}, \text{Animals})$ ,  $\text{Member}(\text{Opus}, \text{Penguins})$ ,  $\text{Rel}(\text{Legs}, \text{Birds}, 2)$ ,  $\text{Rel}(\text{Legs}, \text{Mammals}, 4)$ ,  $\text{Rel}(\text{Flies}, \text{Penguins}, F)$ ,  $\text{Friend}(\text{Opus}, \text{Bill})$ .
- The answer structure that scores:** draw the small net, write the same knowledge as two or three frames beside it, then write one FOPL line per arc underneath. Three formalisms, one set of facts, and the marks are for showing they are interchangeable.
- Close with what FOPL has that neither picture does:** quantifiers, negation and disjunction – and what the pictures have that FOPL does not: inheritance for free, defaults, and a shape a human can check by eye. That is the “why they are important in AI” half of the question.

**Coverage.** All 12 papers that set a sentence-set conversion are worked above: 72 Ma, 78 Po (§5.N.1); 73 Bh, 79 Ch (§5.N.2); 82 Bh, 81 Ch, 79 Bh (§5.N.3); 82 Ka (§5.N.4); 81 Bh (§5.N.5); 82 Ba (§5.N.6); 81 Ash (§5.N.7); 77 Ch (§5.N.8). §5.N.9 to §5.N.11 work the three remaining drawing questions: the open-ended net 72 Ma, the net-to-frame conversion 81 Ba, and the net/frame/FOPL question 75 Ba. · **Three papers print a defect in their own sentence list**, all reproduced above as printed and all flagged: 78 Po names Sakti Gauchan in a Sandeep Lamichane set, 82 Ka leaves a stray “10.” inside a sentence, and 81 Bh calls the dog Figo once and Fido twice. · **Nothing in this file is machine-checkable:** a semantic net has no arithmetic to recompute, so every arc list above was checked by hand against the OCR archive, sentence by sentence, and the arc counts are printed with each answer so the reader can check them too.

## 6. Machine Learning – Solved Problems

Every problem opens with the question **exactly as the paper prints it**, data table and all, so it can be attempted from this document alone. · Three ID3 data tables over six papers, three genetic-algorithm runs, and the fuzzy arithmetic. · Every entropy, gain, fitness value and membership number printed here is recomputed independently by `verify.py`; 242 checks pass across the subject.

### Two things to know before you start.

- **The two “play” tables have different answers.** They look the same – fourteen rows, same four attributes, nine Yes and five No. But 79 Jth changes four rows, and that moves the root attribute from **Outlook** to **Windy**. Copying the golf answer into the cricket question loses the whole 7 marks. Problems 1.1 and 1.2 are printed side by side for exactly this reason.
- 75 Bh **prints no chromosomes**. The question sets up the fitness function, says “the initial population consist of four individuals with the following chromosomes” – and then the page runs on to Q7. Problem 2.2 reproduces the defect as printed and works the standard four chromosomes that the textbook version of this question carries.

### 6.N.1 Decision trees: the method, used identically in all four problems

#### The recipe

- **Step 1.** Count the classes in the whole set  $S$ . Write  $P$  (positives) and  $N$  (negatives), then  $H(S) = -\frac{P}{P+N} \log_2 \frac{P}{P+N} - \frac{N}{P+N} \log_2 \frac{N}{P+N}$ .
- **Step 2.** For each attribute, split  $S$  by its values and fill *one row per value* of the table below: the counts, that branch’s entropy, and the branch weighted by its share of  $S$ .
- **Step 3.** Sum the weighted column. That is the **residual entropy** after splitting on that attribute.
- **Step 4.**  $\text{Gain} = H(S) - \text{residual}$ . Compare the gains; the largest wins.
- **Step 5.** Recurse on each branch that is not already pure.

#### The table idiom, used unchanged in every problem below

| Attribute | Value | $P$ | $N$ | $ S_v $ | $H(S_v)$ | $\frac{ S_v }{ S } H(S_v)$ |
|-----------|-------|-----|-----|---------|----------|----------------------------|
|-----------|-------|-----|-----|---------|----------|----------------------------|

#### Two logarithms are all you ever need

|                                                                                      | Split $P:N$ | 1:1    | 1:2 or 2:1 | 1:3 or 3:1 |
|--------------------------------------------------------------------------------------|-------------|--------|------------|------------|
| $\log_2 x = \frac{\ln x}{\ln 2}$ , and these five cover every split in this chapter. | $H$         | 1.0000 | 0.9183     | 0.8113     |

Also  $H(1:6) = 0.5917$ ,  $H(2:5) = 0.8631$ ,  $H(5:9) = 0.9403$ ,  $H(3:5) = 0.9544$ , and  $H = 0$  whenever one side is empty.

### 6.N.2 Problem 1.1 – the “play golf” table (3 papers)

#### Select the root attribute of the decision tree from the given sample data 8 2+7

PIN 3/41 74 Bh 2+7, 74 Ma 8, 72 Ash 8 · all three print the *same* fourteen rows · 74 Bh asks for the ID3 approach first, which is \$6.5

#### QUESTION AS PRINTED 74 Bh · Q4 · 2+7

State the approach for learning using ID3 and Select the root attribute of the decision-tree from given sample data.

| Outlook  | Temperature | Humidity | Windy | Play golf |
|----------|-------------|----------|-------|-----------|
| Rainy    | Hot         | High     | False | No        |
| Rainy    | Hot         | High     | True  | No        |
| Overcast | Hot         | High     | False | Yes       |
| Sunny    | Mild        | High     | False | Yes       |
| Sunny    | Cool        | Normal   | False | Yes       |
| Sunny    | Cool        | Normal   | True  | No        |
| Overcast | Cool        | Normal   | True  | Yes       |
| Rainy    | Mild        | High     | False | No        |
| Rainy    | Cool        | Normal   | False | Yes       |
| Sunny    | Mild        | Normal   | False | Yes       |
| Rainy    | Mild        | Normal   | True  | Yes       |
| Overcast | Mild        | High     | True  | Yes       |
| Overcast | Hot         | Normal   | False | Yes       |
| Sunny    | Mild        | High     | True  | No        |

#### Step 1 Entropy of the whole set

14 rows: 9 Yes, 5 No.

$$H(S) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.4098 + 0.5305 = \mathbf{0.9403}$$

### Steps 2 and 3 Split on each attribute in turn

| Attribute         | Value    | P | N | S <sub>v</sub> | H(S <sub>v</sub> ) | $\frac{ S_v }{14} H(S_v)$              |
|-------------------|----------|---|---|----------------|--------------------|----------------------------------------|
| Outlook           | Rainy    | 2 | 3 | 5              | 0.9710             | $\frac{5}{14}(0.9710) = 0.3468$        |
|                   | Overcast | 4 | 0 | 4              | <b>0</b>           | 0                                      |
|                   | Sunny    | 3 | 2 | 5              | 0.9710             | $\frac{5}{14}(0.9710) = 0.3468$        |
| residual = 0.6935 |          |   |   |                |                    | <b>Gain = 0.9403 – 0.6935 = 0.2467</b> |
| Temperature       | Hot      | 2 | 2 | 4              | 1.0000             | 0.2857                                 |
|                   | Mild     | 4 | 2 | 6              | 0.9183             | 0.3936                                 |
|                   | Cool     | 3 | 1 | 4              | 0.8113             | 0.2318                                 |
| residual = 0.9111 |          |   |   |                |                    | Gain = 0.0292                          |
| Humidity          | High     | 3 | 4 | 7              | 0.9852             | 0.4926                                 |
|                   | Normal   | 6 | 1 | 7              | 0.5917             | 0.2958                                 |
| residual = 0.7885 |          |   |   |                |                    | Gain = 0.1518                          |
| Windy             | False    | 6 | 2 | 8              | 0.8113             | 0.4636                                 |
|                   | True     | 3 | 3 | 6              | 1.0000             | 0.4286                                 |
| residual = 0.8922 |          |   |   |                |                    | Gain = 0.0481                          |

### Step 4 Compare

| Attribute      | Gain          |
|----------------|---------------|
| <b>Outlook</b> | <b>0.2467</b> |
| Humidity       | 0.1518        |
| Windy          | 0.0481        |
| Temperature    | 0.0292        |

**Answer.** The root attribute is **Outlook**, with the highest information gain, **0.2467**.

- **Why, in words:** Outlook is the only attribute with a **pure branch** – all four Overcast days are Yes – so it removes more uncertainty than any other. Temperature removes almost none: its three branches are as mixed as the parent.

### Step 5 Finish the tree (not asked, but it is two more lines and it proves the root was right)

- **Overcast branch** – 4 rows, all Yes. **Leaf: Yes.**
- **Rainy branch** – 5 rows, 2 Yes 3 No,  $H = 0.9710$ .
  - Humidity: High → 0 Yes, 3 No ( $H = 0$ ); Normal → 2 Yes, 0 No ( $H = 0$ ). Residual 0, **Gain = 0.9710**.
  - Windy gives 0.0200, Temperature 0.5710. So **Humidity** splits it perfectly: High → No, Normal → Yes.
- **Sunny branch** – 5 rows, 3 Yes 2 No,  $H = 0.9710$ .
  - Windy: False → 3 Yes, 0 No; True → 0 Yes, 2 No. Residual 0, **Gain = 0.9710**. So **Windy**: False → Yes, True → No.
- **The tree**, three levels and five leaves:
  - Outlook = Overcast → Yes
  - Outlook = Rainy, Humidity = High → No
  - Outlook = Rainy, Humidity = Normal → Yes
  - Outlook = Sunny, Windy = False → Yes
  - Outlook = Sunny, Windy = True → No
- Temperature never appears. **ID3 has decided it is irrelevant**, which is the feature-selection side of the algorithm.

## 6.N.3 Problem 1.2 – the “play cricket” table, whose answer is different

How is the best attribute selected in a decision tree? Select the root attribute from the given sample data **1+7**

79 Jth **1+7** · four rows differ from Problem 1.1 and the answer changes · rows 1 and 11 carry a different label, row 8 a different Windy value, and the target is renamed *Play cricket*

QUESTION AS PRINTED 79 Jth · Q8b · **1+7**

How best attribute is selected in a decision-tree? Select the root attribute of the decision-tree from given sample data.

| Outlook  | Temperature | Humidity | Windy | Play cricket |
|----------|-------------|----------|-------|--------------|
| Rainy    | Hot         | High     | False | Yes          |
| Rainy    | Hot         | High     | True  | No           |
| Overcast | Hot         | High     | False | Yes          |
| Sunny    | Mild        | High     | False | Yes          |
| Sunny    | Cool        | Normal   | False | Yes          |
| Sunny    | Cool        | Normal   | True  | No           |
| Overcast | Cool        | Normal   | True  | Yes          |
| Rainy    | Mild        | High     | True  | No           |
| Rainy    | Cool        | Normal   | False | Yes          |
| Sunny    | Mild        | Normal   | False | Yes          |
| Rainy    | Mild        | Normal   | True  | No           |
| Overcast | Mild        | High     | True  | Yes          |
| Overcast | Hot         | Normal   | False | Yes          |
| Sunny    | Mild        | High     | True  | No           |

### Step 1 Entropy of the whole set

Still 9 Yes and 5 No out of 14, so  $H(S) = 0.9403$  again – which is exactly why the table looks identical.

### Steps 2 and 3

| Attribute         | Value    | P | N | $ S_v $ | $H(S_v)$ | $\frac{ S_v }{14} H(S_v)$       |
|-------------------|----------|---|---|---------|----------|---------------------------------|
| Windy             | False    | 7 | 0 | 7       | 0        | 0                               |
|                   | True     | 2 | 5 | 7       | 0.8631   | $\frac{7}{14}(0.8631) = 0.4316$ |
| residual = 0.4316 |          |   |   |         |          | Gain = 0.9403 – 0.4316 = 0.5087 |
| Outlook           | Rainy    | 2 | 3 | 5       | 0.9710   | 0.3468                          |
|                   | Overcast | 4 | 0 | 4       | 0        | 0                               |
|                   | Sunny    | 3 | 2 | 5       | 0.9710   | 0.3468                          |
| residual = 0.6935 |          |   |   |         |          | Gain = 0.2467                   |
| Temperature       | Hot      | 3 | 1 | 4       | 0.8113   | 0.2318                          |
|                   | Mild     | 3 | 3 | 6       | 1.0000   | 0.4286                          |
|                   | Cool     | 3 | 1 | 4       | 0.8113   | 0.2318                          |
| residual = 0.8922 |          |   |   |         |          | Gain = 0.0481                   |
| Humidity          | High     | 4 | 3 | 7       | 0.9852   | 0.4926                          |
|                   | Normal   | 5 | 2 | 7       | 0.8631   | 0.4316                          |
| residual = 0.9242 |          |   |   |         |          | Gain = 0.0161                   |

**Answer.** The root attribute is **Windy**, gain **0.5087** – more than twice Outlook's 0.2467.

- **Compare with Problem 1.1, where Windy was the second worst attribute at 0.0481.** Three changed labels turned it into the best one.
- **Why:** in this table **every one of the seven non-windy days is a Yes**. One branch of Windy is completely pure and holds half the data, which is the largest single reduction in uncertainty available.
- **The lesson worth writing in the answer:** never carry a remembered root across from a similar-looking table. Recompute.

### Finishing the tree (it is only two levels deep)

- **Windy = False** – 7 rows, all Yes. **Leaf: Yes.**
- **Windy = True** – 7 rows, 2 Yes 5 No,  $H = 0.8631$ . Split on Outlook:
  - Rainy → 3 rows, all No ( $H = 0$ ).
  - Sunny → 2 rows, all No ( $H = 0$ ).
  - Overcast → 2 rows, all Yes ( $H = 0$ ).
  - Residual 0, so **Gain = 0.8631**: a perfect split.
- **The tree**, four leaves and only two tests:
  - Windy = False → Yes
  - Windy = True, Outlook = Overcast → Yes
  - Windy = True, Outlook = Rainy → No
  - Windy = True, Outlook = Sunny → No
- Neither Temperature nor Humidity is ever tested. The cricket tree is **smaller than the golf tree** – two tests against three – which is what a higher root gain buys you.

## 6.N.4 Problem 1.3 – the mushroom island

### Which attribute should you choose as the root of a decision tree? Give reason 8

77 Ch 8 · the attributes are already 0/1, so no entropy is wasted on multi-way splits · **only A to H are training data**; U, V and W carry ? in the Edible column and are the cases to be classified afterwards

QUESTION AS PRINTED 77 Ch · Q7 · 8

You are locked on a deserted island. Mushrooms of various types grow widely all over the island but no other food is anywhere to be found. Some of the mushrooms have been determined as poisonous and others as not (Determined by your former companions' trial and error). You are the only one remaining on the island. You have the following data to consider;

| Example | NotHeavy | Smelly | Spotted | Smooth | Edible |
|---------|----------|--------|---------|--------|--------|
| A       | 1        | 0      | 0       | 0      | 1      |
| B       | 1        | 0      | 1       | 0      | 1      |
| C       | 0        | 1      | 0       | 1      | 1      |
| D       | 0        | 0      | 0       | 1      | 0      |
| E       | 1        | 1      | 1       | 0      | 0      |
| F       | 1        | 0      | 1       | 1      | 0      |
| G       | 1        | 0      | 0       | 1      | 0      |
| H       | 0        | 1      | 0       | 0      | 0      |
| U       | 0        | 1      | 1       | 1      | ?      |
| V       | 1        | 1      | 0       | 1      | ?      |
| W       | 1        | 1      | 0       | 0      | ?      |

You know whether or not mushroom A through H are poisonous, but you do not know about U through W. Which attributes should you choose as the root of a decision tree? Give reason.

The paper's own wording, slips included: "the only one remaining" for *the only one remaining*.

#### Step 1 Entropy of the training set

Rows A–H only. Edible = 1 for A, B, C →  $P = 3$ ; Edible = 0 for D, E, F, G, H →  $N = 5$ .

$$H(S) = -\frac{3}{8} \log_2 \frac{3}{8} - \frac{5}{8} \log_2 \frac{5}{8} = 0.5306 + 0.4238 = \mathbf{0.9544}$$

#### Steps 2 and 3 All four attributes

| Attribute         | Value (rows)  | $P$ | $N$ | $ S_v $ | $H(S_v)$ | $\frac{ S_v }{8} H(S_v)$ |
|-------------------|---------------|-----|-----|---------|----------|--------------------------|
| Smooth            | 1 (C,D,F,G)   | 1   | 3   | 4       | 0.8113   | 0.4056                   |
|                   | 0 (A,B,E,H)   | 2   | 2   | 4       | 1.0000   | 0.5000                   |
| residual = 0.9056 |               |     |     |         |          | Gain = 0.0488            |
| NotHeavy          | 1 (A,B,E,F,G) | 2   | 3   | 5       | 0.9710   | 0.6069                   |
|                   | 0 (C,D,H)     | 1   | 2   | 3       | 0.9183   | 0.3444                   |
| residual = 0.9512 |               |     |     |         |          | Gain = 0.0032            |
| Smelly            | 1 (C,E,H)     | 1   | 2   | 3       | 0.9183   | 0.3444                   |
|                   | 0 (A,B,D,F,G) | 2   | 3   | 5       | 0.9710   | 0.6069                   |
| residual = 0.9512 |               |     |     |         |          | Gain = 0.0032            |
| Spotted           | 1 (B,E,F)     | 1   | 2   | 3       | 0.9183   | 0.3444                   |
|                   | 0 (A,C,D,G,H) | 2   | 3   | 5       | 0.9710   | 0.6069                   |
| residual = 0.9512 |               |     |     |         |          | Gain = 0.0032            |

**Answer.** Choose **Smooth** as the root: gain **0.0488**, fifteen times the gain of any other attribute.

### The reason the question asks for

- Smooth is the only attribute that produces an **uneven** class split: of the four smooth mushrooms only one is edible (3:1), while the four non-smooth ones are an even 2:2.
- The other three attributes split the data identically** – each one puts 1 edible and 2 inedible on one side and 2 edible and 3 inedible on the other, so all three have exactly the same entropy and exactly the same gain, 0.0032. Say this: it is the observation the examiner is looking for, and it means the choice between those three is arbitrary.

### Two things worth adding for the last marks

- No attribute here is much use.** The best gain is 0.0488 against a starting entropy of 0.9544, so the root removes only about 5 % of the uncertainty. With eight examples and four binary attributes the data simply does not determine a good tree – and on this question that is worth saying, because the story is about eating something poisonous.
- Classifying U, V and W**, which the story implies even though the question stops at the root:
  - U: Smooth = 1 → the 3:1 inedible branch → predict **not edible**.
  - V: Smooth = 1 → predict **not edible**.
  - W: Smooth = 0 → the 2:2 branch, **a tie** – the tree cannot decide, and a majority-class leaf would be a coin toss.
- So the honest answer ends: **do not eat W**. A model this uncertain should abstain, not guess.

## 6.N.5 Problem 1.4 – the profit table (extra practice, not a PYQ)

**PROBLEM** not a PYQ · *Insights* Example 6.4, the worked example the textbook uses

Build a decision tree for the following data using ID3.

| Age | Competition | Type     | Profit |
|-----|-------------|----------|--------|
| Old | Yes         | Software | Down   |
| Old | No          | Software | Down   |
| Old | No          | Hardware | Down   |
| Mid | Yes         | Software | Down   |
| Mid | Yes         | Hardware | Down   |
| Mid | No          | Hardware | Up     |
| Mid | No          | Software | Up     |
| New | Yes         | Software | Up     |
| New | No          | Hardware | Up     |
| New | No          | Software | Up     |

### Level 1

10 rows, 5 Up and 5 Down, so  $H(S) = 1.0000$  exactly – the easiest possible starting point.

| Attribute | Value | P | N | $ S_v $ | $H(S_v)$ | $\frac{ S_v }{10} H$ |
|-----------|-------|---|---|---------|----------|----------------------|
| Age       | Old   | 0 | 3 | 3       | <b>0</b> | 0                    |
|           | Mid   | 2 | 2 | 4       | 1.0000   | 0.4000               |
|           | New   | 3 | 0 | 3       | <b>0</b> | 0                    |

residual = 0.4000 **Gain = 0.6000**

|             |     |   |   |   |        |        |
|-------------|-----|---|---|---|--------|--------|
| Competition | Yes | 1 | 3 | 4 | 0.8113 | 0.3245 |
|             | No  | 4 | 2 | 6 | 0.9183 | 0.5510 |

residual = 0.8755 **Gain = 0.1245**

|      |          |   |   |   |        |        |
|------|----------|---|---|---|--------|--------|
| Type | Software | 3 | 3 | 6 | 1.0000 | 0.6000 |
|      | Hardware | 2 | 2 | 4 | 1.0000 | 0.4000 |

residual = 1.0000 **Gain = 0**

- Type has gain exactly zero:** both its branches are split 50/50, the same as the parent, so it tells you nothing at all. That is the cleanest illustration in this chapter of what a useless attribute looks like.
- Root = **Age**. Old → Down, New → Up, both pure. Only *Mid* needs a second level.

### Level 2, the Mid branch

4 rows, 2 Up and 2 Down,  $H = 1$ .

| Attribute   | Value    | P | N | H | Gain          |
|-------------|----------|---|---|---|---------------|
| Competition | Yes      | 0 | 2 | 0 | <b>1.0000</b> |
|             | No       | 2 | 0 | 0 |               |
| Type        | Software | 1 | 1 | 1 | 0             |
|             | Hardware | 1 | 1 | 1 |               |

**Answer.** Age = Old → Down;  
 Age = New → Up;  
 Age = Mid, Competition = Yes → Down;  
 Age = Mid, Competition = No → Up.

- Two tests, four leaves, and Type is never used – the tree *Insights* prints, and its arithmetic (0.6, 0.13, 0) checks out to the second decimal.
- The general shape to remember:** an attribute whose branches are all pure has residual 0 and therefore gain  $= H(S)$ , the maximum possible. Whenever you see one, stop comparing and take it.

## 6.N.6 Genetic algorithms: the method, then three runs

### The recipe every GA problem follows

- **1. Encode.** Say what one chromosome means, gene by gene. Almost every exam problem is a binary string with one bit per yes/no decision, or a string of digits.
- **2. Initialise.** Take the population the question gives you. If it gives none, write out four or five random strings and say they are random.
- **3. Evaluate.** Apply the fitness function to every chromosome, showing the substitution. **An infeasible chromosome scores 0**, and saying *why* it is infeasible earns the mark.
- **4. Select.** Rank them; keep the best, drop the worst.
- **5. Crossover.** State the point, draw the cut with a bar, write both offspring, and **evaluate them too**.
- **6. Mutate.** Flip one gene, say which, and evaluate.
- **7. Form the next generation** and compare its best and mean fitness against the previous one. **That comparison is the answer**; a GA solution without it is only half a solution.

## 6.N.7 Problem 2.1 – the knapsack, one full generation

**PROBLEM** not a PYQ · *Insights* Example 6.1, the standard GA demonstration

A thief has a knapsack of limited capacity that can hold almost 12 kg. Which items should be kept in the knapsack so that its value is maximized? Solve using a genetic algorithm.

| Item | Weight | Value |
|------|--------|-------|
| A    | 5 kg   | \$12  |
| B    | 3 kg   | \$5   |
| C    | 7 kg   | \$10  |
| D    | 2 kg   | \$7   |

### Step 1 Chromosome encoding

- Four items, so a **4-bit chromosome** in the fixed order  
A B C D.
- Gene = 1 means the item is in the knapsack, 0 means it is not. The search space is  $2^4 = 16$  chromosomes.

### Step 2 Initial population and fitness

Fitness = total value if total weight  $\leq 12$  kg, otherwise 0.

| Chromo     | Items   | Weight       | Value | Fitness   |
|------------|---------|--------------|-------|-----------|
| $C_1$ 0010 | C       | 7 kg         | 10    | 10        |
| $C_2$ 0110 | B, C    | 10 kg        | 15    | 15        |
| $C_3$ 1001 | A, D    | 7 kg         | 19    | <b>19</b> |
| $C_4$ 1111 | A,B,C,D | <b>17 kg</b> | 34    | <b>0</b>  |

- $C_4$  is the heaviest *and* the most valuable, and it scores **zero**: 17 kg will not fit in a 12 kg sack, so it is not a solution at all.
- Generation 1: best = 19, mean =  $\frac{10+15+19+0}{4} = 11.0$ .

### Step 3 Selection

- Rank:  $C_3$  (19),  $C_2$  (15),  $C_1$  (10),  $C_4$  (0).
- $C_4$  has zero fitness so a roulette wheel gives it **no slice at all** and it never reproduces. Take  $C_3$  and  $C_2$  as parents.

### Step 4 Single-point crossover after gene 2

|                 | String    | Weight | Fitness   |
|-----------------|-----------|--------|-----------|
| Parent $C_3$    | 1 0   0 1 | 7 kg   | 19        |
| Parent $C_2$    | 0 1   1 0 | 10 kg  | 15        |
| Offspring $O_1$ | 1 0   1 0 | 12 kg  | <b>22</b> |
| Offspring $O_2$ | 0 1   0 1 | 5 kg   | 12        |

–  $O_1 = 1010$  is A and C:  $5 + 7 = 12$  kg, exactly the capacity, value  $12 + 10 = 22$ . **Better than either parent**, which is crossover doing its job.

### Step 5 Mutation

- Flip **gene 2** of  $C_3 = 1001$ : 1001  $\rightarrow$  1101.
- 1101 is A, B and D:  $5 + 3 + 2 = 10$  kg, value  $12 + 5 + 7 = 24$ .

**Step 6 Generation 2**

| Chromosome     | Weight | Value | Fitness   |
|----------------|--------|-------|-----------|
| 1101 (mutant)  | 10 kg  | 24    | <b>24</b> |
| 1010 ( $O_1$ ) | 12 kg  | 22    | 22        |
| 1001 ( $C_3$ ) | 7 kg   | 19    | 19        |
| 0110 ( $C_2$ ) | 10 kg  | 15    | 15        |

- Best 19 → **24**; mean 11.0 → **20.0**. Both rose, which is what you must state.
- The two weakest of generation 1 ( $C_1$  and the infeasible  $C_4$ ) have been replaced by the offspring and the mutant.

**Answer.** Take A, B and D: 10 kg, value **\$24**. Chromosome **1101**.

**Termination, and the point of the whole problem**

- An exhaustive check of all 16 chromosomes confirms 24 is the **global optimum**; the next best feasible sacks are 1010 and 0111, both worth 22. So the run stops here, having converged.
- **Notice which operator found it.** Crossover took 19 to 22. It could never have reached 24, because *no chromosome in the population carried a 1 in gene B and gene D together* – crossover can only shuffle genes that already exist. Mutation supplied the missing bit.
- That is the complete answer to 81 Ba's "justify why mutation is important", and it is far stronger than quoting the definition.

**6.N.8 Problem 2.2 – fitness, ranking and crossover on eight-gene chromosomes****Rank the chromosomes by fitness, then perform the crossover operations** 2+8

75 Bh 2+8 · **the paper prints no chromosome list** – the sentence ends and Q7 follows · the four chromosomes below are the ones the standard version of this question carries (*Insights* Example 6.2) · the swarm-intelligence half of the question is answered in §6.9 of the theory

**QUESTION AS PRINTED** 75 Bh · Q6 · 2+8

What do you understand by swarm intelligence? Suppose chromosomes are of the form  $x = a b c d e f g h$  with a fixed length of eight genes. Each gene can be any digit between 0 and 9. Let the fitness of individual  $x$  be calculated as:

$$f(x) = (a + b) - (c + d) + (e + f) - (g + h)$$

and let the initial population consist of four individuals with the following chromosomes.

**DEFECT IN THE PAPER:** the sentence ends there and no chromosome list is printed. Confirmed against the scan. Worked below with the standard four.

**SUPPLIED** the four chromosomes the textbook version of this question prints

$x_1 = 65413532$   $x_2 = 87126601$   $x_3 = 23921285$   $x_4 = 41852094$

(a) Evaluate the fitness of each, and rank them fittest first. (b) Perform single-point crossover of the two fittest at the middle point, and a two-point crossover of  $x_1$  and  $x_3$ .

**(a) Step 1 Fitness of each chromosome**Read the eight digits as  $a$  to  $h$  in order, then substitute.

|       | Chromosome         | Substitution                                                | $f(x)$ |
|-------|--------------------|-------------------------------------------------------------|--------|
| $x_1$ | 6 5 4 1 3 5 3<br>2 | $(6+5) - (4+1) +$<br>$(3+5) - (3+2)$<br>$= 11 - 5 + 8 - 5$  | 9      |
| $x_2$ | 8 7 1 2 6 6 0<br>1 | $(8+7) - (1+2) +$<br>$(6+6) - (0+1)$<br>$= 15 - 3 + 12 - 1$ | 23     |
| $x_3$ | 2 3 9 2 1 2 8<br>5 | $(2+3) - (9+2) +$<br>$(1+2) - (8+5)$<br>$= 5 - 11 + 3 - 13$ | -16    |
| $x_4$ | 4 1 8 5 2 0 9<br>4 | $(4+1) - (8+5) +$<br>$(2+0) - (9+4)$<br>$= 5 - 13 + 2 - 13$ | -19    |

**Step 2 Rank****Answer (a).** Fittest first:  $x_2$  (23),  $x_1$  (9),  $x_3$  (-16),  $x_4$  (-19).Mean fitness of generation 1 =  $\frac{9+23-16-19}{4} = -0.75$ .

- **Read the fitness function before you compute:** genes  $a, b, e, f$  are *added* and  $c, d, g, h$  are *subtracted*, so a fit chromosome is large in positions 1, 2, 5, 6 and small in 3, 4, 7, 8.  $x_2$  is exactly that shape;  $x_4$  is its mirror image, and scores -19.
- The theoretical maximum is  $f = 36$  at 99009900, the minimum -36.

**(b) Step 3 Single-point crossover of the two fittest, at the middle**

Eight genes, so the middle point is after gene 4.

|       | String      | Substituti             | $f$ |
|-------|-------------|------------------------|-----|
| $x_2$ | 8712   6601 |                        | 23  |
| $x_1$ | 6541   3532 |                        | 9   |
| $O_1$ | 8712   3532 | $15 - 3 +$<br>$8 - 5$  | 15  |
| $O_2$ | 6541   6601 | $11 - 5 +$<br>$12 - 1$ | 17  |

- Both offspring beat  $x_1$ ; neither beats  $x_2$ . **Mean rises from -0.75 to 16** if the two weakest are replaced by  $O_1$  and  $O_2$ .

**Step 4 Two-point crossover of  $x_1$  and  $x_3$ , cuts after genes 2 and 6**

|       | String         | Substitu                    | $f$ |
|-------|----------------|-----------------------------|-----|
| $x_1$ | 65   4135   32 |                             | 9   |
| $x_3$ | 23   9212   85 |                             | -16 |
| $O_1$ | 65   9212   32 | $11 -$<br>$11 +$<br>$3 - 5$ | -2  |
| $O_2$ | 23   4135   85 | $5 - 5 +$<br>$8 - 13$       | -5  |

- In a two-point crossover the **middle segment** is exchanged and the two outer segments stay put.
- **Both offspring are worse than  $x_1$** , and that is worth stating: crossover does not guarantee improvement. It is *selection*, applied next generation, that throws these away and keeps the gains.

**6.N.9 Problem 2.3 – reaching a target fitness, and a textbook crossover that does nothing****PROBLEM** not a PYQ · *Insights* Example 6.3, reworked – see the note

You are given a coin. Find the generations using a genetic algorithm so that you can get the number of heads above 30. Take 5 chromosomes made up of 10 genes to proceed with the solution.

**Steps 1 and 2 Encoding and the initial population**

- Ten tosses per chromosome: gene 1 = **head**, 0 = **tail**.
- Fitness of a chromosome = its number of heads. The *objective* is on the population: **total heads** > 30 across all five.

|              | Chromosome          | $f$ = heads                 |
|--------------|---------------------|-----------------------------|
| $C_1$        | 1 0 0 0 1 0 1 1 0 0 | 4                           |
| $C_2$        | 1 1 1 0 0 1 0 1 1 0 | 6                           |
| $C_3$        | 1 0 1 0 1 1 0 0 1 0 | 5                           |
| $C_4$        | 0 1 1 1 1 0 1 0 1 1 | 7                           |
| $C_5$        | 0 1 1 1 0 0 1 1 1 1 | 7                           |
| <b>Total</b> |                     | <b>29</b> < 30, so continue |

**Step 3 Selection**

- The two fittest are  $C_4$  and  $C_5$ , both 7. Take them as parents.

**Step 4 Choosing the crossover point matters here**

- $C_4$  and  $C_5$  **share their first four genes**, 0111. Cut anywhere in the first four positions and the “offspring” are just the two parents swapped – no new chromosome exists and the population cannot improve.
- So cut **after gene 6**, where the parents actually differ.

|       | String        | $f$      |
|-------|---------------|----------|
| $C_4$ | 011110   1011 | 7        |
| $C_5$ | 011100   1111 | 7        |
| $O_1$ | 011110   1111 | <b>8</b> |
| $O_2$ | 011100   1011 | 6        |

**Step 5 Generation 2: keep the best five of the seven**

| Chromosome       | $f$       |          |
|------------------|-----------|----------|
| $O_1$ 0111101111 | <b>8</b>  | new best |
| $C_4$ 0111101011 | 7         |          |
| $C_5$ 0111001111 | 7         |          |
| $C_2$ 1110010110 | 6         |          |
| $O_2$ 0111001011 | 6         |          |
| <b>Total</b>     | <b>34</b> |          |

**Answer.** Total heads 29 → **34**, and the best individual 7 → **8**. The objective is reached in the **second generation**.

**Where the textbook goes wrong, and why it matters**

- *Insights* crosses the same two parents *inside* the four genes they share, so its  $O_1$  and  $O_2$  come out equal to  $C_5$  and  $C_4$ . Its total still reaches 34, but only because the best chromosome has been **counted twice**; the best individual never improves past 7.
- Cutting where the parents differ produces a genuinely new chromosome with 8 heads. **An examiner reading the two solutions can see which one understood crossover.**

**6.N.10 Problem 3.1 – fuzzy set operations**

**PROBLEM** not a PYQ directly, but the arithmetic behind every “explain fuzzy operations” answer

For the fuzzy sets  $A = \{(10, 0.2), (20, 0.3), (30, 0.4), (40, 0.5)\}$  and  $B = \{(10, 0.3), (20, 0.4), (30, 0.1), (40, 0.2)\}$  over the universe  $U = \{10, 20, 30, 40\}$ , find  $A \cup B$ ,  $A \cap B$ ,  $A'$ , the bold union  $A \oplus B$  and the bold intersection  $A \odot B$ .

**Work element by element**

Every fuzzy operation is defined *pointwise*: apply it to the membership values at each  $x$  independently. There is nothing else to it.

| $x$                                      | 10  | 20  | 30         | 40  |
|------------------------------------------|-----|-----|------------|-----|
| $\mu_A(x)$                               | 0.2 | 0.3 | 0.4        | 0.5 |
| $\mu_B(x)$                               | 0.3 | 0.4 | 0.1        | 0.2 |
| $A \cup B = \max$                        | 0.3 | 0.4 | 0.4        | 0.5 |
| $A \cap B = \min$                        | 0.2 | 0.3 | 0.1        | 0.2 |
| $A' = 1 - \mu_A$                         | 0.8 | 0.7 | <b>0.6</b> | 0.5 |
| $A \oplus B = \min(1, \mu_A + \mu_B)$    | 0.5 | 0.7 | 0.5        | 0.7 |
| $A \odot B = \max(0, \mu_A + \mu_B - 1)$ | 0   | 0   | 0          | 0   |

- **Bold intersection is empty** because no pair of memberships sums past 1. Say that; it is the check that you applied the formula rather than guessing.
- **Note**  $A \cap A' \neq \emptyset$ : at  $x = 40$  both are 0.5, so their intersection is 0.5. In crisp set theory that is impossible. This one line is the sharpest possible answer to “how does fuzzy logic differ from classical logic”, and the same goes for  $A \cup A' \neq U$ .

**Answers.**

|                      |                                                  |
|----------------------|--------------------------------------------------|
| $A \cup B$           | $\{(10, 0.3), (20, 0.4), (30, 0.4), (40, 0.5)\}$ |
| $A \cap B$           | $\{(10, 0.2), (20, 0.3), (30, 0.1), (40, 0.2)\}$ |
| $A' = \{1 - \mu_A\}$ | $\{(10, 0.8), (20, 0.7), (30, 0.6), (40, 0.5)\}$ |
| $A \oplus B$         | $\{(10, 0.5), (20, 0.7), (30, 0.5), (40, 0.7)\}$ |
| $A \odot B$          | $\{(10, 0), (20, 0), (30, 0), (40, 0)\}$         |

**One correction to the textbook**

- *Insights* p167 prints the complement as  $A' = \{(10, 0.8), (20, 0.7), (30, \mathbf{0.5}), (40, 0.5)\}$ .
- $\mu_A(30) = 0.4$ , so  $1 - 0.4 = \mathbf{0.6}$ , not 0.5. Every other value in that box is right; only this one is a slip.
- The check that catches it in one second:  $\mu_A(x) + \mu_{A'}(x)$  **must equal exactly 1 at every element**. Run it down the column before you hand the paper in.

### 6.N.11 Problem 3.2 — fuzzification: turning a crisp reading into a membership

**PROBLEM** not a PYQ · *Insights* p166, the step every fuzzy question needs

The linguistic term *medium speed* has a membership function that rises linearly from 0 at 7.5 km/h to 1 at 12.5 km/h. Fuzzify the readings 10 km/h and 9 km/h.

#### The one formula

On a rising limb from  $a$  to  $b$ :

$$\mu(x) = \frac{x - a}{b - a} \quad a \leq x \leq b$$

- **At 10 km/h:**  $\mu = \frac{10 - 7.5}{12.5 - 7.5} = \frac{2.5}{5} = 0.5$ , so the fuzzy pair is (10, 0.5).
- **At 9 km/h:**  $\mu = \frac{9 - 7.5}{5} = \frac{1.5}{5} = 0.3$ , so the pair is (9, 0.3).
- On a *falling* limb from  $b$  to  $c$  the formula is  $\mu(x) = \frac{c - x}{c - b}$ , and a triangle is one of each.

#### Two things this small problem teaches

- **The first number in the pair is the element, not the peak.** *Insights* writes the second answer as (10, 0.3), but the reading was 9 km/h, so the pair is (9, 0.3). The membership value is right; the label is not.
- A crisp reading normally fuzzifies to **more than one term at once**, because the membership functions overlap: 10 km/h might be *slow* to degree 0.5 and *medium* to degree 0.5 at the same time. That overlap is the whole reason fuzzy control is smooth, so never report only one term unless the reading really is outside every other set.

### 6.N.12 Problem 3.3 — a complete Mamdani fuzzy inference

**Which fuzzy answer does this paper want?** Sixteen papers ask something about fuzzy logic and they do not want the same answer. Decide from *what is asked for*, not from the word *fuzzy*. “explain fuzzy inference / the Mamdani method / the steps, with an example”, 5 marks or more → §6.N.12, the worked run below. “the architecture”, or “the major steps for developing a system based on fuzzy logic” → §6.N.13: those ask you to *build* the variables, membership functions and rule base that §6.N.12 is handed for free. A 4-mark short note, or a 2-mark “what is fuzzy logic” → neither of them. That is Ch 6 §6.7, and running a defuzzification for it spends twenty minutes earning four marks.

**Explain fuzzy inference / the Mamdani method / the steps of a fuzzy inference system, with an example** 1+2+5 2+6 3+7

8

**HOT 6/41** 73 Ma 3+7, 80 Bh 8, 79 Bh 2+6, 70 Ma 2+6, 80 Ash 1+2+5, 79 Ash 1+2+5 · the papers never supply numbers, so **you have to invent the example** — this one is small enough to write out in full in about ten minutes · 73 Ma is the only paper in the 41 that names **Mamdani**; the other five say only *fuzzy inference* or *the steps*, and this one run answers all six

**QUESTION AS PRINTED** 73 Ma · Q7 · 3+7

What is Fuzzy Logic and why is it important? Explain the Mamdani Fuzzy Inference Method with example.

**PROBLEM** the worked example, chosen so every number is checkable

A fan-speed controller has two inputs, temperature  $T$  (0–40 °C) and humidity  $H$  (0–100 %), and one output, fan speed  $S$  (0–100 %).

**Membership functions.** *Cold*: 1 up to 10, falling to 0 at 20. *Warm*: triangle 10–20–30. *Hot*: 0 at 20, rising to 1 at 30 and staying there. *Low humidity*: 1 up to 30, falling to 0 at 60. *High humidity*: 0 at 30, rising to 1 at 60. Output sets *Slow*, *Medium*, *Fast* are triangles peaking at 0, 50 and 100.

**Rules.** R1: IF  $T$  is Cold AND  $H$  is Low THEN  $S$  is Slow. R2: IF  $T$  is Warm THEN  $S$  is Medium. R3: IF  $T$  is Hot OR  $H$  is High THEN  $S$  is Fast.

Find the fan speed for  $T = 26$  °C and  $H = 45$  %.

**Step 1 Fuzzify both inputs**

Read each crisp value off every membership curve it touches.

**Term**      **Working at**  $T = 26$ ,  $\mu$   
 $H = 45$

Cold       $26 > 20$ , outside the set      0

Warm      falling limb:  $\frac{30-26}{30-20}$       **0.4**

Hot      rising limb:  $\frac{26-20}{30-20}$       **0.6**

Low      falling limb:  $\frac{60-45}{60-30}$       0.5

High      rising limb:  $\frac{45-30}{60-30}$       0.5

– 26 °C is **both warm and hot**, and 45 % is equally low and high. That is the point of fuzzy logic: no reading has to pick a side.

– Check:  $\mu_{Warm} + \mu_{Hot} = 1$  and  $\mu_{Low} + \mu_{High} = 1$ , as they must be where exactly two curves overlap linearly.

**Step 2 Fire the rules** (AND = min, OR = max)

**Rule**      **Strength**       $w$

R1       $\min(\mu_{Cold}, \mu_{Low}) = \min(0, 0.5)$       **0**

R2       $\mu_{Warm}$       0.4

R3       $\max(\mu_{Hot}, \mu_{High}) = \max(0.6, 0.5)$       0.6

– **R1 does not fire at all** ( $w = 0$ ), so *Slow* contributes nothing. Saying so explicitly is worth a mark.

**Answer.** Run the fan at **about 62 %**. Rules R2 and R3 fired at 0.4 and 0.6; R1 did not fire.

The six Mamdani operations, named against the steps above

- **1. Determine the rules** – R1, R2, R3, given.
- **2. Fuzzify the inputs** – Step 1.
- **3. Combine to get rule strength** – Step 2, fuzzy AND = min, OR = max.
- **4. Implication** – Step 3, clip each consequent set at its strength.
- **5. Aggregation** – Step 4, union the clipped sets with max.
- **6. Defuzzification** – Step 5, centre of gravity.

**Step 3 Implication: clip each output set at its rule strength**

– *Medium* is clipped at **0.4**, *Fast* at **0.6**. Mamdani implication is the **min** operator, so the top of each triangle is simply cut flat.

**Step 4 Aggregation: take the max of the clipped sets**

Sample the aggregated set every 10 units of  $S$ .

|                  |     |     |            |            |            |     |
|------------------|-----|-----|------------|------------|------------|-----|
| $S$              | 0   | 10  | 20         | 30         | 40         | 50  |
| clipped          | 0   | 0.2 | 0.4        | 0.4        | 0.4        | 0.4 |
| Medium           |     |     |            |            |            |     |
| clipped Fast     | 0   | 0   | 0          | 0          | 0          | 0   |
| <b>aggregate</b> | 0   | 0.2 | 0.4        | 0.4        | 0.4        | 0.4 |
| $S$              | 60  | 70  | 80         | 90         | 100        |     |
| clipped          | 0.4 | 0.4 | 0.4        | 0.2        | 0          |     |
| Medium           |     |     |            |            |            |     |
| clipped Fast     | 0.2 | 0.4 | 0.6        | 0.6        | 0.6        |     |
| <b>aggregate</b> | 0.4 | 0.4 | <b>0.6</b> | <b>0.6</b> | <b>0.6</b> |     |

**Step 5 Defuzzify by the centre of gravity**

$$S^* = \frac{\sum S_i \mu(S_i)}{\sum \mu(S_i)} = \frac{272}{4.4} = \mathbf{61.8\%}$$

$$\sum \mu = 0 + 0.2 + 0.4 + 0.4 + 0.4 + 0.4 + 0.4 + 0.4 + 0.6 + 0.6 + 0.6 = 4.4;$$

$$\sum S\mu = 2 + 8 + 12 + 16 + 20 + 24 + 28 + 48 + 54 + 60 = 272.$$

Integrating the true area instead of sampling gives 58.8 %, so the ten-point sum is accurate to about 3 %, which is well inside what any examiner expects.

**The shortcut, and when it is safe**

– If you are short of time, the **weighted average of the consequent peaks** is one line:

$$S^* = \frac{0.4(50) + 0.6(100)}{0.4 + 0.6} = \frac{80}{1.0} = 80\%$$

- **It is not the same answer**, because it ignores the shape and the overlap of the two output sets. Use it only when the question says *weighted average* or *centre of sums*; if it says **centroid** or **centre of gravity**, do the sampled sum above.
- **Two checks before you write the final number**: the answer must lie *inside* the output range (0–100 here), and it must sit between the peaks of the sets that fired (50 and 100). Both 61.8 and 80 pass; a number outside that band means an arithmetic slip.

**6.N.13 Problem 3.4 – designing the fuzzy system that Problem 3.3 runs**

**What are the major steps for developing a system based on fuzzy logic? Explain the architecture and working mechanism of a fuzzy inference system** 2+5 2+6

**PIN 2/41** 80 Ch 2+5, 82 Ba 2+6 · **these two do not want a defuzzification**: §6.N.12 is handed its membership functions and its three rules, and these papers ask *where those came from* · the 2-mark opener, *when do we use fuzzy logic*, is the list in Ch 6 §6.7

**QUESTION AS PRINTED** 80 Ch · Q7 · 2+5

When shall we use Fuzzy Logic? Explain with an appropriate example. What are the major steps for developing a system based on Fuzzy logic?

## QUESTION AS PRINTED 82 Ba · Q7 · 2+6

When do we use fuzzy logic? Explain the architecture and working mechanism of a fuzzy inference system.

### The seven design steps, in the order you do them

- 1. Confirm fuzzy logic fits.** Expert rules exist, no exact plant model does, and the boundaries are linguistic. If a threshold already works, do not fuzzify it.
- 2. Name the variables and their universes of discourse.** One per sensor and one per actuator, each with the range it physically spans.
- 3. Choose the linguistic terms for each variable.** Three to five per variable; **the rule count is the product**, so five terms on three inputs is 125 rules and unmaintainable.
- 4. Draw one membership function per term.** Triangles and trapezoids, and **they must overlap**: every point of the universe belongs to at least one set, and neighbours cross at about 0.5.
- 5. Build the rule base** as a matrix of every input-term combination against the action to take. The domain expert fills the cells, not the engineer.
- 6. Fix the operators:** AND = min, OR = max, implication = min (Mamdani), aggregation = max, and pick the defuzzifier (centroid unless the question says otherwise).
- 7. Simulate, tune, iterate.** Run known cases, move the membership breakpoints until the outputs match the expert's judgement, then deploy.

Steps 1–5 are design and happen once. Steps 4 and 6 are what the running system executes on every reading — that is the *working mechanism* half of 82 Ba, and its block diagram is the architecture figure in §6.7.

### The compression §6.N.12 uses is not a relabelling

- Whole rows and columns of that matrix agree, so it is tempting to write *Hot OR High* → *Fast* as one rule and be done. That is what §6.N.12's R3 does.
- It changes the numbers.** A matrix cell fires at  $\min(\mu_T, \mu_H)$ , so a fully high humidity is dragged down by a half-cold temperature. An OR rule takes max, and lets either antecedent carry the rule on its own.

| At $T = 26, H = 45$ | 3 OR rules | 6 cells |
|---------------------|------------|---------|
| Slow                | 0          | 0       |
| Medium              | 0.4        | 0.4     |
| Fast                | 0.6        | 0.5     |
| centroid            | 61.8 %     | 59.8 %  |

### Worked step 2–4 for the fan controller of §6.N.12

| Variable | Universe | Terms, and where they cross                                                                 |
|----------|----------|---------------------------------------------------------------------------------------------|
| $T$ in   | 0–40 °C  | Cold, Warm, Hot. Cold/Warm cross at <b>15</b> , Warm/Hot at <b>25</b> , both at $\mu = 0.5$ |
| $H$ in   | 0–100 %  | Low, High, crossing at <b>45</b>                                                            |
| $S$ out  | 0–100 %  | Slow, Medium, Fast, peaking at 0, 50, 100                                                   |

- This is why §6.N.12 read  $\mu_{Low} = \mu_{High} = 0.5$ :** its 45 % is exactly the crossover you placed here. A reading that lands on a crossover is the best sanity check a design gets.

### Step 5 The rule matrix, all six cells

|          | $H$ Low | $H$ High        |
|----------|---------|-----------------|
| $T$ Cold | Slow    | Fast            |
| $T$ Warm | Medium  | Medium and Fast |
| $T$ Hot  | Fast    | Fast            |

- The Warm/High cell naming two actions is not a bug.** Both fire, at their own strengths, and aggregation settles it. A crisp rule base would have to pick one and would jerk at the boundary.

- Over the whole input grid the two rule bases give different firing strengths at **1386 of the 4141** integer  $(T, H)$  points, and they are furthest apart at  $T = 15, H = 60$ : the OR rule fires Fast at **1.0** because the air is fully humid, the matrix at **0.5** because min sees a temperature that is only half cold.
- Which one to write:** either, but **say which**. The marks are for naming the operators and using them consistently, not for the crisp number — no paper in the 41 supplies data, so no paper has a right answer to compare against.
- Do not print the matrix and then solve with the OR rules without a word**, which is the one version that is simply inconsistent.

**Answer.** Seven steps: confirm the fit, name the variables and their ranges, choose linguistic terms, draw overlapping membership functions, fill the rule matrix, fix the operators and the defuzzifier, then simulate and tune. **Design ends where §6.N.12 begins.**

### What loses marks here

- **Answering with a defuzzification.** Neither paper gives numbers, and neither asks for a crisp output. Five or six marks are for the *procedure*.
- **Membership functions that do not overlap.** Then the system is a lookup table with fuzzy notation, and it jerks at every boundary. Say the overlap is deliberate.
- **Forgetting the 2-mark opener.** Both papers open with *when do we use fuzzy logic*, and it is a separate list: vague boundaries, expert rules available, no mathematical model, tolerance for imprecision. **§6.7.**

### Sources and verification

- All three PYQ data tables are transcribed from `AI/ocr/CT-653_CT-710_CT-78506_NewScan_OCR.md`, cell by cell.
- Method throughout follows *Insights on Artificial Intelligence* ch 6, checked independently. **Two of its numbers are wrong and are corrected here:** the complement in Problem 3.1 and the element label in Problem 3.2. **One of its methods is wrong:** the crossover in Problem 2.3.
- Every entropy, gain, fitness, membership and defuzzified value on these pages is recomputed from the problem statement by `verify.py`, which also proves the knapsack optimum by exhaustive search over all 16 chromosomes and confirms that *Insights'* crossover point produces offspring identical to its parents.

## 7. Applications of AI – Solved Problems

Every problem opens with the question **exactly as the paper prints it**, the paper's own spelling and slips included. · The gate networks (17 papers), the Hebbian AND net (4), the Hopfield weight matrix (1 at 6 marks), the printed forward-propagation network (2 at 8 marks each) and the parse tree (1). · Every weight, net input, activation and matrix entry printed here is recomputed independently by `verify.py`.

### Three things to know before you start.

- **Use bipolar values,  $\pm 1$ , not 0 and 1.** With binary inputs a 0 contributes nothing to any weight update, so the Hebb net for AND *cannot be learnt at all* and the perceptron takes six epochs instead of two. Every worked answer below uses  $\pm 1$  and says so in its first line.
- **“Design” and “train” are different questions.** Design a net for AND: choose weights, verify the four rows, done in five lines. Construct it using the perceptron / Hebb algorithm and show each calculation: run the table. Read which one the paper asked – 77 Ch says “show each calculation performed” and will not accept a designed answer.
- **Always finish with the verification table.** Four rows, *net* and *y* against the target. It is two minutes, it catches every arithmetic slip, and the marks for it are free.

### 7.N.1 Problem 1.1 – McCulloch-Pitts neuron for AND and OR

What is a McCulloch/Pitts neural network? Explain it with reference to the AND gate 8 10

PIN 2/41 75 Ba 8, 76 Ba 10 · both papers then ask *why it cannot do XOR*, which is Problem 1.4 · the MP neuron's weights are chosen, not learnt, so this problem is solved by inequalities, never by a training table

QUESTION AS PRINTED 75 Ba · Q9 · 8

What is a McCulloch/Pitts neural network? Explain it with reference to AND gate. Justify that it cannot be applied to Exclusive OR gate.

#### Step 1 Write the network and the firing condition

- Two inputs  $x_1, x_2$ , weights  $w_1, w_2$ , threshold  $T$ . The neuron fires ( $y = 1$ ) exactly when

$$x_1 w_1 + x_2 w_2 > T$$

- **Nothing is learnt.** The task is to find one  $(w_1, w_2, T)$  that makes the four rows come out right.

#### Step 2 One inequality per row of the truth table

| $x_1$ | $x_2$ | AND target | Condition       |       |
|-------|-------|------------|-----------------|-------|
| 0     | 0     | 0          | $0 \leq T$      | (i)   |
| 0     | 1     | 0          | $w_2 \leq T$    | (ii)  |
| 1     | 0     | 0          | $w_1 \leq T$    | (iii) |
| 1     | 1     | 1          | $w_1 + w_2 > T$ | (iv)  |

#### Step 3 Solve them

- (ii) and (iii) say each weight alone must not reach the threshold; (iv) says the two together must pass it. So pick  $w_1 = w_2 = w$  and **any  $T$  with  $w \leq T < 2w$** .
- Take  $w_1 = w_2 = 1$  and  $T = 1.5$ . Or, as the deck does,  $w_1 = w_2 = 0.7$  and  $T = 1$ .

#### Step 4 Verify all four rows

| $x_1$ | $x_2$ | $net = x_1 + x_2$ | $net > 1.5?$ | $y$ |
|-------|-------|-------------------|--------------|-----|
| 0     | 0     | 0                 | no           | 0 ✓ |
| 0     | 1     | 1                 | no           | 0 ✓ |
| 1     | 0     | 1                 | no           | 0 ✓ |
| 1     | 1     | 2                 | yes          | 1 ✓ |

**AND.  $w_1 = w_2 = 1, T = 1.5$ .**

#### The OR gate, same method

- Rows:  $0 \leq T; w_2 > T; w_1 > T; w_1 + w_2 > T$ . Now *either weight alone must pass*, so  $0 \leq T < w$ .
- Take  $w_1 = w_2 = 1, T = 0.5$ :  $net = 0$  (no), 1 (yes), 1 (yes), 2 (yes). ✓

#### NOT, for completeness

- One **inhibitory** input:  $w = -1, T = -0.5$ .  
 $x = 0 \rightarrow net = 0 > -0.5 \rightarrow y = 1$ ;  
 $x = 1 \rightarrow net = -1 \rightarrow y = 0$ . ✓
- **AND, OR and NOT together make every Boolean function**, so a *network* of MP neurons is universal even though one neuron is not. That sentence is what 76 Ba's “propose a neural network model” wants.

### 7.N.2 Problem 1.2 — train a perceptron to be an AND gate

Write down the perceptron algorithm and use it to construct a network which performs like a logical AND operation, showing each calculation **2+6 2+2+4**

**HOT 4/41** 77 Ch **2+6**, 73 Bh **2+6**, 78 Po **2+2+4**, 79 Jth **1+6** · 77 Ch says “show each calculation performed”, so the table below is the answer and a designed set of weights is not · 79 Jth says “acts as logic gates”, so give AND and add OR from Problem 1.3

**QUESTION AS PRINTED** 77 Ch · Q9 · **2+6**

Why do you need Multilayer Perceptron Neural Networks? Write down the perceptron algorithm and use it to Construct Network which perform like Logical AND Operation and show each calculation performed.

#### Step 1 Set up

- **Bipolar** inputs and targets,  $\pm 1$ . Bias input fixed at  $+1$ .
- Learning rate  $\alpha = 1$ , threshold  $\theta = 0$ .
- Start with  $w_1 = w_2 = b = 0$ .
- **Activation:**  $y = 1$  if  $net > 0$ ,  $y = 0$  if  $net = 0$ ,  $y = -1$  if  $net < 0$ .
- **Update, only when  $y \neq t$ :**

$$w_i \leftarrow w_i + \alpha t x_i, \quad b \leftarrow b + \alpha t$$

- **$net = 0$  counts as an error**, because  $y = 0$  never equals a target of  $\pm 1$ . That is what makes the very first row update.

Truth table, in bipolar form

| $x_1$ | $x_2$ | AND | $t$ |
|-------|-------|-----|-----|
| 1     | 1     | 1   | +1  |
| 1     | -1    | 0   | -1  |
| -1    | 1     | 0   | -1  |
| -1    | -1    | 0   | -1  |

#### Step 2 Epoch 1

| $(x_1, x_2)$ | $net$ | $y$ | $t$ | Update                    | $w_1$ | $w_2$ | $b$ |
|--------------|-------|-----|-----|---------------------------|-------|-------|-----|
| start        |       |     |     |                           | 0     | 0     | 0   |
| (1, 1)       | 0     | 0   | +1  | $y \neq t$ , add $t x_i$  | 1     | 1     | 1   |
| (1, -1)      | 1     | 1   | -1  | $y \neq t$                | 0     | 2     | 0   |
| (-1, 1)      | 2     | 1   | -1  | $y \neq t$                | 1     | 1     | -1  |
| (-1, -1)     | -3    | -1  | -1  | <b>correct, no change</b> | 1     | 1     | -1  |

Sample working, row 2:  $net = b + w_1 x_1 + w_2 x_2 = 1 + 1(1) + 1(-1) = 1$ , so  $y = 1$  but  $t = -1$ . Then  $w_1 = 1 + (-1)(1) = 0$ ,  $w_2 = 1 + (-1)(-1) = 2$ ,  $b = 1 + (-1) = 0$ .

#### Step 3 Epoch 2 — the verification pass

| $(x_1, x_2)$ | $net = -1 + x_1 + x_2$ | $y$ | $t$ |   |
|--------------|------------------------|-----|-----|---|
| (1, 1)       | $-1 + 1 + 1 = 1$       | 1   | +1  | ✓ |
| (1, -1)      | $-1 + 1 - 1 = -1$      | -1  | -1  | ✓ |
| (-1, 1)      | $-1 - 1 + 1 = -1$      | -1  | -1  | ✓ |
| (-1, -1)     | $-1 - 1 - 1 = -3$      | -1  | -1  | ✓ |

**Answer.** No weight changed in epoch 2, so the perceptron has converged after one epoch of updates:  $w_1 = 1$ ,  $w_2 = 1$ ,  $b = -1$ . Decision line  $x_1 + x_2 = 1$ .

### 7.N.3 Problem 1.3 — train a perceptron to be an OR gate

How can we design a neural network that acts as an OR gate? **2+7**

72 Ma **2+7** · **OR converges even faster than AND** — one row of updates and it is done · only the target column changes from Problem 1.2, so if you have written that one, say so and change the column

**QUESTION AS PRINTED** 72 Ma · Q8 · **2+7**

What do you understand by a perceptron? How can we design a neural network that acts as an OR gate.

## Same setup, one column different

| $(x_1, x_2)$ | $net$ | $y$ | $t$ | $w_1$ | $w_2$ | $b$ |
|--------------|-------|-----|-----|-------|-------|-----|
| start        |       |     |     | 0     | 0     | 0   |
| (1, 1)       | 0     | 0   | +1  | 1     | 1     | 1   |
| (1, -1)      | 1     | 1   | +1  | 1     | 1     | 1   |
| (-1, 1)      | 1     | 1   | +1  | 1     | 1     | 1   |
| (-1, -1)     | -1    | -1  | -1  | 1     | 1     | 1   |

- **Only the first row ever updates.** After it, every remaining pattern is already classified correctly, so the weights are untouched.
- Epoch 2 confirms:  $net = 3, 1, 1, -1$  against targets  $+1, +1, +1, -1$ . ✓

**Answer.**  $w_1 = 1, w_2 = 1, b = +1$ . Decision line  $x_1 + x_2 = -1$ .

## AND and OR side by side, which is worth a sentence

|                   | AND                | OR                 |
|-------------------|--------------------|--------------------|
| $w_1, w_2$        | 1, 1               | 1, 1               |
| Bias $b$          | -1                 | +1                 |
| Fires when        | both inputs are +1 | at least one is +1 |
| Epochs of updates | 1 (three updates)  | 1 (one update)     |

- **The weights are identical; only the bias moves.** The bias slides the same decision line across the square: with  $b = -1$  it cuts off the single corner (1, 1), with  $b = +1$  it cuts off the single corner (-1, -1).
- That is the cleanest statement of what a bias is for, and it is the answer to 79 Jth's "can we design a neural network that acts as logic gates": yes, and the same network does both, with one number changed.
- **NAND** is AND with the targets flipped, giving  $w_1 = w_2 = -1, b = +1$ ; and **NOR** is OR flipped. Both are needed in Problem 1.4.

## 7.N.4 Problem 1.4 – XOR: the proof, and the network that does work

Is a single perceptron sufficient to realize a network behaving as an XOR gate? Justify. How can we design one that does?

6 10 1+8 2+6 1+7

**TOP 8/41** 73 Ma 1+7, 74 Ma 10, 74 Bh 1+8, 82 Bh 2+6, 79 Ash 1+8, 77 Ch 2+6, 76 Ba 10, 75 Ba 8 · eight papers, and the answer has two halves: prove one neuron cannot, then build one that can. Doing only the first half costs half the marks, and doing only the second costs more

**QUESTION AS PRINTED** 79 Ash · Q8 · 1+8

What is perceptron? Is single perceptron sufficient to realize a network behaving as XOR gate? Justify.

**Part 1 Why one perceptron cannot: the geometric answer**

- Plot the four inputs on the unit square. XOR outputs 1 at (0, 1) and (1, 0), and 0 at (0, 0) and (1, 1).
- The two 1s are at **diagonally opposite** corners, and so are the two 0s. **No straight line puts one pair on one side and the other pair on the other.**
- A single neuron computes  $w_1x_1 + w_2x_2 + b$  and thresholds it, and the boundary of a threshold is exactly a straight line. So it cannot be done. **Draw the square with the two attempted lines.**

**Part 1 again The algebraic proof, if the paper says “justify”**

- Assume  $w_1, w_2, b$  exist with output 1 iff  $w_1x_1 + w_2x_2 + b > 0$ :
  - (i)  $(0, 0) \rightarrow 0: b \leq 0$
  - (ii)  $(0, 1) \rightarrow 1: w_2 + b > 0$
  - (iii)  $(1, 0) \rightarrow 1: w_1 + b > 0$
  - (iv)  $(1, 1) \rightarrow 0: w_1 + w_2 + b \leq 0$
- Adding (ii) and (iii):  $w_1 + w_2 + 2b > 0$ , so  $w_1 + w_2 > -2b$ .
- But (iv) gives  $w_1 + w_2 \leq -b$ . Hence  $-2b < -b$ , i.e.  $b > 0$ , contradicting (i).
- No such weights exist. ■**
- Say what this does not mean:** it is a limit on *one* neuron, not on neural networks. XOR is not a hard function; it is a *non-linearly-separable* one.

**Part 2 The two-layer network that does work**

- The idea in one line:**  
 $x_1 \oplus x_2 = (x_1 \vee x_2) \wedge \neg(x_1 \wedge x_2)$ . So build OR and AND in a hidden layer and combine them at the output.
- The network** (binary inputs, step activation, fire if  $net > 0$ ):
  - Hidden  $h_1 = \text{OR}$ :  $w = (1, 1), b = -0.5$
  - Hidden  $h_2 = \text{AND}$ :  $w = (1, 1), b = -1.5$
  - Output  $y$ :  $w = (+1 \text{ from } h_1, -1 \text{ from } h_2), b = -0.5$

**Verification, all four rows**

| $x_1$ | $x_2$ | $net_1$ | $h_1$ | $net_2$ | $h_2$ | $y$      |
|-------|-------|---------|-------|---------|-------|----------|
| 0     | 0     | -0.5    | 0     | -1.5    | 0     | -0.5 → 0 |
| 0     | 1     | 0.5     | 1     | -0.5    | 0     | 0.5 → 1  |
| 1     | 0     | 0.5     | 1     | -0.5    | 0     | 0.5 → 1  |
| 1     | 1     | 1.5     | 1     | 0.5     | 1     | -0.5 → 0 |

**Answer.** A single perceptron cannot. A **2-2-1 multilayer perceptron** can: hidden units computing OR and AND, output computing  $h_1 - h_2 - 0.5$ . Output column matches XOR exactly.

- Why it works, geometrically:** each hidden unit draws one straight line, and the output unit takes the strip *between* them. Two lines can isolate a diagonal pair; one cannot.
- And why it needs back-propagation to be learnt:** the hidden units have no targets of their own, so no perceptron rule can train them. This is the answer to 77 Ch's “why do you need multilayer perceptron networks” and to 79 Ch's “when do we need the back propagation algorithm”. One step of it is Problem 2.2.

**7.N.5 Problem 1.5 – a Hebbian network for AND**

**Define Hebbian learning. Use the Hebbian learning algorithm to construct a network which performs the AND function**

2+6 1+7 3+7 6+3

**HOT 4/41** 73 Bh 3+7, 81 Ba 2+6, 78 Ba 1+7, 78 Ch 6+3 · **one pass, no iteration, no error term** — that is what makes it different from Problem 1.2 and it is what the examiner checks · 78 Ch also asks *is Hebbian learning supervised? Justify*

**QUESTION AS PRINTED** 81 Ba · Q8 · 2+6

What is perception? Construct a Hebbian Network that performs like AND Gate.

**Step 1 The rule**

- Bipolar inputs and targets; bias input fixed at +1; all weights start at 0.
- For each training pair, **set the output to the target** and update:

$$w_i \leftarrow w_i + x_i t, \quad b \leftarrow b + t$$

- **No learning rate, no comparison, no second pass.** Every pattern updates, whether the current weights would have got it right or not.

**Step 2 One pass through the four patterns**

| $(x_1, x_2)$ | $t$ | Working                        | $w_1$ | $w_2$ | $b$ |
|--------------|-----|--------------------------------|-------|-------|-----|
| start        |     |                                | 0     | 0     | 0   |
| (1, 1)       | +1  | $0+(1)(1), 0+(1)(1), 0+1$      | 1     | 1     | 1   |
| (1, -1)      | -1  | $1+(1)(-1), 1+(-1)(-1), 1-1$   | 0     | 2     | 0   |
| (-1, 1)      | -1  | $0+(-1)(-1), 2+(1)(-1), 0-1$   | 1     | 1     | -1  |
| (-1, -1)     | -1  | $1+(-1)(-1), 1+(-1)(-1), -1-1$ | 2     | 2     | -2  |

**Step 3 Verify**

| $(x_1, x_2)$ | $net = 2x_1 + 2x_2 - 2$ | $y$ | $t$ |   |
|--------------|-------------------------|-----|-----|---|
| (1, 1)       | $2 + 2 - 2 = 2$         | 1   | +1  | ✓ |
| (1, -1)      | $2 - 2 - 2 = -2$        | -1  | -1  | ✓ |
| (-1, 1)      | $-2 + 2 - 2 = -2$       | -1  | -1  | ✓ |
| (-1, -1)     | $-2 - 2 - 2 = -6$       | -1  | -1  | ✓ |

Answer.  $w_1 = 2, w_2 = 2, b = -2$ , from one pass. Decision line  $x_1 + x_2 = 1$  – the same line the perceptron found in Problem 1.2, at twice the scale.

**Is Hebbian learning supervised? (78 Ch 3)**

- **Yes, as used here.** The update contains  $t$ , the target, so the network is being told the right answer for every pattern. Without  $t$  there is no AND net.
- **But Hebb's original postulate is not.** *Neurons that fire together wire together* describes a purely local, unsupervised change in a synapse; nothing in it mentions a desired output.
- So the honest answer is both sentences: the principle is unsupervised, the *Hebb net for pattern association* is supervised because it substitutes the target for the output.
- **Why bipolar is compulsory here.** With binary values, the pattern (0, 0) contributes  $0 \times t = 0$  to both weights, so it can never push the weights away from a wrong answer. Run the same table in binary and the AND net comes out wrong. Say this if you have room; it is the trap the question is built on.

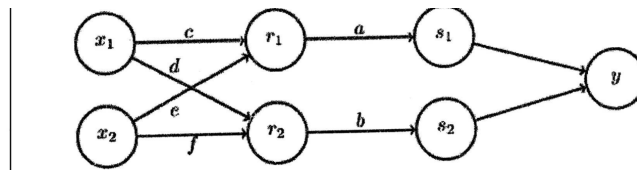
**7.N.6 Problem 2.1 – forward propagation through the printed network**

Perform forward propagation, and update the weights of the second hidden layer, a and b 8

PIN 2/41 82 Ka 8, 81 Ch 8 · the same question printed twice, with the same figure · the paper gives no target value and no learning rate, so both are assumed below and stated · the interesting part is what happens to c, d, e and f

QUESTION AS PRINTED 82 Ka · Q7 · 8 · identical at 81 Ch · Q9

Below is a neural network with weights a, b, c, d, e, and f. The input are  $x_1$  and  $x_2$ . The first hidden layer computes  $r_1 = \max(c \cdot x_1 + e \cdot x_2, 0)$  and  $r_2 = \max(d \cdot x_1 + f \cdot x_2, 0)$ . The second hidden layer and third layer computes  $s_1, s_2$  and  $y$  using sigmoid activation function. Suppose the network has inputs  $x_1 = -1$  and  $x_2 = -1$ . The weight values are  $a = 1, b = 1, c = 4, d = 1, e = 2$  and  $f = 2$ . Perform forward propagation. And update weight for second hidden layer for a and b.



**Step 1 Read the network off the figure**

- $x_1 \rightarrow r_1$  with weight  $c$ ,  $x_2 \rightarrow r_1$  with weight  $e$ ;  $x_1 \rightarrow r_2$  with  $d$ ,  $x_2 \rightarrow r_2$  with  $f$ .
- $r_1 \rightarrow s_1$  and  $r_2 \rightarrow s_2$  apply the **sigmoid**.
- $s_1 \rightarrow y$  with weight  $a$ ,  $s_2 \rightarrow y$  with  $b$ , then sigmoid again.  
**No biases anywhere**, which the figure confirms.
- $\max(z, 0)$  is the **ReLU**;  $\sigma(z) = 1/(1 + e^{-z})$ .

**Step 2 First hidden layer**

$$r_1 = \max(cx_1 + ex_2, 0) = \max(4(-1) + 2(-1), 0) \\ = \max(-6, 0) = 0$$

$$r_2 = \max(dx_1 + fx_2, 0) = \max(1(-1) + 2(-1), 0) \\ = \max(-3, 0) = 0$$

**Step 3 Second hidden layer and output**

$$s_1 = \sigma(r_1) = \sigma(0) = \frac{1}{1+e^0} = 0.5$$

$$s_2 = \sigma(r_2) = \sigma(0) = 0.5$$

$$net_y = as_1 + bs_2 = 1(0.5) + 1(0.5) = 1$$

$$y = \sigma(1) = \frac{1}{1+e^{-1}} = 0.7311$$

**Forward pass.**  $r_1 = r_2 = 0$ ,  $s_1 = s_2 = 0.5$ ,  $y = 0.7311$ .

**Step 4 Update a and b**

The paper gives no target and no learning rate. Assume  $t = 1$  and  $\eta = 0.1$ , and say so – the method is what is marked, and any other pair of values substitutes into the same three lines.

- Loss  $E = \frac{1}{2}(y - t)^2$ . Output delta:

$$\delta = (y - t)y(1 - y)$$

$$= (0.7311 - 1)(0.7311)(0.2689) = -0.05288$$

- Gradients, since the input to both weights is 0.5:

$$\frac{\partial E}{\partial a} = \delta s_1 = -0.05288 \times 0.5 = -0.02644$$

and  $\partial E/\partial b$  is the same, because  $s_1 = s_2$ .

- Update:

$$a \leftarrow a - \eta \frac{\partial E}{\partial a} = 1 - 0.1(-0.02644) = 1.00264$$

$$b \leftarrow b - \eta \frac{\partial E}{\partial b} = 1.00264$$

- **Both weights rise**, and by the same amount, because the output was below the target and both units fed it the same 0.5.

**The part worth extra marks: c, d, e and f cannot be updated at all**

- The gradient reaching  $c$  runs through  $r_1 = \max(-6, 0)$ . The derivative of ReLU at a negative input is 0, so  $\partial E/\partial c = \partial E/\partial d = \partial E/\partial e = \partial E/\partial f = 0$ .
- Both hidden units are **dead for this input**: no gradient flows past them, so the first layer learns nothing from this example. This is the textbook **dying-ReLU** problem, and it is precisely why the question asks you to update only  $a$  and  $b$ .
- Saying that explicitly is what turns a correct arithmetic answer into a complete one.

**7.N.7 Problem 2.2 – one full step of back-propagation**

**Explain the back-propagation algorithm with an example** 4 2+6 3+5 4+4

**TOP 8/41** 70 Bh 4+4, 72 Ash 2+4+4, 70 Ma 3+5, 80 Bh 2+2+4, 74 Bh 4, 79 Ch 2+6, 71 Ma 3+5, 72 Ash 3+5 · **no paper supplies numbers**, so this is the standard XOR network with the weights every deck uses · one input pattern and one step is enough for 6 marks; do not attempt a full epoch

**PROBLEM** not a PYQ · the 2-2-1 XOR network, with the weights the lecture decks use

Train the network below on the XOR pattern  $x = (1, 1)$ , target  $t = 0$ , for one step. Sigmoid activation, loss  $E = \frac{1}{2}(y - t)^2$ , learning rate  $\eta = 0.1$ . Initial weights  $w_{13} = 0.5$ ,  $w_{14} = 0.9$ ,  $w_{23} = 0.4$ ,  $w_{24} = 1.0$ ,  $w_{35} = -1.2$ ,  $w_{45} = 1.1$ ; biases  $b_3 = -0.8$ ,  $b_4 = 0.1$ ,  $b_5 = -0.3$ .

**Step 1 Forward pass**

$$net_3 = x_1 w_{13} + x_2 w_{23} + b_3 = 0.5 + 0.4 - 0.8 = 0.1$$

$$h_3 = \sigma(0.1) = \frac{1}{1+e^{-0.1}} = \mathbf{0.5250}$$

$$net_4 = x_1 w_{14} + x_2 w_{24} + b_4 = 0.9 + 1.0 + 0.1 = 2.0$$

$$h_4 = \sigma(2.0) = \mathbf{0.8808}$$

$$net_5 = h_3 w_{35} + h_4 w_{45} + b_5$$

$$= 0.5250(-1.2) + 0.8808(1.1) - 0.3 = 0.0389$$

$$y_5 = \sigma(0.0389) = \mathbf{0.5097}$$

$$\text{Error } E = \frac{1}{2}(0.5097 - 0)^2 = \mathbf{0.1299}.$$

**Step 2 Output delta**

$$\delta_5 = (y_5 - t) y_5 (1 - y_5)$$

$$= (0.5097)(0.5097)(0.4903) = \mathbf{0.12738}$$

$y(1 - y)$  is the derivative of the sigmoid, and the forward pass already gave you  $y$  – that is the whole reason sigmoid is used.

**Step 3 Hidden deltas**

$$\delta_3 = \delta_5 w_{35} h_3 (1 - h_3)$$

$$= 0.12738(-1.2)(0.5250)(0.4750) = \mathbf{-0.03812}$$

$$\delta_4 = \delta_5 w_{45} h_4 (1 - h_4)$$

$$= 0.12738(1.1)(0.8808)(0.1192) = \mathbf{0.01471}$$

- $\delta_3$  is negative because  $w_{35}$  is: unit 3 pushes the output down, so to reduce an output that is too high you must push unit 3 up. The sign carries real information; do not drop it.

**Step 4 Update every weight,  $w \leftarrow w - \eta \delta$  (its input)**

| Weight   | Working                       | Old    | New                |
|----------|-------------------------------|--------|--------------------|
| $w_{35}$ | $-1.2 - 0.1(0.12738)(0.5250)$ | $-1.2$ | $\mathbf{-1.2067}$ |
| $w_{45}$ | $1.1 - 0.1(0.12738)(0.8808)$  | $1.1$  | $\mathbf{1.0888}$  |
| $b_5$    | $-0.3 - 0.1(0.12738)$         | $-0.3$ | $\mathbf{-0.3127}$ |
| $w_{13}$ | $0.5 - 0.1(-0.03812)(1)$      | $0.5$  | $\mathbf{0.5038}$  |
| $w_{23}$ | $0.4 - 0.1(-0.03812)(1)$      | $0.4$  | $\mathbf{0.4038}$  |
| $b_3$    | $-0.8 - 0.1(-0.03812)$        | $-0.8$ | $\mathbf{-0.7962}$ |
| $w_{14}$ | $0.9 - 0.1(0.01471)(1)$       | $0.9$  | $\mathbf{0.8985}$  |
| $w_{24}$ | $1.0 - 0.1(0.01471)(1)$       | $1.0$  | $\mathbf{0.9985}$  |
| $b_4$    | $0.1 - 0.1(0.01471)$          | $0.1$  | $\mathbf{0.0985}$  |

**Answer.**  $y_5$  moved from 0.5097 towards the target 0, and every one of the nine parameters changed in the direction that reduces  $E$ . Repeat for the other three XOR patterns to complete one epoch.

**Three checks before you stop**

- **The output-layer weights move most.** They are nearest the error, and  $\delta$  shrinks as it is multiplied by weights and by  $h(1 - h) \leq 0.25$  on the way back. Ten layers of that is the **vanishing gradient** problem, named in §7.3.
- Every  $\Delta w$  here is under 0.02, which is what  $\eta = 0.1$  is supposed to give. A change of order 1 means a sign or a factor is wrong.
- Nothing outside the weights and biases changed. If your answer alters an activation or an input, you have misread the algorithm.

**7.N.8 Problem 2.3 – design a Hopfield network and recall from noise**

**Design a Hopfield network to store three binary patterns of length 4. Compute the weight matrix and demonstrate recall from a noisy input** 6

82 Bh 6 · the paper chooses no patterns, so choose three and say why · three patterns in four units is above the usual  $0.15N$  capacity, so pick orthogonal ones – that choice is the first mark

QUESTION AS PRINTED 82 Bh · Q7 · 6

Design a Hopfield network to store three binary patterns of length 4. Show how to compute the weight matrix and demonstrate the network's ability to recall a pattern from a noisy input.

**Step 1 Choose the patterns, and convert to bipolar**

|       | Binary  | Bipolar $s$    |
|-------|---------|----------------|
| $P_1$ | 1 1 1 1 | [1, 1, 1, 1]   |
| $P_2$ | 1 0 1 0 | [1, -1, 1, -1] |
| $P_3$ | 1 1 0 0 | [1, 1, -1, -1] |

- Convert with  $s_i = 2b_i - 1$ : a 1 stays +1, a 0 becomes -1. The training rule needs bipolar values or the outer products come out wrong.
- Why these three:** they are mutually orthogonal,  $P_i \cdot P_j = 0$  for  $i \neq j$ . Check:  $P_1 \cdot P_2 = 1 - 1 + 1 - 1 = 0$ ,  $P_1 \cdot P_3 = 1 + 1 - 1 - 1 = 0$ ,  $P_2 \cdot P_3 = 1 - 1 - 1 + 1 = 0$ .
- Orthogonal patterns do not interfere, so all three are stable even though  $3 > 0.15 \times 4$ . **Say this:** it is the design decision the question is testing.

**Step 2 The weight matrix**

$$w_{ij} = \sum_{p=1}^3 s_i^{(p)} s_j^{(p)} \quad (i \neq j), \quad w_{ii} = 0$$

- Worked entry,  $w_{12}$ :  
 $(1)(1) + (1)(-1) + (1)(1) = 1 - 1 + 1 = \mathbf{1}$ .
- Worked entry,  $w_{14}$ :  
 $(1)(1) + (1)(-1) + (1)(-1) = 1 - 1 - 1 = \mathbf{-1}$ .
- $w_{ji} = w_{ij}$ , so only six entries have to be computed.

$$W = \begin{bmatrix} 0 & 1 & 1 & -1 \\ 1 & 0 & -1 & 1 \\ 1 & -1 & 0 & 1 \\ -1 & 1 & 1 & 0 \end{bmatrix}$$

**Step 3 Check that all three patterns are stable**

$net_i = \sum_j w_{ij} s_j$ , then  $y_i = +1$  if  $net_i > 0$ ,  $-1$  if  $net_i < 0$ .

|       | net vector     | Sign             |
|-------|----------------|------------------|
| $P_1$ | [1, 1, 1, 1]   | [1, 1, 1, 1] ✓   |
| $P_2$ | [1, -1, 1, -1] | [1, -1, 1, -1] ✓ |
| $P_3$ | [1, 1, -1, -1] | [1, 1, -1, -1] ✓ |

Sample working for  $P_2$ , unit 1:

$net_1 = 0(1) + 1(-1) + 1(1) + (-1)(-1) = -1 + 1 + 1 = 1 > 0 \rightarrow +1$ , which is what  $P_2$  has. Every pattern reproduces itself, so all three are stable states.

**Step 4 Recall from a noisy input**

- Corrupt  $P_2 = 1010$  in its first bit: present  $x = 0010$ , i.e.  $y = [-1, -1, 1, -1]$ .

| Update | $net_i = \sum_j w_{ij} y_j$   | New $y_i$          | State          |
|--------|-------------------------------|--------------------|----------------|
| $y_1$  | $1(-1) + 1(1) + (-1)(-1) = 1$ | +1<br><b>flips</b> | [1, -1, 1, -1] |
| $y_2$  | $1(1) + (-1)(1) + 1(-1) = -1$ | -1                 | unchanged      |
| $y_3$  | $1(1) + (-1)(-1) + 1(-1) = 1$ | +1                 | unchanged      |
| $y_4$  | $(-1)(1) + 1(-1) + 1(1) = -1$ | -1                 | unchanged      |

**Answer.** The state converged to [1, -1, 1, -1], i.e. binary **1010** =  $P_2$ . The corrupted bit was repaired in one update, and a second sweep changes nothing, so the network has converged.

- Update one unit at a time and use the new value immediately** – that is what *asynchronous* means, and updating all four together is the commonest error on this question.
- Two honest remarks worth a mark each: this net also has **spurious stable states** it was never taught (the complement of every stored pattern is always one), and four units cannot reliably hold three patterns unless they are orthogonal, which is why they were chosen that way in step 1.

**7.N.9 Problem 3.1 – syntactic analysis and a parse tree****Explain syntactic analysis and draw a parse tree for the given sentence** 5+4

76 Bh 5+4 · the only NLP question in 41 papers that asks for something *drawn* · the sentence is four words and the tree has eight nodes, so this is four marks in three minutes

**QUESTION AS PRINTED** 76 Bh · Q9 · 5+4

Explain the steps of Natural Language Processing. Explain syntactic analysis and draw a parse tree for the following sentence. "A girl has a name."

**Step 1 Tag every word**

| Word | Part of speech       | Symbol  |
|------|----------------------|---------|
| A    | article (determiner) | Article |
| girl | noun                 | Noun    |
| has  | verb                 | Verb    |
| a    | article              | Article |
| name | noun                 | Noun    |

**Step 2 Write the grammar rules you will use**

- Sentence → Noun\_phrase Verb\_phrase
- Noun\_phrase → Article Noun
- Verb\_phrase → Verb Noun\_phrase
- Article → a | the      Noun → girl | name  
Verb → has
- **Print the rules beside the tree.** Each level of the tree is one rule application, and showing the rules is half of what “explain syntactic analysis” means.

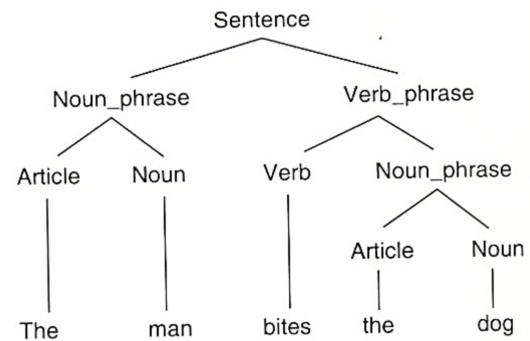
**Step 3 The parse tree**

```

Sentence
|-- Noun_phrase
|   |-- Article ---- "A"
|   `-- Noun ----- "girl"
`-- Verb_phrase
    |-- Verb ----- "has"
    `-- Noun_phrase
        |-- Article - "a"
        `-- Noun ---- "name"

```

Draw it as a tree with the sentence's words along the bottom. The published figure below is the same grammar on “The man bites the dog”, which is the same shape node for node – copy its layout and change the leaves.



**Figure 10.3** Parse tree for the sentence “The man bites the dog.”

- **Two checks:** every word of the sentence appears as exactly one leaf, in the original order; and every internal node is the left-hand side of one grammar rule whose right-hand side is its children.
- **Syntactic analysis in three lines**, which is the other half of the marks: it takes a tagged sentence and produces a representation of its **grammatical structure**; a **grammar** says which parts of speech exist and how they combine into phrases; the **parse tree** records which rule was applied where. It **rejects** “The school goes to boy”, and it **accepts** “Colourless green ideas sleep furiously”, which semantic analysis then rejects. Those two examples together define what syntax does and does not do.

**Sources and verification**

- Question text and the network figure are transcribed from AI/ocr/CT-653\_CT-710\_CT-78506\_NewScan\_OCR.md and AI/ocr/CT-653\_CT-
- Method follows BA Sir’s CH-07 worked examples (the Hebb net, the Adaline run and the back-propagation step) and *Insights on Artificial Intelligence* §7.1.10 for the Hopfield training and testing algorithm, checked independently.
- **Every number on these pages is recomputed by verify.py:** both perceptron runs epoch by epoch, the Hebb weights, an exhaustive search proving no integer weights realise XOR, the two-layer XOR truth table, the whole Hopfield weight matrix with all three stability checks and the recall sweep, the forward propagation, and all nine back-propagation updates.

## Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

| Short | Means                                                | Short   | Means                                                                      |
|-------|------------------------------------------------------|---------|----------------------------------------------------------------------------|
| AI    | artificial intelligence                              | ANI     | artificial narrow intelligence                                             |
| AGI   | artificial general intelligence                      | ASI     | artificial super intelligence                                              |
| KR    | knowledge representation                             | KRR     | knowledge representation and reasoning                                     |
| PEAS  | performance measure, environment, actuators, sensors | GPS     | General Problem Solver                                                     |
| NLP   | natural language processing                          | NLU     | natural language understanding                                             |
| NLG   | natural language generation                          | CAPTCHA | completely automated public Turing test to tell computers and humans apart |
| ML    | machine learning                                     | ANN, NN | artificial neural network                                                  |
| MLP   | multilayer perceptron                                | BP      | back-propagation                                                           |
| SOM   | self-organizing map                                  | GA      | genetic algorithm                                                          |
| FIS   | fuzzy inference system                               | ID3     | Iterative Dichotomiser 3                                                   |
| CSP   | constraint satisfaction problem                      | DFS     | depth first search                                                         |
| BFS   | breadth first search                                 | DLS     | depth limited search                                                       |
| IDS   | iterative deepening search                           | UCS     | uniform cost search                                                        |
| CNF   | conjunctive normal form                              | WFF     | well-formed formula                                                        |
| FOL   | first-order logic                                    | FOPL    | first-order predicate logic                                                |
| RRS   | resolution refutation system                         | BBN     | Bayesian belief network                                                    |

| Short   | Means                      | Short  | Means                                                                                       |
|---------|----------------------------|--------|---------------------------------------------------------------------------------------------|
| ES      | expert system              | KB     | knowledge base                                                                              |
| CD      | conceptual dependency      | LISP   | list processing (the language)                                                              |
| MOP     | memory organization packet | isa    | the class-to-class link of a semantic net ( <i>instance</i> is the individual-to-class one) |
| attr(s) | attribute(s)               | repo   | repository                                                                                  |
| freq    | frequency                  | config | configuration                                                                               |
| info    | information                | no.    | number                                                                                      |
| Eg      | example                    | Vs     | compare / differentiate                                                                     |
| w/, w/o | with, without              | +Fig   | draw the figure                                                                             |
| →       | leads to, therefore        | =      | is, equals                                                                                  |

## Reading the year tags and chips

**Exam-paper month codes:** Ba = Baishakh · Jth = Jestha · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Po = Poush · Ma = Magh · Ch = Chaitra. Years are Bikram Sambat, so 80 Ch is 2080 Chaitra.

**Bold** year tags are Regular exams, plain are Back exams. A header printing “Regular / Back” in one box counts as Regular.

**Three course codes, three faces.** Plain = CT 653 (BCT, III-II, 26 papers) · monospace = CT 710 (BEI, IV-I, 7 papers) · *italic* = CT 78506 (BEX, IV-II, 8 papers). **The same year code names different papers under different codes:** 77 Ch and 77 Ch are two different exams, and only the font tells them apart.

**Tier chips** over all 41 papers, 2069–2082 BS: **TOP /41** = topic asked in **8 or more** papers · **HOT /41** = **4 to 7** papers · **PIN /41** = **1 to 3** papers but worth  $\geq 6$  marks.

**Mark chips** such as **2+6** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic never yet examined but written up anyway. **verified/added** marks anything researched and added that no source note in this folder covers.